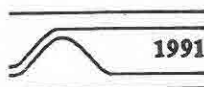


SPUDM - 13 ; 1991 FRIBOURG, SWITZERLAND

thirteenth research conference
on subjective probability,
utility and decision making



fribourg, switzerland

18-23 august

SPUDM



XIII



S P U D M - 1 3

Σ Π Υ Δ M - 1 3

S P U D M - 1 3

S P U D M - 13

PAPERS

**THIRTEENTH RESEARCH CONFERENCE ON
SUBJECTIVE PROBABILITY, UTILITY
AND DECISION MAKING**

18 - 23 AUGUST, 1991 IN FRIBOURG/SWITZERLAND

P A P E R S

INDEX

	page
<u>Major Papers</u>	
BREHMER Berndt (Uppsala University): Dynamic decision making: Human control of complex systems	3
LIEBRAND Wim B.G. (University of Groningen): How to improve our understanding of social decision making with the help of computers	28
MOSHKOVICH Helen M. (Institute Systems Studies Moscow): Qualitative methods in multiattribute decision making	44
PAYNE John W., BETTMAN James R., COUPEY Eloise & JOHNSON Eric J. (University of Durham): Multiple strategies in judgment and choice: A constructive process view of decision making	80
VLEK Charles (University of Groningen) & KEREN Gideon (University of Utrecht): Behavioral decision theory and environmental risk management: What have we learned and what has been neglected?	124
WATSON Stephen R. (University of Cambridge): The presumptions of prescription	125
<u>Workshops</u>	
BROWN Rex V. (USA) & VARI Anna (Hungary): Prescriptive decision science	144
MAULE John A. (UK) The effects of time pressure and stress on judgment/decision making	145
ROHRMANN Bernd (Germany): Risk Communication	146
RUSSO Edward J. (USA): Adaptive and constructive decision making	147
SCHMALHOFER Franz (Germany): Knowledge acquisition for expert systems	148
SHANTEAU James (USA): Decision making perspective on experts	149
<u>Specific Papers</u>	150
<u>Authors List</u>	215

DYNAMIC DECISION MAKING: HUMAN CONTROL OF COMPLEX SYSTEMS*

Berndt Brehmer

Uppsala University

Our research on dynamic decision making has its origin in analyses of decision making in applied contexts, first in analyses of military decision making and later in analyses of the work of process operators, in emergency management, and in intensive care. Two things emerged from these analyses.

The first was that decision making as such was never the primary focus of activity. Instead, decision making was always in aid of some goal. A quote from Klein (personal communication, 1986), describing his work with fire chiefs, summarizes much of our experience too. When asked about their decisions on the fire ground, the fire chiefs said: "*We do not make decisions, we fight fires!*". Our first task, therefore, was to consider the *function* served by decision making in these and other contexts. This led us to think of the function of decision making as part of an attempt to *gain control*, i.e., as an attempt to achieve some desired state of affairs.

The second thing to emerge from our analyses was the *dynamic character* of the tasks facing decision makers in the applied contexts that we analyzed. Thus, the decisions could usually only be understood as part of an ongoing process, and the decision problems conformed to Edwards' (1962) classical description of dynamic decision making in that

- (1) A series of decisions is required to reach the goal. That is, to achieve and maintain control is a continuous activity requiring not one, but many decisions, and each of these decisions could only be understood in the context of the other decisions.
- (2) The decisions are not independent. That is, later decisions are constrained by earlier decisions, and would in turn constrain those that would come after them.
- (3) The state of the decision problem changes, both autonomously and as a consequence of the decision maker's actions.

In an important later paper, Rapoport (1975) provided a more formal definition, but this did not change the general meaning of dynamic decision making compared to Edwards' original definition.

*Grants from the Swedish Council of Research in the Humanities and Social Sciences supported the work on this paper as well as the experiments by the author reviewed herein.

However, the three characteristics mentioned by Edwards did not quite capture the essence of the decision problems we studied. We also had to add another feature:

(4) the decisions have to be made in real time.

Thus, in these decision problems, it was not sufficient to make correct decisions and to make them in the correct order; the decisions also had to be made at the correct point in time. In short, dynamic decisions are decisions in *context* and in *time*. This means that the decision maker must consider the consequences of each decision for future decisions, that he or she is constrained by earlier decisions, and that he may also sometimes be able to correct problems caused by earlier decisions in later decisions. The latter feature of dynamic decisions has also been emphasized by Hogarth (1981).

The real time character of the decisions creates special problems for the decision maker (Brehmer, 1990a). It means, first, that decision makers are not free to make decisions when they themselves feel ready for the decisions. Instead, they have to make the decisions when the environment demands decisions from them. This introduces an element of stress into dynamic decision making, and creates a need to control one's own working situation. Consequently, dynamic decisions will often be a compromise between a good strategy for coping with the decision task, and a strategy that enables the decision maker to exert some measure of control over the rate at which he has to make decisions. This kind of compromise is often observed in process operators (Bainbridge, 1979), but the general idea that cognitive behavior is compromise between the demands of the task and the need to conserve one's cognitive resources goes back to Bruner, Goodnow and Austin (1956) and their careful analysis of the processing demands introduced by different forms of concept attainment strategies. The work of Bruner, et al. must be seen as one of the earliest attempts to study dynamic decision making, although it is generally not thought of in these terms.

A second consequence of the real time feature is that both that which the subjects seeks to control and the means that he uses for control must be seen as *processes*. Indeed, we may define the task facing the decision maker in dynamic decision making as *the problem of finding a way to use one process to control another process* (Brehmer & Allard, 1991a).

A third consequence is the need to consider the various time scales of the dynamic decision task, and to make sure that all the relevant time scales are monitored (see Brehmer, 1990b, for a general discussion of time scales in dynamic tasks, and A. Brehmer, 1989, for an example from intensive care).

All this may seem very abstract, but it is, perhaps, easier to understand if we consider one of the experimental paradigms that we have developed for the study of dynamic decision making (Brehmer & Allard, 1991a). This paradigm requires the subject to assume the role of a fire chief, charged with the task of fighting forest fires. Even though

we have a lot of forest in Sweden, we cannot create a real forest fire every time we want to do an experiment, so we have developed a computer simulation of forest fire fighting, and the subject thus fights computer simulated forest fires. The general characteristics of the experimental paradigm are shown in Figure 1. As shown in that fi-

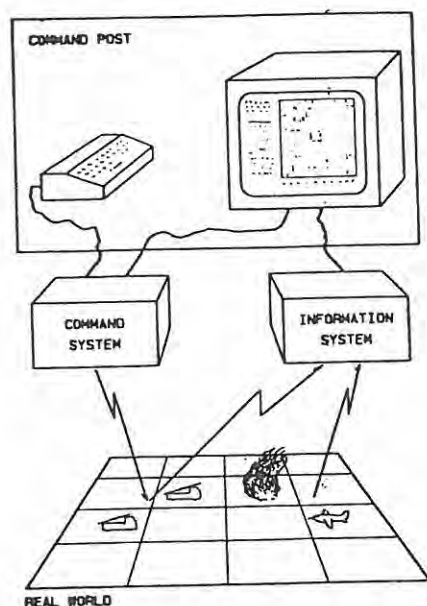


Figure 1. The structure of the fire fighting simulation developed to study dynamic decision making (from Brehmer & Allard, 1991a)

gure, the subject receives information about fires from a spotter plane via an information system, and sends out commands to his fire fighting units (FFUs) via a command system by typing commands on the keyboard. The FFUs report back to the fire chief (= the subject) about their actions and location, i.e., whether they are en route to some destination, fighting fire, or are simply inactive and waiting for new commands. On the basis of these reports and the further information from the spotter plane, the subject then issues new commands until the fires have been extinguished. The progress of the simulation is controlled by the computer's clock and it does not stop and wait for the subject to think or make decisions.

As can be seen from this description, the problem facing the subject in this simulation has exactly the characteristics described above. Thus, (1) it requires a series of decisions, (2) the decisions are not independent for as resources are committed by one decision, they are unavailable in later decisions, and current decisions are similarly constrained by earlier decisions, even though there is also a possibility of correcting errors in earlier decisions, (3) the state of the problem changes, both autonomously (fires start and burn according to prevailing meteorological conditions) and as a function of the decision maker's actions (the subject's attempts to extinguish the fire will change its development), and (4) the decisions have to be made in real time (or at least in simulated real time).

The reader will note that many of the features of real fire fighting are left out of this simulation. This is done on purpose to simplify matters for the subject, so as to cut down the time required for the experiments. It is presumably of little consequence for the results of our experiments, for they are not designed to study fire fighting such (had this been our purpose, we would not have stayed in the laboratory). Instead, we are interested in the more general problem of how subjects cope with spatio-temporal processes of the kind exemplified by the fire fighting task.

It is not difficult to think of a host of decision problems that fit the general description above. We have already mentioned military contexts, emergency management, patient care in hospitals, modern process plants, but they abound also in management of companies, and in our personal decision making in everyday life. Despite this, they have not received much attention from psychologists. There are at least two reasons for this. The first is that it is hard, or impossible, to cast these problems into the standard normative theory of decision making that has guided so much psychological research on decision making, and to find analytical solutions to these decision problems (Rapoport, 1975; Slovic, Fishhoff & Lichtenstein, 1977), and the second, that they require new experimental methods.

The problem for standard decision theory lies in that the decision trees for these kinds of problems rapidly become impossibly large and despite this, such decision trees still fail to incorporate the real time aspect.

As for methods, experiments on dynamic decision making simply cannot be made with the ordinary paper and pencil methods because of the dynamic and interactive nature of the experimental tasks. Experimentation in these areas is impossible without interactive computer programs, and computers powerful enough for this kind of work have only recently become standard equipment in psychological laboratories.

Despite these problems, there have been some attempts to study dynamic decision making. We now turn to a brief review of this work.

puts. Feedback control is possible when there are no significant feedback delays in the system, and it is useful when the system to be controlled changes over time, so that no stable model of the system can be constructed. Feedforward control requires that the system is stable, so that the model remains valid, but it can then use a model to overcome the problems of feedback delays. It is, of course, also possible to use a combination of these two strategies. Modern applications of control theory in the automatic control of systems often rely on a model to produce the actual control inputs, while the feedback information is used to update the model.

Brehmer (1990a) has hypothesized that feedback control, which is cognitively simpler, would be the preferred mode of control in dynamic decision tasks. The results presented in Brehmer (1990a) support the hypotheses, but as we shall see, things are a little more complicated than they first seem.

The strategy issue is important because the kind of model that is needed for control depends on the strategy adopted. Thus, one cannot study the subjects' models without first assessing their strategy.

A general framework for research is only a first step. We also need an experimental paradigm that allows us to study dynamic decision making empirically. This paradigm should allow us to define observability and action possibilities in psychologically meaningful ways, and to observe or infer the subjects' goals and mental models. We now turn to this problem.

Computer simulated microworlds - a research paradigm for the study of dynamic decision making

Dynamic decision making can, of course, be studied in the field, but this is difficult, for the complexity of most dynamic systems outside that laboratory prohibits such an approach. For example, to study the mental models of process operators, the researcher would usually have to spend months and months trying to learn the process of which the operator is supposed to have a model. Moreover, even if one understands the process, the task of collecting data is still formidable, and some form of experimental intervention is usually required to obtain useful data (see Hoc, 1989, for a good example). It is therefore not surprising that no theory of dynamic decision making has emerged from field studies so far. It is therefore necessary to supplement field studies with more laboratory based methods. The remainder of this paper will be concerned with some of these methods and the results obtained with them.

As noted above, dynamic decision problems are characterized by the fact that the state of the problem changes both autonomously and as a consequence of the decision maker's actions, and that the decisions have to be made in real time. To capture these features in an ordinary paper and pencil experiment is not feasible, and laboratory research on dynamic decision making therefore had to await the arrival of a new research tool: the *laboratory computer*.

Laboratory computers make it possible to create dynamic simulations and to study how subjects interact with such simulations. We have termed such simulations "microworlds" (Brehmer & Dörner, 1990). Such microworlds simulate some of the essential features, but not all the detail of some dynamic system; they present a "wood cut" (Holzschnitt), to use Dörner's (1991) term, of systems such as a small town, a developing country, an industrial process or a forest fire and charge the subject with the task of controlling this system for some period of time.

Such microworlds are designed to reflect three intuitively striking characteristics of real world decision problems: *complexity*, *dynamics* and *opaqueness*. They are complex in the sense that they require the subjects to consider many different elements, e.g., many different, and possibly conflicting, goals. They are dynamic in some, or all, of the four senses described above. They are opaque in that they do not reveal all their characteristics automatically to the subject, thus requiring him to form and test hypotheses about their state and characteristics.

Microworlds differ in the extent to which they emphasize the one or the other of these characteristics. Microworlds such as Lohhausen (Dörner, et al., 1983) or Moro (Dörner, Stäudel & Strohschneider, 1986) developed by Dörner and his associates, who work in the German tradition of research on "komplexes Problemlösen", are characterized by high complexity and opaqueness while microworlds such as DESSY (Brehmer & Allard, 1991a) and NEWFIRE (Brehmer & Lövborg, 1991), which are designed to study decision making, play down complexity and opaqueness and emphasize dynamics. One practical consequence of this that the progress of the latter kinds of simulations is controlled by the computer's clock. In simulations of the former kind, the subjects go through a number of moves, where the simulation stops and waits for input from the subject before going on to the next state.

A recent experiment by Brehmer, Allard & Jansson (1991) suggests that this difference in simulations has the same effects as a variation in rate of change in the process to be controlled: overall performance is better in a discrete time simulation, but the relative effects of a factor such as feedback delays remains the same. This suggests that the results from the two kinds of simulation are compatible and we will ignore the distinction between discrete time and continuous time simulations in the remainder of this paper.

Experiments with microworlds

As noted by Brehmer and Dörner (1990), the use of computer simulated microworlds requires us to rethink the way we do experiments. In such microworld experiments, the actual decision problems are not directly controlled by the experimenter. Instead, they depend on the behavior of the subject. The experimenter can create a system with certain characteristics, i.e., a given level of complexity, but the trajectory through this system is determined by the subject: the state of the system at any given moment is dependent both on the general characteristics of the system *and* on the decisions made by

Earlier approaches to the study of dynamic decision making

The problem of dynamic decision making was introduced to psychologists by Edwards and by Toda in two papers published in 1962. As noted by Rapoport (1975) the main purpose of Edwards' paper was to extend SEU theory from static problems to dynamic problems, and to propose a general research strategy: the normative-descriptive approach, for its study. This strategy involves constructing an "ideal decision maker" similar to the "ideal observer" in psychophysics, and then comparing the behavior of real decision makers to that of the ideal decision maker. If systematic discrepancies are found, these may then be interpreted as evidence of the limitations imposed on the actual decision maker, e.g., limitations of memory, limitations in the ability to predict the effects of one's decisions into the future, and so on. Hence, it would become possible to model the human decision maker as a case of constrained optimization. Rapoport (1975) provides examples of how this can be done for some sequential decision problems.

A major problem with this approach is that it is impossible to find analytical solutions for most dynamic decision problems (Rapoport, 1975). An added problem, also discussed by Rapoport (1975), is the "flat maximum problem". That is, for many dynamic problems, even wide discrepancies from the optimal strategy have very little effect on the actual outcomes of the decisions. Hence, it is difficult to find the limitations imposed on the actual decision maker using this strategy. It is therefore not surprising that the normative-descriptive strategy has not attracted the great following among those who study dynamic decision making that it has among those who focus on static problems.

Toda's (1962) paper introduced a one-person game "the fungus eater" as a paradigm for the study of dynamic decision making (among other things). However, it proved possible to find analytical solutions only for very simplified versions of the game, and its empirical potential was never fully realized. The general approach of constructing a computer simulation as a device for studying dynamic decision making was prophetic of what was to come, however.

Subsequent analyses of dynamic decision making have largely abandoned the normative ambitions and concentrated on an empirical analysis. Before turning to a review of these results, we will discuss a general framework for the study of dynamic decision making.

Towards a framework for studying dynamic decision making

As noted above, it is not feasible to analyze dynamic decision problems in terms of classical decision theory. We therefore need an alternative framework to guide research. Here, the observation that decisions are made to achieve some overall goal is helpful. As noted above, a general formulation of this observation is to think of the goal of decision making as that of *achieving control*, i.e., that decisions are made to achieve some desired state of affairs, or to keep some system in some desired state.

This is not an original observation, Broadbent, Fitzgerald and Broadbent (1986) made it earlier in their work with dynamic problems as did Rapoport (1975) and Mackinnon and Wearing (1985). It suggests a general framework for research on dynamic decision making: that of *control theory*. Control theory is a branch of engineering, and this seems quite appropriate; from the functionalist psychological perspective adopted by the present author, it makes more sense to think of man as an intuitive engineer than as an intuitive scientist.

This is, of course, not the first to use control theory as a theory of behavior. This theory has been quite fashionable in studies of manual control (see, e.g., Kelley, 1968; Powers, 1971). However, like Bainbridge (1981), we have not found the mathematical framework of control theory particularly helpful. There is no need to repeat Bainbridge's excellent discussion here, except to mention that a major problem with control theory as a psychological framework lies in its inability to handle the kind of input descriptions that are natural in the analysis of human perception: patterns and Gestalts. Instead, it requires that the input be described in terms of measured individual signals.

Therefore, control theory is useful mainly as metaphor, and its usefulness stems from the fact that it specifies the general conditions for control of any system, regardless of whether control is exercised manually or by some automatic controller. There are four general preconditions:

- there must be a goal
- it must be possible to ascertain the state of the system (the observability condition)
- it must be possible to affect the state of the system
- there must be a model of the system

Properly speaking, two of the conditions pertain to the system to be controlled while the other two are properties of the decision maker. Thus, observability and action possibilities are properties of the system while the goals and the models are properties of the decision maker. The most general formulation of the problem for research, therefore, is that it is concerned with people's formulation of goals and models as a function of the observability and action possibilities of the system to be controlled. This is not quite as straightforward as it seems, however. First, the concepts of observability and action possibilities are quite abstract, and it is a real research problem to ascertain what constitutes observability and action possibilities for a human decision maker. Second, there is also the problem of *strategy*. A system may be controlled either by means of a feedback strategy or a feedforward strategy (or some combination). In feedback control, the controller uses only the current information about the system, and takes this as information about the actual state of the system. In feedforward control, the controller uses a model of the system to predict its state and to select the appropriate control in-

the subject. Thus, we need to think of microworld experiments in cybernetic terms, rather than in the linear causal terms usually preferred by psychologists, a point also made by Rapoport (1975) in an earlier SPUDM major paper on dynamic decision making.

The independent variables these experiments, then, are *system characteristics*, such as complexity, feedback structure, and the like, but not specific aspects of individual decision problems, such as probabilities and utilities. Therefore, it, makes little sense to study individual responses. Instead, we must focus on the decision maker's *tactics* and *strategies*. It is only at this level that we can hope to find regularities in the data from our experiments. This poses special problems of data reduction. It may well be necessary to use simulation to identify these strategies, i.e., some actual trajectory is compared to those expected on the basis of some ideal strategies, and the strategy that exhibits the best fit is taken as that actually describing what the subject did (see Reichert & Dörner, 1987, for an example).

Data reduction is indeed a very pressing problem in these experiments because an experimental run will produce thousands of individual observations. Moreover, the trajectory of each individual in these simulations may well be unique. This has led Dörner (1991) to take the view that we need a *case study approach* to these simulations. That is, we must study the individual subjects in great detail and describe their individual trajectories and use them to infer the relevant psychological processes. To do so, we must construct computer simulations of the subjects' behavior. This is reminiscent of Newell and Simon's (1972) approach to the study of problem solving. So far, there are no examples of this strategy in the study of dynamic decision making, however.

Despite the individual trajectories, it is, however, often quite possible to obtain highly stable data, provided that some care is taken in the data reduction process. Specifically, it is necessary to find the right level of analysis, i.e., a level that reflects the subjects' strategies. This is illustrated in the highly stable and psychologically meaningful relations among various performance measures and measures showing how subjects used their resources in the fire fighting simulation (e.g., Brehmer, 1990a).

Research problems in the study of dynamic decision making

Research on dynamic decision making has followed two different approaches: an individual differences approach and a more standard experimental approach.

The individual differences approach has been pursued mainly by Dörner and his associates. In the typical experiment in approach, all subjects perform in the same complex simulation, and they are then divided into two (or more) groups according to how they performed in the simulation. These groups are then compared with respect to their performance on some test, or with respect to their behavior in the simulation, to find possible explanations for the differences in performance (see Dörner, et al., 1983, and Stäudel, 1987, for examples).

In the experimental approach, subjects' performance in simulations with different characteristics, e.g., feedback delays, side effects or complexity, is compared (see Brehmer & Allard, 1991a, Funke, 1990, for examples).

Such experiments are not only undertaken to investigate the effects of system characteristics, but for a variety of purposes. For example, Broadbent and his associates have used different dynamic simulations to study the issue of awareness and to formulate alternatives to the serial, STM-LTM-model of human information processing (e.g., Berry & Broadbent, 1988, 1989). However, in this paper we will leave these kinds of uses out since they have little to say about dynamic decision making.

In the remainder of this paper, we will review some of empirical results obtained with complex dynamic microworlds.

Empirical examples

This section is in two parts. In the first, we review the results from the individual differences approach, in the second those from the experimental approach.

The individual differences approach

As already noted above, the principal method in this approach has been to have a group of subjects perform in the same microworld, and then divide them up into two extreme groups, one that succeeds and one that does not succeed. These groups are then compared with respect to their behavior, and/or with respect to their scores on various psychological tests.

Research following this approach could have two different goals: to *find ways of predicting behavior* in dynamic decision tasks, so that it would then be possible to select good decision makers, and to *identify the demands* made by these tasks by comparing the behavior of the subjects who are successful in controlling the microworld with that of those who are not so successful.

Prediction of performance

Results from various simulations show that there are wide individual differences in performance. Despite this, it has proved difficult to find any significant correlations between performance in microworld experiments and test performance. Performance in the microworld experiments does not seem to correlate with any standard psychological test, be they tests of intelligence or tests of personality (Dörner, et al., 1983; Stäudel, 1987). There is some evidence that performance in microworlds may correlate with intelligence, but only when the opaqueness is removed (Putz-Osterloh & Lüer, 1987, Schoppek, 1991). Some results indicate that a new variable defined by Dörner and his associates, and operationalized by Stäudel in a questionnaire might correlate with per-

nagers performed better because they approached the decision task in a different way. Compared to students, the managers were less activistic, i.e., they made fewer decisions, they collected more information before they made their decisions, and they checked on the results of their decisions before making new decisions. This is exactly what is required to perform well in a complex dynamic system (the behaviors reflect what Dörner and Schölkopf, 1989, have called "grandmother rules") and may be seen as an expression of greater heuristic competence on the part of the managers, i.e., they had learned some general ways of coping with the demands made by complex dynamic systems.

To be successful, this approach requires that reliable performance measures can be found. A study by Strohschneider (1986) suggests that this is problematic, at least for Moro, one of the commonly used simulations. He found no significant correlations between performance in two consecutive runs with Moro. This suggests that one possible explanation for the failure to find stable correlations between performance in microworlds and test performance may be that the criterion variable lacks reliability. This is hardly surprising. The subjects' performance in a microworld is dependent not only upon their performance, but also on the characteristics of the microworld. In some microworld scenarios, even a small and seemingly unimportant mistake may lead to serious problems later on, from which the subject cannot recover, even if he follows a good strategy. Thus, we should, perhaps, not be too surprised that it has proved so difficult to find correlations between microworld performance and test performance.

The demands that complex, dynamic systems make

The second goal of the individual differences approach is to ascertain the demands made by complex, dynamic microworlds. The first method here has been to compare the behavior of extreme groups, e.g., the best and the worst third of the subjects, on psychological tests. The rationale behind this approach is that if we find, for example, that subjects who score high on spatial ability perform better than those who score low on these tests, we might conclude that the simulation in question requires spatial ability. As already noted above, the search for such correlations has not been very successful.

The second method has been to analyze the differences in actual behavior in the simulation for extreme groups. This has not produced any surprising results. It seems that subjects who collect more information, who collect it more systematically, who construct adequate goals, who evaluate the effects of their decisions, and generally behave in a systematic fashion tend to perform better than those who do not (Dörner, et al, 1983). Thus we see the "grandmother rules" in operation once more. However, the results of Dörner, et al (1983) point to one important conclusion: those subjects who behave in a way that makes it more likely that they will acquire a good model of the task also learn to control the task better (see also Funke & Müller, 1988).

So far, it has not proved possible to improve performance of subjects by training them in the general methods for handling complex systems. For example, in the Lohhausen

study, Dörner et al. (1983) compared two different training programs, one designed to give general strategic insight and one designed to teach useful tactics for coping with complex dynamic systems. Subjects in the control group received no special instructions. There were no differences among the three groups with respect to performance. Since the training programs can be seen as an attempt to induce heuristic competence, these results may, at first, seem to be in conflict with those of Stäudel and of Schaub and Strohschneider with respect to heuristic competence. However, it may well be that the skills involved here cannot be taught in the abstract, i.e., as a set of general rules, but that they can only develop in conjunction with actual practice in controlling systems. Such practice may, for example, teach the decision makers *when* and *how* to apply a given rule. This problem clearly requires further study.

Since simulations such as Moro and Lohhausen are designed to mirror at least some of the characteristics of actual systems such as developing countries (Moro) and towns (Lohhausen), it therefore seems reasonable to expect that *epistemic competence*, i.e., domain knowledge, would be a powerful factor in these experiments. This aspect has, however, not received much attention. Hasselrot and Brehmer (1991) compared a group of university students taking courses in the agriculture, ecology and economics of developing countries with a group of psychology students. They found initial differences favouring the students with high epistemic competence, but in the long run both groups produced similar dismal conditions in Moro. This was due to that the subjects with high epistemic competence did exhibit any evidence of heuristic competence. Specifically, they did not check on the results of their decisions as they should have done, and as a consequence, they ignored the development of one of the crucial variables in Moro: the ground water level. These results suggest, then, that both *epistemic and heuristic competence* are needed, but if a subject can have only one of these, heuristic competence may well be more useful than epistemic competence.

The pathologies of decision making

While the study of successful subjects seems to have uncovered no more than that these subjects, for whatever reason, exhibit the kind of behaviors that would be expected to be useful in these kinds of systems, the study of unsuccessful subjects has led to some novel and interesting results. These studies have uncovered a number of general maladaptive behaviors that could be called *pathologies of decision making*. These pathologies are not necessarily the cause of failure, but are observed as subjects try to cope with their failures (Dörner, 1980). Dörner (1989) theorizes that these behaviors may be the consequence of feelings of low competence that result from failure. They then lead to even greater failure in a self-reinforcing cycle.

The first of these pathologies is called "thematic vagabonding", and it refers to a tendency to shift goals. Thus, a subject in, say, Moro, may start out doing something about the water supply. When this fails to solve all the problems, he may then concentrate on improving education, then shift back to working on the water supply, and then move on to do something about storage, and so on. Thus, subjects who exhibit thematic

vagabonding shift from one subgoal to another, but fail to work on the problem as a whole, which is what is needed in a complex system such as Moro.

The second pathology is called "encystment". It involves sticking to one specific goal that the subject feels competent to achieve. Thus, a subject who has great faith in education may stick to this method as the method which is to solve all problems in Moro (see Dörner, et al., 1983, for examples).

A third pathology is a general refusal to make any decisions at all.

A fourth pathology involves blaming others for one's failures (Many of these programs have simulated others who have to carry out the subject's commands). One of the most likely candidates here is, of course, the simulation itself, a response similar to that of blaming "the system" for one's failures.

A fifth pathology is that of delegating responsibility that cannot, or should not, be delegated.

It is interesting to note that these pathologies can be divided into two groups. The first group comprises the first two pathologies. These pathologies can be seen as failures of the goal formulation process. The second group comprises the last three pathologies, and the pathologies in this group may be seen as signs of a refusal to learn from experience. Thus, the first of the three pathologies in this group, refusal to make decisions obviously means that the subject will not receive any new experience to learn from, while the last two involves rejecting whatever experience could have been used for learning.

The results with respect to pathologies so far consist mainly of a set of examples. We do not know the relative frequency of these pathologies, nor do we know if there are individual differences among people with respect to which of these pathologies they will exhibit. Indeed, we not even know if the pathologies are common. This is an important problem for future studies.

The experimental approach

Research following the experimental approach aims to assess the effects of the characteristics of the system on the behavior of subjects in dynamic tasks in order to enable the investigator to make inferences about how the subjects develop mental models and formulate goals. This requires a taxonomy of the task that would describe it in terms of those aspects that affect performance. The problem of taxonomies for complex dynamic microworlds has been discussed by (Brehmer, 1990b, 1991) and Funke (1990). However, we will not dwell on the general problems of taxonomy here but use a simple set of concepts proposed by Brehmer and Allard (1991a) to organize the results obtained so far.

The characteristics of dynamic tasks

A taxonomy of dynamic systems is interesting only if it helps predict the behavior or performance of the subjects. This means that the taxonomy is a first step towards a theory. It is therefore important that the set of dimensions chosen for the taxonomy has some psychological meaning.

Brehmer and Allard (1991a) have proposed such a set of dimensions to be used to describe dynamic tasks, and to guide experimentation. They proposed six dimensions.

The first of these is *complexity*. This is a difficult concept, for it cannot be given an absolute definition, nor does it have an obvious metric. Instead, it is a relative concept, and it must be defined in relation to something or someone for which the system is complex. In the present context, our interest is in control, and it is therefore reasonable to define complexity in relation to the capacity of a controller. This is the approach proposed by Ashby (1956) in introducing the concept of *requisite variety*. In this case, the controllers of interest are humans, and it is in relation their limitations that a task may be seen as complex. Since one of the fundamental limitations of humans concerns the number of items they can process at a given time, it makes sense to define complexity in terms of the number of systems elements. However, there can be no simple relation between the number of elements of a system and difficulties in controlling it. We must also consider the relations among the elements and the nature of these elements. For example, if a system has many elements and the relations are negative feedback loops, the system will be very stable, and it will not develop in a catastrophic direction regardless of what the decision maker does. If the relations are positive feedback loops, on the other hand, even a system with few elements may be hard to control. A mixture of positive and negative feedback loops may lead to system behavior that is hard to understand and predict, and may make control very difficult (see Mackinnon & Wearing, 1980, for a discussion of these problems). Moreover, not all elements may contribute equally to the difficulty of a system. Brehmer and Allard (1991a) distinguished among four kinds of elements: goals, processes to be controlled to reach the goals, control processes, and side effects. This represents an attempt to distinguish among different elements in relation to the subjects' purposes. Whether this also makes empirical sense is still an open question.

The second dimension is *feedback quality*. In dynamic tasks, all information is feedback information, but this feedback information may nevertheless vary in quality, leading to variation in the observability of the system.

The third concept is *feedback delays*. This is the time interval between the point in time when an action is taken and that when information about the effects of that action arrives. In complex systems, feedback delays are inevitable, for everything takes time. The concept of delay is a complex one, however. First, feedback delays may occur anywhere in the feedback loop. Thus, they may occur in the response to commands (what control engineers call *dead time*), in the execution of commands (usually called

the *time constants* in control theory), or in the transmission of information about the results of the action taken (*feedback delay*). Second, the information about these delays may vary. Thus, the concept of observability is relevant here too.

The fourth concept is *rate of change*. This refers to the rate of change in the process to be controlled. Dynamic tasks vary widely with respect to rate of change, from the very rapid tasks facing a fighter pilot making a low level attack, to the very slow tasks facing the minister of finance of a country trying to control the economy of that country. Presumably, there is a range of values here with which a person may feel comfortable. This is probably the range within which it is possible to control the tasks by being "in the loop". As tasks become very slow, they can only be controlled via representations. Here, the dynamic character may well be ignored, and the task seen as a series of independent decisions, or, at best as a set of sequential decisions. As the rate of change becomes very fast, it is no longer possible to control the system on the basis of feedback. Instead, the controller must develop some means of feedforward or open loop control.

The fifth concept is *the relation between the characteristics of the processes to be controlled and those of the process used for control*. This defines the strategy for the task. For example, fire fighting of the kind represented in Figure 1 is distinguished by the fact one linear process: the fire fighting process, is used to fight another linear process: the spreading fire, but where these processes differ with respect to their rate of change: the fire grows faster than the fire fighting units are able to extinguish it. Therefore, to succeed in extinguishing the fire, the subject has to attack it massively and rapidly, i.e., the subject must send so many fire fighting units to a given fire that they cover at least the area that will be on fire when the units arrive at the fire. Matching the number of fire fighting units to the fire as it is when the units are sent out will not be sufficient. The subjects quickly learn the elements of this strategy in the fire fighting simulation (see, e.g., Brehmer & Allard, 1991a). This concept encompasses both the "Eigendynamik" of the system, i.e., the extent to which it changes autonomously, and the feedback structure, i.e., whether the feedback is positive or negative. Both of these will be aspects of the characteristics of the process to be controlled. Thus, this concept will need to be differentiated into a number of subclasses.

The sixth concept refers to *the extent to which the decision making power in the system can be delegated*, or distributed, among the persons in the system. This is especially important in systems where there are long delays. In this case, delegating the decision making to lower levels in the system that are closer to the actual events may increase efficiency, a fact very well recognized in military contexts (see Brehmer, 1988).

This list of concepts is, of course, only a preliminary one, to be replaced by a new list as research progresses. From a systems point of view, it is a problem that the concepts are not independent. For example, there is a close relation between the relation between the characteristics of the process used for control and the characteristics of the process to be controlled and the concept of feedback delay. Our problem, however, is not only to characterize systems, but to find psychologically meaningful concepts for

describing systems. Whether the one concept can be subsumed under the other is an important question for future research.

Experimental results

Research with computer simulated microworlds is still in its infancy, and there are as yet few well established findings concerning the effects of various system characteristics. However, even the small body of results that are available shed some interesting light on people's ability to control system. In this section, we will review these results, using the list of concepts discussed above to organize them.

Complexity

Despite the central position of the concept of complexity in the German tradition of "Komplexes Problemlösen", there have been few attempts to study the effects of complexity systematically. Instead, the strategy has been to create a simulation that can be said to be complex, and then use this to compare the behavior of different groups of subjects as described above.

One reason why there have been few studies of the effects of this factor may well be that there is something trivial about this variable as an independent variable in experimental research. Since there is no metric for complexity, it must be defined in relation to the capacity of the subjects. That is, either the subjects can mobilize the requisite variety needed to control the task or they cannot. Moreover, their ability to produce requisite variety cannot be seen as a fixed property. Instead, it is likely to develop with practice. However, even though the demonstrating effects of complexity may be uninteresting, assessing the variety a person can produce, to track its development, to find ways of speeding up this development or to provide means for augmenting it, are far from trivial research problems.

Because we have so little experimental work on the effects of complexity factors, it is difficult to interpret some of the results that we have. For example, there is some consensus that people have problems taking the side effects of their decisions into account, a conclusion based both on analyses following the individual differences approach (e.g., Dörner, et al, 1983) and the experimental approach (Müller, Funke, & Rasche, 1988). This is also confirmed by results obtained by Sterman (1989) which show that subjects had problems understanding the feedback between their responses and the environment in an economic simulation. Specifically, his results show that the subjects did not understand that their demand for capital would affect the total demand for capital, leading to at least temporary problems in getting the capital they needed. This may, of course, be seen as a side effect, but it is also evidence that one of the fundamental characteristics of a dynamic decision task: that the state of the task changes both autonomously and as a consequence of one's own behavior, is difficult to grasp.

It is not clear whether the problems people have with side effects should be seen as an inherent characteristic or whether the results simply express the difficulties naive subjects have when working with systems about which they know little. That is, the subjects in the microworld experiments may simply not have received enough experience to learn about the important side effects of the decisions they have made. But it is, of course, also possible that the neglect of side effects is a natural consequence of people's limited attentional resources, and that at least some side effects are likely to be ignored in most complex situations.

Another problem making it difficult to assess the effects of complexity is due to the interaction between different factors in the system. This is illustrated by some results by Mackinnon and Wearing (1980). They failed to demonstrate any effects of complexity defined in terms of the number of elements in the system when the system was characterized by strong negative feedback loops. Under these conditions, the system becomes inert and self stabilizing. That is, even though that the strategy that the subjects use to control the system is inappropriate, the negative feedback loops of the system may save the decision makers from the worst consequences of their inappropriate strategies. On the other hand, they will also be unable to achieve the kind of change that they strive for.

The results from Dörner's studies suggest that subjects have considerable problems finding adequate goals in complex simulations (Dörner, et al., 1983). However, it is hard to decide whether this is due to the complexity of the simulation. There are no direct studies on, e.g., the effects of the number of goals on subject performance. Again, this is due to the research strategy of constructing a system that is complex enough, and then using it without any experimental variation of the number of goals or the number of goal conflicts. Moreover, in these experiments subjects usually only receive very general and diffuse goals in the instructions (such as to "care for the welfare of the moros"), and they have to derive a series of more concrete goals before they are able to act. This emphasizes the close relation between the goal formulation process and the mental models that subjects may have. If these models are inadequate, the goals formulated on the basis of them will be inadequate also. Consequently, it is not possible to decide whether the goal formulation problems observed in e.g., the Lohhausen study are due to inadequate models or to problems in the goal formulation process as such.

Feedback quality

As noted above, microworlds are usually designed to be opaque, i.e., they are designed so that the feedback quality is less than perfect. That is, the subjects are not informed about all of the consequences of their actions automatically. Instead, they must often use more indirect information to infer the state of the system, or at least actively search for information about some of the relevant aspects of the system. This requires that they know, and remember, what these aspects are.

There are no experimental results on the effects of feedback quality as such. However, the importance of this factor is illustrated in studies using Moro. Here, one of the main problems is the ground water level, a variable that is not displayed automatically to the subject. To control this factor, the subject must remember to ask for information about it. The failure to ascertain the ground water level is one of the main explanations why so many subjects fail in Moro (see e.g., Hasselrot & Brehmer, 1991).

The subjects' ability to develop the appropriate strategy of rapid and massive response to a fire in the fire fighting simulation and their inability to cope with delays in reports from the fire fighting units (Brehmer & Allard, 1991a, b) may also be seen as a matter of differences in feedback quality. Thus, the delays caused by the time constants of the system, which are due to the speed at which the fire fighting units move, are directly observable, but the report delays have to be inferred.

Feedback delays

Feedback delays are interesting for at least two different reasons. First, they are inevitable in most complex systems, for everything takes time, and a decision maker cannot expect to have the results of his actions displayed immediately. Whether he can cope with these delays is a central feature of his ability to cope with complex dynamic systems. Second, the effects of feedback delays give us information about the subjects' general strategies. For example, if they do not compensate for the delays, this suggests that they employ a feedback control strategy, but if they compensate adequately for the delays, this suggests some form of feedforward strategy, or at least some open loop strategy.

Feedback delays may occur in different parts of the feedback loop and depending on where in the feedback loop that the delays occur, different models are required to cope with the delays. Thus, dead time and time constants will require a model that makes it possible to predict when an action is needed before the need has arisen. Delays in the reports about the effects, on the other hand, requires a model which makes it possible to predict when a given action has taken effect and when the resources will be available again. Developing the former kind of model seems inherently more difficult, unless there is some regularity in the need for decisions about actions. We would therefore expect that dead time would cause more difficulties for decision makers than report delays do.

A number of studies show that feedback delays have negative effects on performance in a variety of complex dynamic systems (Brehmer & Allard, 1991a, b; Dörner & Preussler, 1990; Reichert & Dörner, 1987). However, as noted above, the negative effects may be limited to delays that have to be inferred (Brehmer, 1990a).

As expected, there are also differences in the effects of delays that occur in different locations in the feedback loop. In the fire fighting simulation, dead time causes greater problems than report delays (Brehmer & Allard, 1991c). This is hardly surprising. If

the subjects fail to compensate for both forms of delays, as they seem to do, they will be sending fewer fire fighting units to the fire under dead time conditions (because the fire will spread even further during the dead time) than they will under report delay conditions, where the subjects have an adequate picture of the fire at the moment of decision. The predicted differences in use of resources between the two kinds of delay conditions were also found by Brehmer and Allard (1991c). Similar results were obtained when the subjects were informed about the delays.

As noted above, we would expect that dead time would be more difficult than report delays in the general case, (this is true also in engineering applications of control theory to real systems). Once, the need for a decision is apparent, on the other hand, dead time will have very much the same effects as report delays, provided that the process to be controlled does not change too rapidly in relation to the dead time. That is, in both cases, the main effect will be that it will take some time before the decision maker can see the effects of his actions.

Subjects do not use the strategy of delegating decision making power to local units to help them cope with the delays. Thus in experiments using the fire fighting simulation, Brehmer and Allard (1991a, b) gave subjects the opportunity to delegate decision making to the fire fighting units. Specifically, they could delegate the decision when to start fighting a fire to the (simulated) fire units commanders, so that they would not have to report from them about their location to decide whether they needed to change their destination to where the fire was most threatening. However, subjects did not delegate decision making power more when there were delays. Indeed, they were generally unwilling to delegate any decision making to the unit commanders, and they generally became less willing to delegate as the experiment progressed, despite that they were not doing particularly well when using their centralized decision making policy.

With respect to strategy, the results reviewed in this section suggest that subjects will adopt an open loop control strategy when the delays will have to be inferred, and an open loop strategy when the delays can be seen to happen (Brehmer, 1990a). Thus, subjects seem to develop different models of the task depending on the observability. The results also show that the subjects seem to adopt the same feedback control strategy for dead time and report delays when they are told about these delays as when they are not told about them and when sufficient information to infer the delays and their causes is provided in displays in the experiment. This shows that observability from the subjects' point of view is not the same as from an control engineer's point of view.

Rate of change

The effect of increasing the rate of change in a dynamic decision task is to increase the work load of the subject. Predictably, this leads to decreased performance (Brehmer & Allard, 1991d; Brehmer, Allard & Jansson, 1991).

The relation between the characteristics of the process to be controlled and the characteristics of the control process

There are no systematic studies of this factor as yet. In most simulations, the control process tends to be a linear one. Perhaps this is typical of man made control processes. However, we could easily think of exponential control processes, e.g., when one biological process is used to control another biological process. Dörner (1980) reports that exponential processes cause difficulties for the subjects, but this conclusion is not based on any systematic comparison between different processes to be controlled or between different relations between the control process and the process to be controlled. Funke (1990) reports that process with "Eigendynamik" are difficult to control, and more difficult the more pronounced the "Eigendynamik". Again, the difficult of controlling a process with "Eigendynamik" may be due to the characteristics of the control process. Thus, we should reserve judgment about the relative difficulty of different processes until we have investigated its difficulty relative to different control processes.

The fire fighting simulation of Brehmer and Allard (1991a) involves using a linear process (fire fighting) to control another linear process (the fire), where the slope of the latter exceeds that of the former. As shown by Brehmer and Allard's results, subject master this relation, at least if they are able to see the processes, as they can in the fire fighting simulation where both the spread of the fire and the movements of the fire fighting units are displayed to them.

These results also show the close relation between the characteristics of the process to be controlled, the characteristics of the control process and the concepts of time constants and dead time. The slope the function describing the process to be controlled is obviously related to the time constant of the process. Dead time introduces a nonlinearity in the function describing the control process. This raises the problem of whether it is meaningful to retain the distinctions among dead time, time constants and the functional characteristics of the processes involved. This is a question that can only be answered by further psychological research. Only if subjects behave differently to the nonlinearity introduced by dead time and that introduced by other factors, it is psychologically meaningful to make the distinction.

Summary

The results reviewed here are probably reflect only one aspect of the subjects' performance: their nature of their mental models. They suggest that these mental models do not include all aspects of a complex dynamic task. Thus, side effects and delays that have to be inferred seem to be problematic, and there is little evidence from the subjects' performance in the microworld experiments that they have incorporated them in their models. For delays, this is true, despite even when the subjects claim to have detected them (Brehmer & Allard, 1991b), and even when they are told about these delays Brehmer & allard, 1991b, c).

Both side effects and feedback delays are common features of real world systems. Our conclusion, therefore, would be that we would expect that decision makers working with complex dynamic systems would show a level of performance that is far from what it could be.

Conclusions

This paper had three purposes: to introduce a general paradigm for the study of dynamic decision making: that of control theory, to describe a general methodology for the laboratory study of dynamic decision making: that of computer simulated microworlds, and to review results obtained with this paradigm.

Generally speaking, the research described here can be seen as an extension of the ideas presented by Rapoport (the use of control theory) and Toda (the use of computer simulations as research tools), but it differs from Rapoport's and Toda's original papers in that it has abandoned the normative-descriptive comparisons for empirical analysis and psychological theorizing. This is because the analytical analysis of complex dynamic tasks has proved impossible, a conclusion already drawn by Rapoport (1975).

So far, the total body of research is rather small, and much remains to be done. However, the results so far demonstrate that it is possible to investigate even complex dynamic decision making phenomena in experimentally. Moreover, a number of stable results have been obtained. These results suggest that people may have problems developing the mental models required by complex dynamic tasks. Some of these results are corroborated by observations from field studies, such as studies of operators in process plants, who have also been found to have problems with feedback delays and with side effects (e.g., Crossman, 1974). This suggests that the results obtained in the microworlds are not simply laboratory artifacts, but that they may express some generally valid characteristics of people trying to control complex dynamic systems.

The present results are limited in one important respect, however. The limitation is illustrated by the fact that there has been little need to invoke the concept of decision making. This is, of course, because there has been very few goal conflicts in the tasks used so far. Therefore, the subjects are not required to use their models for different purposes, and the results can generally be interpreted as an indication of the characteristics of the subjects' mental models.

The next step in research on dynamic decision making therefore must be to introduce motivational factors in a more systematic way in the experiments. For this purpose, we have introduced a new version of the fire fighting task called NEWFIRE (Brehmer & Lövborg, 1991) which makes it possible to vary cost factors in the fire fighting task in a systematic way. This should make it possible to assess whether subjects are able also to modify their strategies in response to economic considerations and not only in response to system characteristics such as time constants.

This should enable us, not only to create a firm connection between dynamic decision making and general cognitive psychology, as is done via the interest in mental models, but also between dynamic decision making and the more traditional decision literature via the new possibilities of studying the effects of cost factors.

References

- Ashby, W. R. (1956). *An introduction to cybernetics*. London: Chapman & Hall.
- Bainbridge, L. (1979). Forgotten alternatives in skill and work load. *Ergonomics*, 21, 169-185.
- Bainbridge, L. (1981). Mathematical equations or processing routines? In J. Rasmussen & W. B. Rouse (Eds.), *Human detection and diagnosis of systems failures*. New York: Plenum.
- Berry, D. C. & Broadbent, D. E. (1987). The combination of explicit and implicit learning processes in task control. *Psychological Research*, 49, 7-15.
- Berry, D. C. & Broadbent, D. E. (1988). Interactive tasks and the implicit-explicit distinction. *British Journal of Psychology*, 79, 251-272.
- Brehmer, A. (1989). *Brännskadevård som ett hierarkiskt organiserat system*. Uppsala University: CMD Report No. 4.
- Brehmer, B. (1988). Organization for decision making in complex systems. In L. Goodstein, H. Andersen & S. E. Olsen (Eds.), *Tasks, errors and mental models*. London: Taylor & Francis.
- Brehmer, B. (1990a). Strategies in real-time, dynamic decision making. In R. Hogarth (Ed.), *Insights in decision making*. Chicago: University of Chicago Press.
- Brehmer, B. (1990b). Time scales, distributed decision making, and modern information technology. In J. Rasmussen, B. Brehmer & J. Leplat (Eds.), *Distributed decision making: Cognitive models for cooperative work*. Chichester: Wiley.
- Brehmer, B. (1990c). Dynamic decision making. *In press*.
- Brehmer, B. (1990d). Towards a taxonomy for microworlds. In J. Rasmussen, B. Brehmer, M. de Montmollin & J. Leplat (Eds.), *Taxonomy for analysis of work domains. Proceedings of the first Mohawc workshop*. Roskilde. Risø National Laboratory.
- Brehmer, B. (1991). *Human control of complex systems: A view from research with microworlds*. Paper prepared for the Third European Meeting on Cognitive Science Approaches to Process Control. Cardiff, September 2-5.
- Brehmer, B. & Allard, R. (1991a). Real-time dynamic decision making. Effects of task complexity and feedback delays. In J. Rasmussen, B. Brehmer & J. Leplat (Eds.), *Distributed decision making: Cognitive models for cooperative work*. Chichester: Wiley.
- Brehmer, B. & Allard, R. (1991b). *Feedback delays in real-time dynamic decision making*. Uppsala Psychological Reports.
- Brehmer, B. & Allard, R. (1991c). *Dead time and report delays in dynamic decision making*. Uppsala Psychological Reports.

- Brehmer, B. & Allard, R. (1991d). *Rate of change in dynamic decision making*. Uppsala University, Department of Psychology: Manuscript.
- Brehmer, B. & Allard, R. & Jansson, A. (1991). *Discrete time and continuous time simulations in the study of dynamic decision making*. Uppsala Psychological Reports.
- Brehmer, B. & Dörner, D. (1991). Research with computer simulated microworlds: Escaping the narrow straits of the laboratory as well as the deep blue sea of the field study. In J. Rasmussen, B. Brehmer, M. de Montmollin & J. Leplat (Eds.), *Taxonomy for analysis of work domains. Proceedings of the first MO-HAWC workshop*, Liege May 15-16, 1990.
- Brehmer, B. & Löwborg, L. (1991). *NEWFIRE*. Roskilde: Risø National Institute. M-Report.
- Broadbent, D. E., Fitzgerald, P. & Broadbent, M. H. P. (1986). Implicit and explicit knowledge in the control of complex systems. *British Journal of Psychology*, 77, 33-50.
- Bruner, J. S., Goodnow, J. J. & Austin, G. A. (1956). *A study of thinking*. New York: Wiley.
- Crossman, (1974). Automation and skill. In E. Edwards & F. P. Lees (Eds.), *The human operator in process control*. London: Taylor & Francis.
- Dörner, D. (1980). On the problems people have in dealing with complexity. *Simulation and Games*, 11, 87-106.
- Dörner, D. (1989). *Die Logik des Misslingens. Strategisches Denken in komplexen Situationen*. Reinbek bei Hamburg: Rowohlt.
- Dörner, D. (1991). *Über die Philosophie der Verwendung von Mikrowelten oder "Computerszenarios" in der psychologischen Forschung*. Working Paper No. 7. Berlin: Projektgruppe Kognitive Anthropologie. Max-Planck-Gesellschaft.
- Dörner, D. & Schölkopf, J. (1989). *Controlling complex systems or: Expertise as "Grandmother's Wisdom"*. Memorandum 68, Lehrstuhl Psychologie II, Bamberg University.
- Dörner, D., Kreuzig, Reither & Stäudel, T. (1983). *Lohhausen: Vom Umgang mit Unbestimmtheit und Komplexität*. Bern: Huber.
- Dörner, D., Stäudel, T. & Strohschneider, S. (1986). *Moro - Programmdokumentation*. Memorandum 23. Lehrstuhl Psychologie II, Bamberg University.
- Edwards, W. (1962). Dynamic decision theory and probabilistic information processing. *Human Factors*, 4, 59-73.
- Funke, J. (1990). Systemmerkmale als Determinanten des Umgangs mit dynamischen Systemen. *Sprache & Kognition*, 9, 143-154.
- Funke, J. & Müller, H. (1988). Eingreifen und Prognostizieren als Determinanten von Systemidentifikation. *Sprache & Kognition*, 7, 176-186.
- Hasselrot, A. & Brehmer, B. (1991). *Epistemic competence in dynamic decision making*. Uppsala University, Department of Psychology: Unpublished manuscript.
- Hoc, J. M. (1989). Strategies in controlling a continuous process with long response latencies: Needs for computer support to diagnosis. *International Journal of Man-Machine Studies*. 30, 47-67.

- Kelley, C. R. (1968). *Manual and automatic control. A theory of manual control and its application to manual and automatic control systems*. New York: Wiley.
- Mackinnon, A. J. & Wearing, A. J. (1980). Complexity and decision making. *Behavioral Science*, 25, 285-296.
- Mackinnon, A. J. & Wearing, A. J. (1985). Systems analysis and dynamic decision making. *Acta Psychologica*, 58, 159-172.
- Müller, H., Funke, J. & Rasche, B. (1988). *Wechselseitige Abhängigkeiten: Zum Einfluss von Eigendynamik und Nebenwirkungen auf der Bearbeitung dynamischer Systeme*. Berichte aus dem Psychologischen Institut der Universität Bonn, 14, No. 1.
- Powers, W. (1971). *Behavior: The control of perception*. New York: Addison-Wesley.
- Putz-Osterloh, W. & Lüer, G. (1981). Über die Vorhersagbarkeit komplexer Problemlöseleistungen durch Ergebnisse in einem Intelligenztest. *Zeitschrift für Angewandte und Experimentelle Psychologie*, 28, 309-334.
- Rapoport, A. (1975). Research paradigms for the study of dynamic decision behavior. I D. Wendt & C. Vlek (Red.), *Utility, probability and human decision making*. Dordrecht: Reidel.
- Reichert, U. & Dörner, D. (1987). Heuristiken beim Umgang mit einem "einfachen" dynamischen System. *Sprache & Kognition*, 7, 12-24.
- Schaub, H. & Strohschneider, S. (1989). *Die Rolle heuristischen Wissens beim Umgang mit einem komplexen Problem oder: Können Manager wirklich besser managen?* Memorandum No. 71, Lehrstuhl Psychologie II, Bamberg University.
- Schoppek, W. (1991). Spiel und Wirklichkeit - Reliabilität und Validität von Verhaltensmustern in komplexen Situationen. *Sprache & Kognition*, 10, 15-27.
- Slovic, P., Fishhoff, B. & Lichtenstein, S. (1977). Behavioral decision theory. *Annual Review of Psychology*, 28, 1-39.
- Sterman, J. D. (1989). Misperception of feedback in dynamic decision making. *Organizational Behavior and Human Decision Processes*, 43, 301-335.
- Strohschneider, S. (1986). Zur Stabilität und Validität von Handeln in komplexen Realitätsbereichen. *Sprache & Kognition*, 5, 42-48.
- Stäudel, T. (1987). *Problemlösen, Emotionen und Kompetenz. Die Überprüfung eines integrativen Konstrukts*. Regensburg: Roderer.
- Toda, M. (1962). The design of a fungus-eater: A model of human behavior in an unsophisticated environment. *Behavioral Science*, 7, 164-183.

How to improve our understanding of social decision making
with the help of computers

Wim B. G. Liebrand

University of Groningen, the Netherlands ¹

Understanding social decision making has proven to be an important and complex task for many researchers within various disciplines. Previous research has described both the ways in which people go about to solve problems and make decisions as well as how they should have done that.

In the area of decision making we observe major departures from what in terms of SEU theory would be called rational decision making. The elegant mathematical world of decision making obviously does not sufficiently match reality. In light of the rather stringent assumptions being made it seems not that surprising that we cannot call man a 'SEU-rational' decision maker. People normally do not have available complete information about all utility functions involved, and as a consequence they are not in the position to carry out the

¹ I would like to thank dr. Ellen Boekkooi, dr. David Messick and dr. Tetsuo Takigawa for their comments on an earlier version of this paper.

required numerical operations. What is more, even if all the necessary information would have been available, their limited capacity for information processing would impede such operations.

On the other hand, empirical research on decision making and problem solving shows that people transform the structure of the decision setting according to their individual outcome preferences, their short and long term strategic goals, and their selective perception of complex structures. These numerous and diverse empirical research findings, however, adds to the complexity of the field and as a consequence we currently lack a comprehensive theory of human decision making which integrates the variety of theoretical and empirical research findings.

The major aim of the present paper is to provide a possible way to integrate previous research findings in a rather pragmatic way. We will try to accomplish that by using recent developments in social science information technology. Although the present line of reasoning is applicable to various research areas, we will focus primarily on the domain of decision making in groups in which the well being of the group members is affected by the decisions taken by the groupmembers.

Social Science Information Technology

In the last decade we have observed an ever increasing role of

computers in empirical research. Whereas the computer primarily was used to carry out numerical operations, after the introduction of the personal computer in the early eighties we have witnessed an exponential growth in the capabilities and availability of desktop computers. The computer is now in use in every phase of empirical research and a regular desktop computer is capable to carry out millions of instructions per second. This provides a powerful tool for analyzing highly complex situations. As an example one might think of the allocation of passengers to planes and planes to flight schedules all under the objective to minimize costs. The human mind is not very well equipped to perform such a task within a reasonable amount of time.

Despite this brute computational force, and despite several predictions (e.g. Simon and Newell, 1958), the computer is still not able to beat the world's chess champion. In general experts are capable of finding intuitively and almost instantaneously the relevant patterns out of may thousands of different configurations. These experts tend to have a so called "tuned awareness" from which they selectively perceive, process, store and recall information. They have strategies for storing and retrieving information or knowledge sets in short- and long-term memory (Tuthill, 1990). To be able to use these strategies would of course be very beneficial to computer applications as well as to non-experts in the field. Next we will discuss this possibility in more detail.

Ordered levels of knowledge

Knowledge may be defined as the result of a process of a synthesis in which information is compared to other information and combined into meaningful links (Tuthill, 1990). In this view, incoming information is interpreted within an already established personal knowledge framework. This knowledge framework forms a scheme which serves as a model for matching and comparing the incoming information with the current problem set. Within this framework Tuthill distinguishes four levels of knowledge.

Facts constitute of relationships between objects, symbols and events. They can be represented by traditional data structures and are considered to be the lowest level of knowledge. Concepts are facts which are grouped with respect to common attributes. Concepts are hierarchically organized in such a way that the lower-level concepts inherit attributes of the ancestor concepts. Rules are the next level of knowledge. They operate on facts and concepts. Rules can be conceived of as guides for action and consist of IF-THEN statements representing conditions and actions to take in deductive problem solving. The highest level of knowledge is described in Tuthill's taxonomy as Heuristic Knowledge. Heuristic knowledge is especially relevant in problem solving. In facing new problem situations, rather than using rule algorithms, shortcuts and combinations of rules are applied. In fact, heuristic knowledge is the personal synthesis of facts, concepts and rules.

The usage of heuristic knowledge from experts

The importance of heuristic knowledge might grab us when we realize that our common understanding of decision making processes consists of facts, concepts and rules. These levels of knowledge are easiest to transfer by means of articles, books and lectures. Heuristic knowledge, or the set of personal rules of thumb is not normally transferred to others as acquired knowledge. There are several reasons to call this a highly undesirable situation. The most important of them might be the potential of heuristic strategies to efficiently find solutions to problems. Knowledge from problem solving strategies is essential because it provides insight into the domain itself. Experts in the field of, for example, social decision making, not only possess knowledge of a large amount of facts, concepts and rules in that research area, they also have heuristic knowledge which transcends rule knowledge, and is used in case novel situations are encountered. What is applied in these novel problem situations is a selective synthesis of facts concepts and rules previously acquired. All in all they have access to a very complex personal knowledge base which is constantly evolving and which they constantly use in their reasoning about the field.

The question which arises now, of course, is whether it is possible to profit from the heuristic knowledge acquired by others. A tentative answer might be confirmative. In a way that is exactly what 'knowledge engineers' are trying to accomplish. A knowledge engineer tries to emulate expert problem solving in a computer program. In theory their task is simple: elicitate knowledge from experts and then organize and

represent it in such a way that it can be used by a computer. In practice knowledge engineers have encountered several problems and 'bottle necks'. However, it is only fair to note that substantial progress has been made in this field. Knowledge engineers have now access to several techniques for knowledge elicitation, including structured interviews, protocol analysis, repertory grid and various psychometric scaling techniques. For representing the expert's knowledge in a knowledge based system also several methods have been developed by now. Among the commonly used approaches we find semantic networks, production rules, scripts, frames, repertory grids and object-oriented approaches. All these approaches have their specific advantages and disadvantages. It is not my intention to discuss this in further detail (see for example: Gammack, 1987; Greenwell, 1988; Hart, 1989; Parsaye & Chignell, 1988). For present purposes it is sufficient to allude to the possibility of representing heuristic knowledge of experts in the field of social decision making.

The domain of social decision making

Decision making in situations of outcome interdependency involves actors whose decisions have reciprocal consequences on their well being. A convenient way to display different classes of decision making within this domain is shown in Figure 1.

Insert Figure 1 about here

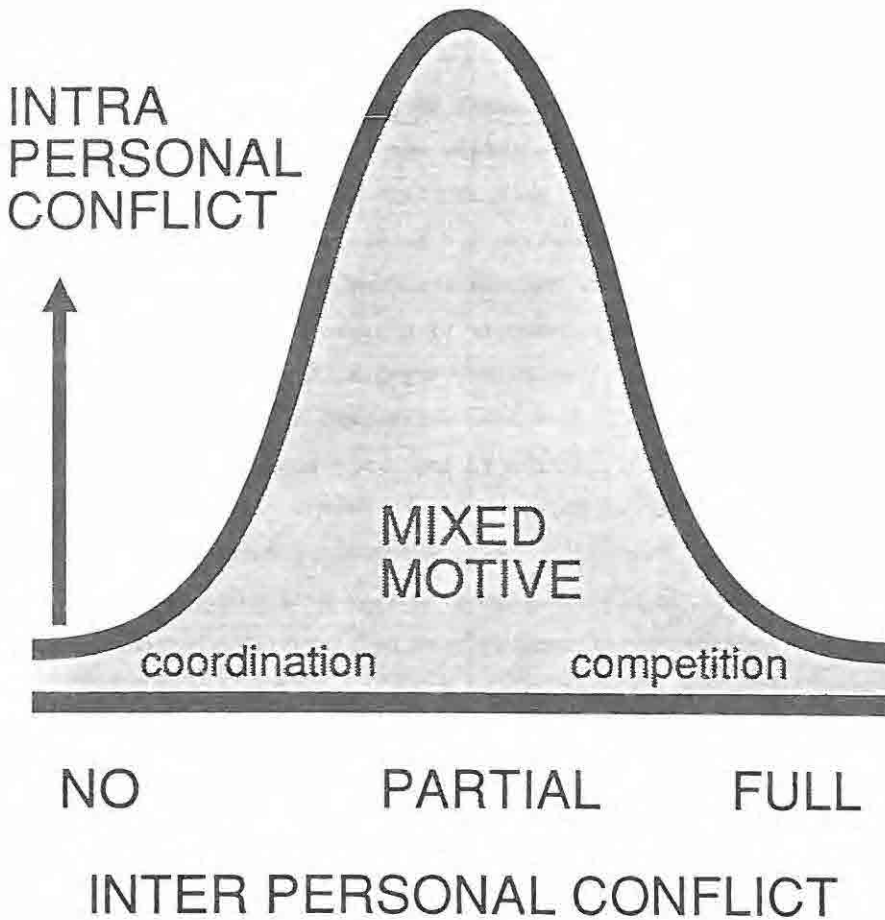


Figure 1: A taxonomy of decision making under settings of outcome interdependency (after Liebrand, 1989)

The classification is based on whether the own interests of the actors consists of a harmonious set of overlapping interests, a set of competing interests, or consists of a mixture of overlapping and competing interests. The assumed amount of intrapersonal conflict is depicted on the vertical axis. That is, we assume that in case the actors' interests coincide, there will be no ambiguity in what one ought to do, hence actors know for sure that it is their own interest to cooperate. Real world examples consists of sailing or problem solving by groups. On the opposite side of the horizontal dimension of Figure 1, we again expect low levels of intrapersonal conflict. Here we are dealing with zero sum settings in which gains for the one implies equal losses for the other party. Hence, one ought to compete in these situations. Chess playing and the 'battle of the Bismarck Sea' are the often used real life examples.

Intrapersonal levels of conflict are high in between strictly coordinative and competitive forms of interpersonal relationships. These are ones of partial conflict and they involve relationships within which one has the opportunity to cooperate and to compete. Such relationships of partial conflict turn out to be somewhat difficult to describe for reasons of terminology. As Schelling (1973) observes, it is both interesting and puzzling that though we have words to describe individuals in relationships of coordination, e.g., "partners", "team members", "cooperators" and "collaborators", as well as ones in relationships of strict competition, e.g.,

"enemies", "adversaries", and "opponents", we do not appear to have similar terms that describe relationships of the partial conflict variety. This strangely obtains even though such forms of relationships are highly dominant in wars, in strikes, in athletic contests, in classrooms, in the relationships between faculty members, and even in marriages. In each of the latter, the participants find themselves in relationships in which their goals are partly correspondent, and partly in competition with interdependent others. In short the participants are in a mixed-motive form of relationship. If each wishes to maintain this relationship, then they must both receive and deliver rewards to the other, activities generally associated with coordination. On the other hand, they are also likely to want to obtain the most out of a relationship at the least cost. To the extent that most is defined either by the situation, or by their own values as attaining an increased share of a jointly controlled resource at some cost to the other, then the relationship also has a strictly competitive or a conflictual component.

This situation in which the decision maker is caught in the dilemma to either cooperate or compete has been a central research topic in the social and behavioral sciences after the important book of Luce and Raiffa (1957). They laid the foundation for the development of experimental games which can be used as research tools to test predictions derived from formal game theory. The most famous experimental game is the Prisoner's Dilemma Game (PDG). It concerns a situation in which each decision maker is best off acting in their own self

interest, regardless of what the other persons do. Each self-interested action, however, creates a negative outcome or cost for the other people who are involved. When a large number of people make the self-interested choice, the costs or negative outcomes accumulate creating a situation in which everybody would have done better had they decided not to act in their own short term interest. One of the reasons for its popularity is that the PDG resembles various real life situations of conflict.

An example of a real life PDG is the decision to pollute. Pollution problems can be found at various levels of decision making, ranging from individuals to nations. For example at the industrial level, no matter what other chemical industries may do to get rid of their chemical waste, it is cheaper to have the waste dumped at some rubbish-dump, or alternatively, in the ocean, than to take care of an adequate decomposition of the waste. The ultimate long-term consequences of this selfish act have to be shared by all individuals. At the individual level the slogan "every litter bit hurts" nicely reflects the negative consequences accompanying the decision to pollute. Though all individuals would like to avoid the long-term negative consequences, it remains cheaper and simpler for them to keep polluting as anonymous individuals, no matter what the others are doing. As a consequence "rational behavior" leads to a collective disaster.

The major advantage in using experimental games lies in the amount of control the experimenter has over the decision making situation. This advantage combined with the possibility

to study very appealing situations of conflict resulted in a veritable flood of experimental gaming studies. By now, several thousands of studies have been published in this area, a number which makes it effectively impossible to review them all.

Towards a comprehensive view of research on social decision making

Within science we primarily obtain knowledge of a specific field by reading the published results of theoretical and empirical studies. As long as we can be sure that the body of knowledge for that research area is adequately covered in the literature, there seems to be nothing wrong with this approach. On second thought, however, this approach may lead to a biased and incomplete perspective.

It might be biased because studies who failed to produce significant results, tend not to be published. Speaking from my own experience in the field of social dilemma research, I know of several studies from different laboratories, which failed to find systematic differences for a specific type of framing effect. Almost all of these studies never got published, a few which found such differences were. It would therefore be unlikely that somebody draws a valid conclusion with regard to the effects of this specific manipulation, based on what has been published in this area. The example might be non significant, I am afraid, however, that it is not the only one.

The perspective of a research area gained by studying

published research findings might be incomplete because, as indicated earlier, it primarily contains the lower levels of knowledge. It concerns mainly, facts, objects and rule related knowledge while heuristic knowledge is underrepresented severely. This kind of high level knowledge normally is transferred in small workgroups of expert researchers only.

Incomplete or not, the sheer number of published research findings in the area of social decision making renders it almost impossible to identify and integrate the major features of these findings into a theoretical model. What we need here are some inspired guesses or heuristics to search for recognizable patterns. It is therefore desirable to have access to the heuristic knowledge of expert researchers. What we would like to know is how the expert identifies threads in research findings on the basis of which a specific interpretation is chosen out of all possible interpretations. And, in addition, what we also would like to know is how experts go about and solve those highly complicated problems to which we confront our undergraduates.

A proposal for research

Experts can be used as providers of information, problem-solvers and explainers (Hart, 1990). More specific to the area of decision making in situations of outcome interdependency we might use experts to specify a comprehensive theoretical framework. Such a technique is a major departure from the usual way to develop theories in this area. First, the development of that theoretical framework will become a task for multiple experts instead of one. From a methodological

point of view, the advantage is obvious: it is more reliable to have several estimators instead of one. No matter how good one expert might be both in terms of declarative and procedural knowledge, the combination of expertise from several individuals is an efficient way to get rid of idiosyncracies.

Eliciting knowledge from experts in the proposed way has the advantage that the resulting knowledge base is highly efficient. The knowledge engineer knows that the acquired knowledge has to be represented in such a way that it can be used in a computer program. As a consequence the knowledge has to be functional with respect to the requirements of the system agreed upon earlier. Irrelevant facts, concepts and rules therefore tend to disappear.

The third advantage of the proposed way to construct a theoretical framework lies in the broader domain covered by the experts' knowledge, and more important, the broader domain to which their common heuristic knowledge applies. As a consequence, a group of experts is more flexible in dealing with highly complex situations than a single expert will be. In addition, small group judgment tends to be better than individual judgment and offers the possibility of selecting the optimum solution to a set problem (Greenwell, 1988).

On the negative side it is obvious that the logistics of dealing with several experts are far from being trivial. The use of multiple experts with slightly different fields of expertise requires a study of the overlap of the different domains as well as a thorough study of the ultimate goal of

this knowledge elicitation proces.

With respect to this goal we have to be modest. One of the prominent criticisms of artificial intelligence is that AI-researchers promised far too much. What we obviously will not obtain in the near future is an expert system for the construction of theoretical frameworks in the area of decision making. However, in my opinion it is realistic to expect a sizable progress by using knowledge elicitation techniques on a small group of experts in a well defined domain of decision making. For the present field of research I would suggest the domain of cooperative decision making, the domain of competitive decision making and finally the mixed-motive domain of decision making.

For each domain no more than 5 experts could with the help of a knowledge engineer specify concepts, rules and higher-order rules. Next with the use of one or more representation schemes, this team could validate, review and synthesize the knowledge. If necessary, the experts could participate themselves in decision making experiments, so as to get more insight into decision making and problem solving by experts.

After implementation, the knowledge based system could then assist in decision making in novel situations, based on the synthesis of the experts-knowledge of relevant research findings. The major advantage for the user would be the immediate access to the existing high quality knowledge in that field, without the need to have to study irrelevant material. In addition the system is capable of keeping a

record of the user requirements which could be used to direct future research.

Conclusion

It was observed that the area of social decision making currently lacks a comprehensive theoretical framework which integrates the numerous theoretical and empirical research findings. As a potential solution to this problem we proposed that a small team of experts under the guidance of a knowledge engineer attempt to elicit and represent their knowledge of facts, concepts rules and especially their heuristic knowledge in such a way that it could be implemented in a computer program. It is concluded that SWAT (specialists with advanced tools) teams for the areas of cooperative, competitive and mixed-motive decision making, have a much better starting position than individual experts who try to develop such comprehensive frameworks, without the help of recent computer techniques.

References

- Gammack, J.G. (1987). Different techniques and different aspects on declarative knowledge. In A.L. Kidd (Ed.), Knowledge acquisition for expert systems. New York: Plenum Press.
- Greenwell, M. (1988). Knowledge engineering for expert systems. Chicester: Ellis Horwood Limited.
- Hart, A. (1989). Knowledge acquisition for expert systems. London: Kogan Page Ltd.
- Luce, R.D., & Raiffa, H. (1957). Games and decisions: Introduction and critical survey. London: John Wiley and sons.
- Parsaye, K., & Chignell, M. (1988). Expert systems for experts. New York: John Wiley.
- Schelling, (1973). Hockey helmets, concealed weapons, and daylight saving; a study of binary choices with externalities. Journal of Conflict Resolution, 17, 381-428.
- Simon, H.A., & Newell, A. (1958). Heuristic problem solving: the next advance in operations research. Operations Research, January-February, 1-10.
- Tuthill, G.S. (1990). Knowledge engineering: Concepts and practices for knowledge-based systems. Blue Ridge Summit, PA: TAB BOOKS Inc.

QUALITATIVE METHODS IN MULTIATTRIBUTE DECISION MAKING

Moshkovich H.M. (Institute for Systems Studies, Moscow, USSR)

INTRODUCTION

The majority of the current decision methods are based on the concepts of utility theory (von Neumann & Morgenstern, 1947). The most widely accepted approaches are multiattribute utility theory (MAUT) (Keeney & Raiffa, 1976) and decision trees (Raiffa, 1968).

The necessary element of multiattribute decision methods is the process of information elicitation from decision-makers and experts. Only this information is able to eliminate the uncertainty, connected with the presence of multiple criteria, to elaborate the necessary compromise and find good decisions.

The classical theory of von Neumann and Morgenstern (1947) passes from qualitative determination of utility to quantitative determination of utilities and probabilities, necessary to find the best solution within the framework of the theory of expected utility (Savage, 1954, Luce & Raiffa, 1957).

Numerous investigations of psychologists and specialists in decision making (see, for example, reviews of Larichev, 1987, Slovic et al., 1977, Kahneman et al., 1982), indicate the real complexity of the problem to receive valid data while eliciting information from human beings. There are rather many publications in which it is stated that people may be inconsistent and feature stable biases in their estimates. There exist facts about limitations in human beings' possibilities in evaluation and comparison of multiattribute options. These limitations lead to inconsistencies in people's judgements (Tversky, 1969, Russo & Rosen, 1975, Hoffman et al., 1968), or to implementation of simplified rules, which do not take into account some essential aspects of options under consideration (Payne, 1976, Larichev and Moshkovich, 1988, Montgomery, 1977).

Such situation leads to a number of problems, which are to be taken into account while designing methods for decision making. First of all, it is desirable to analyse what type of information and in what form people are able to give enough reliably, and try to use just these types and forms in the

methods. For the second, it is necessary to develop means for verification of the received information for consistency, and correction of the detected errors. For the third, it is useful to have explanations of the result in terms, understandable for a decision-maker.

The pointed out problems are under consideration in the following report. In the second part of the report the examples of two methods will be given, in which an attempt has been made to meet these requirements.

TYPES OF INFORMATION USED IN DECISION MAKING METHODS

The majority of decision problems are usually analysed within the following framework. It is assumed that there is a decision-maker (DM), possessing, in general, some preference system (may be an implicit one) for options, each of which may be estimated upon a number of attributes (criteria). It is also supposed that the initial task of decision making is almost adequate to the choice, ranking or grouping of alternatives, estimated upon a set of criteria. Within the framework of DM's preference system, alternatives are compared on the base of their utility by implementing some rule for aggregation of alternative's estimates upon separate criterion.

Problems of elicitation and implementation of information about DM's preferences for comparison (evaluation) of alternatives have become very important in the process of developing methods for multicriteria decision making.

If we analyse the existing ways for solution of these problems (see, for example, reviews of von Winterfeldt & Fischer, 1975, Humphreys, 1977, von Winterfeldt, 1975), it is easy to see that in decision methods we can find three types of the information used:

- 1) on the importance (value, utility, preferability) of the criteria used for estimation of alternatives;
- 2) on the importance (value, utility, preferability) of separate estimates upon criterion scales;
- 3) on the model of aggregation of the previous information (in the form of alternatives' value or utility functions, rules for

comparison of alternatives and so on).

Certainly, in different decision methods there may be used (explicitly) not all of the mentioned types of information, but the task solution usually reflects all three types of information.

It is worth mentioning the crucial difference between the first two and the third type of information. The information of the first two types is received usually through the dialogue with a DM, the way (model) of implementation of this information (form of value or utility functions, some rules for comparison or estimation of alternatives and so on) are rather often proposed by the authors of the decision method (axiomatically or heuristically). Nevertheless there exist systems, in which the DM is also asked to select the rule for aggregation of criterion estimates (see, for example, Khaynish & Vlasov, 1983).

Let us consider the way of eliciting judgements from people. The following ways methods the mostly popular in decision making:

- 1) pairwise comparisons on the base of importance (preferability, utility, valuability and so on) of different objects;
- 2) direct evaluation of importance (preferability, utility, value, probability and so on) of different objects;
- 3) ranking on the base of importance (preferability, utility, value, probability and so on) of different objects;
- 4) choice of the best object from the set on the base of importance (preferability, utility, value, probability, and so on).

Objects in these operations may be viewed as attributes (criteria), alternatives (options), estimates upon criterion scales, lotteries, probabilities and other concepts of decision tasks.

In (Larichev et al., 1987) there was made an attempt to collect and classify all elementary operations in information processing, used in normative decision making. 23 such operations were defined and analysed from the point of view of their complexity for human beings. The main conclusion which it is possible to make analysing the data, states that quantitative evaluation and comparison of different objects is much more difficult for subjects than to carry out the same operations in a qualitative form.

This conclusion is rather popular in nowadays and is based not only on the data of descriptive investigations, but also on the experience in real decision problems (Larichev, 1984). Of course, some of the analysed operations have been rather thoroughly investigated (as, for example, pairwise comparisons of alternatives and estimates upon criterion scales; assessing of subjective probabilities; judgements on importance of criteria; comparison of lotteries and some others). Other operations have not been explored to the necessary extent. But on the whole, the idea that it is more preferable to use qualitative information (from the point of view of the appropriateness for people), must not face severe opposition. That is why further material of the report will be concentrated on the qualitative information about DM's preferences and the possibilities for its implementation.

Direct evaluation

The operation of direct evaluation of objects in decision making procedures is considered to be one of the most unpopular and its inappropriateness in decision methods is argued much (see, for example, Larichev et al., 1987, Larichev, 1984). At the same time if this operation is not a desirable one in assessing weights to criteria, utility to alternatives, value to estimates upon criterion scales, value to lotteries, its implementation for estimation of alternatives upon criteria with continuous scales may be justified (for example, for evaluation of price, profit, weight and so on). But to our mind, even when you have continuous scales, in many cases in multicriteria decision task, it is more preferable to transform them into verbal (ordinal) ones.

In fact estimates upon criterion scale must be of the type to reflect the differences in value (utility, preference, importance and so on) of alternatives. People poorly evaluate and compare alternatives with similar utilities. While using continuous scales, unessential changes in data for one criterion may lead to an inconsistency in preferences, as the alternative's preferability will be caused by this unessential difference in a pair of alternatives with all other equal estimates, but when estimates upon other criteria change, people will easily neglect

this difference upon the criterion under consideration (as on the whole the quality (utility) of alternatives having this difference is practically the same for the DM).

Just this property was the basis for Tversky's experiment (Tversky A., 1969), in which the stable intransitivity of preferences was demonstrated: the subjects were presented sequentially with pairs of lotteries, in which the award and the loss were unessentially enlarged. Subjects were stable in preferring the enlargement of the award even with some small increase in a possible loss. But when lotteries from the first and the last pairs were presented, just the same subjects were stable in preferring the first lottery as they did not want to agree with so large increase in losses even with the corresponding growth of the award. To our mind, the used continuous scale for estimates did not allow the subjects to recognize the "evolutional transformation of quantity into quality". The implementation of ordinal (qualitative, verbal) scales for losses and awards in the same task with the same subjects might lead to transitive preferences.

The uncomfortability of continuous scales for a DM was demonstrated also in one of our experiments in which subjects had to choose a summer house to rent. In this experiment the variants of summer houses were estimated by two criteria with continuous scales (other criteria had verbal ordinal scales): cost and area. The main goal of the experiment was to compare sequential and parallel processes in a choice task. At the same time, it was found out that subjects were not always able to see dominating alternatives, when this dominance was due to small differences in cost. Some times subjects rejected dominating alternative from a rather small subset (4-7) of the "best" alternatives. We think that such facts may also be explained by small differences in cost, which do not influence the value of alternatives on the whole.

Rather often the direct quantitative evaluation is used in assessing probabilities to some events. It is known from special literature (Slovic et al., 1977, Kahneman et al., 1982) that people are "poor statisticians" and that their judgements suffer from stable biases.

These problems are under thorough investigation for many years. In spite of it, the implementation of quantitative estimates of probabilities is rather popular, as the idea to use the expected utility theory (Savage, 1954, Luce & Raiffa, 1957) for problem solution is a very attractive one. Such estimates are usually used even by specialists in expert systems, who traditionally try to base only on qualitative concepts and logical rules to inference the result (these estimates are used to mark the assurance in the system's answer) (Johnson & Keravnov, 1984).

But, as in a previous case, it is possible to use qualitative terms to assess subjective probability to different events. In fact, subjects make are not very good in quantitative judgements of probability (even if they have enough knowledge in the probability theory). At the same time people rather often and with enough certainty use such definitions as: "very probable", "rather likely", "most unlikely" and so on. If you ask people, what numbers they think of, using such terms, the difference in them may be rather significant. But sometimes experts estimating the probabilities of the same objects would give one and the same definitions (of the previous type), though numbers might be different. What does it mean?

To our point of view, this means just that people make numerical estimations rather poorly. In (Dror, 1989) while analysing tasks of political choice, it is regretted that people give different numbers for such verbal definitions (and so, there is no valid transition from words to numbers). To our mind this must not obligatory mean that, one considers this event to be of probability 0.7 and considers this 0.7 to be "rather probable" and the other assesses to this event the probability 0.9 and considers it to be "rather probable". It is possible that both experts are sure that this event is "rather probable", but when they are asked to estimate this probability in numbers (in percents, or as part of a line or in some other way), they replace their difficulty in fulfilling such job with some (enough random) number. In such case the estimate "rather probable" may be much more reliable than any numerical equivalent.

B (Gallhofer & Saris, 1989) while analysing old political

decisions on the base of documents (about discussions, argumentation and so on) there has been given an example, in which the adopted decision is explained in terms of several criteria, one of which has been the probability of different events. This criterion had a verbal scale. Why was such scale used? Just because of the fact that in the real discussion, judgements and arguments were given verbally. It is natural for human beings. But in this work it is shown that people not only naturally use such estimates, but that the decision may be also explained on this verbal level (choice among alternatives with verbal estimates upon criteria, one of which is probability of events).

All the previous discussions were concerned mostly with the direct evaluation of simple objects (alternatives upon separate criteria or separate attributes and so on). Even in such more simple cases it has been shown that it is more preferable to use rather limited number of verbal estimates. This means that we replace direct evaluation of objects, by their direct classification (when a priori we know the definition of the classes). This task is rather thoroughly investigated in (Larichev & Moshkovich, 1986, Clancey, 1984). Even the implementation of scales with more than 5-7 separate numbers (or ranks) leads to much more unstable information.

There are some works in which it is proposed to carry out quantitative evaluation of multiattribute objects. It is evident that assessment of value (utility) to an alternative or criterion is too complicated operation and must be replaced by a more simple one. What holistic estimation is possible for complex objects? Just the same as for simple ones (ordinal classification). But there are some limitations for the dimensionality of objects. This problem has been studied in (Larichev & Moshkovich 1988a, Larichev, Moshkovich, Rebrik 1988).

So, we are able to say that there are arguments for implementation of a limited number of verbal estimates for criterion scales as well as for estimation of multiattribute objects.

Ranking and choice

Different decision methods use subjective rankings of

different objects and also choice of the best one from some (not large) set of alternatives. Though both of these operations deal with receiving qualitative judgements, the complexity of such operations is well known. This is possibly due to the fact that they are not elementary. As a rule the ranking of objects may be received as a result of their pairwise comparisons, and choice of the best alternative may be carried out on the base of objects' ranking. Why should we study them by themselves?

At different stages of decision methods such modes of DM's preferences' elicitation are rather often used. For example, in many decision methods ordinal criterion scales are used. The process of such scales' formation usually deals with ranking of possible estimates upon the criterion. In multicriteria mathematical programming procedures a DM is often faced with the problem of choosing one variant out of several effective points (Korhonen, 1986, Roy 1968). That is why it is useful to realize the capacities of these ways of information elicitation.

Our experience shows that people are good enough at ranking several simple objects (estimates upon a criterion scale, criteria, decision classes and so on). If it is necessary to rank complex objects (for example, multiattribute alternatives) the task becomes much more sophisticated, and information becomes much more inconsistent. But even if you deal with rather simple objects it is necessary to take into account the possibility of errors in one's answers (as in any subjective information).

The way how people decide a choice problem is a traditional subject for psychological investigations. It is known that this problem is enough complicated one, even when you have rather small number of objects (Payne, 1976, Russo & Rosen, 1975, Svenson & Montgomery, 1983). In (Montgomery et al. 1990) it is pointed out that in an experiment the task "to choose out of two objects" and the task "to mark the more preferable of two objects" may lead to different results. In (Korhonen, 1986) the essential inconsistency in the choice of the best alternative from a subset of effective decisions is reported to be observed. So, we have to understand that choice is too complicated for people and try to avoid it in the proposed methods.

Pairwise comparisons

Pairwise comparisons of different objects is one of the most popular ways of information elicitation. It is possible to carry out pairwise comparisons of simple and complex objects, in a quantitative or qualitative form. Quantitative comparison of any objects is difficult for people (even using rank estimation). It is also difficult to compare complex objects (multiattribute alternatives, lotteries, and etc.) even in a qualitative form (Tversky, 1969, Larichev, 1984, John et al., 1983).

As it has been marked, pairwise comparisons are often used to solve ranking problems (as this operation is considered to be more simple). For example, to form ordinal scales for attributes in a medical diagnostic task a special procedure, based on pairwise comparisons is implemented (Furems & Mechitov, 1990), in (Podinovsky, 1978) it is proposed to carry out pairwise comparisons of criteria importance, a number of methods is based on pairwise comparison of multiattribute alternatives (Larichev, 1982, Keeney, 1980, Edwards, 1983). We can find different publications, devoted to investigation of how people carry out pairwise comparisons (Payne, 1976, Russo & Rosen, 1975, Larichev, 1984).

Out of the results it is possible to say that people are good enough at comparison of alternatives differing by not more than two-three criteria.

While comparing complex objects even in a qualitative form people tend to make mistakes and contradictions in their judgements. So, it is more preferable to use information on pairwise comparisons of simple objects.

As a result we can say that it is more preferable to use qualitative judgements and the tendency to use such type of information dominates in decision procedures during last years. Nevertheless this information is usually transformed into a quantitative form (Humphreys & Wishudha, 1982, Edwards et al., 1977).

IMPLEMENTATION OF INFORMATION ABOUT DECISION-MAKER'S PREFERENCES

Almost in any decision method some function of utility

(value, importance, preference or any other) for alternatives is used. It is necessary to note that further we shall interpret utility function in a more general way - as some rule for evaluation or comparison of multiattribute alternatives.

For many years the task of a decision aid was meant as a task to find an appropriate rule for aggregation of criterion estimations. So, efforts in this field were concentrated on analysis of advantages and disadvantages of different aggregation rules to be used (Keeney & Raiffa, 1976, Rivett, 1977, Svenson, 1979, Svenson, 1983)).

The implementation of an additive utility function is most popular among specialists as it is the most simple and understandable model for both a DM and a consultant. Sometimes multiplicative and combinations of additive and multiplicative forms of utility functions are used. In (Humphreys, 1983) there is review of utility functions, means for determination of their parameters and of axioms forming their basis. It is shown that different forms of utility function are based on the axioms of independence of different subsets of criteria from their supplement.

In spite of a large popularity of these methods, there are problems connected with their application (Montgomery, 1983): complicated judgements, needed from a DM (comparison of lotteries, construction of indifference curves, assessment of criterion weights and so on); impossibility to explain the results on the base of the made judgements; implementation of quantitative measures for qualitative notions. But it is not always necessary to replace qualitative information by quantitative measurements (Larichev & Moshkovich, 1988b).

Why are researchers try to build formal models with quantitative parameters for alternatives' utility? The answer is rather simple. The implementation of such models makes it possible to carry out "exact" (quantitative) evaluation of alternatives, to fulfil complete ranking of them and choose the best one and in such a way to solve all problems at hand. Just very "strong" result is so attractive for specialists in decision methods. But there is an opinion that such result is not always necessary in real decision tasks (Berkeley et al., 1990).

In some real tasks, in which implementation of logical rules and understandable reasoning is not enough to choose the best alternative, the possibility to achieve the decision by introducing numerical estimations and formal aggregation rules, may not be convenient for a DM. For example, if alternatives are equal for a DM in quality and according to the carried out measurements their utilities (in a numerical form) are different, then the DM may not want to use such model in a real task. In this case we achieve only the impression that we have the decision as we substitute a decision maker by some (smaller) decisions of a consultant (or the author of the method). In general, unessential differences in numerical expression of utilities, as a rule, do not confirm a DM in a validity of the measurements made, but on the contrary, may cause the occurrence of some doubts in them (Larichev, 1979).

So, we can conclude that implementation of numerical measurements and simple aggregation rules are caused mostly by the possibility to guarantee the task solution. But it is necessary to understand that in many cases we are able to obtain task solution without the transition from qualitative information to a quantitative one.

Implementation of qualitative information for evaluation and comparison of multiattribute alternatives

The most simple rule, based on qualitative judgements is the rule of dominance, that is, component comparison of multiattribute alternatives (selection of non-dominated alternatives according to the Pareto principle). This rule is implemented in many decision methods at the first stage to contract the initial set of alternatives. It is necessary to note, that such rule is correct if each criterion is independent (preferentially) from all the others. But this rule usually lacks the possibility to solve the task.

Next examples of decision rules based on qualitative judgements, are connected with ranking of criteria according to their importance for a DM. Lexicographical rule for comparison of alternatives is a well known example of such rule. According to

this rule first we select alternatives possessing the best estimates upon the mostly important criterion, in the resulting subset we select alternatives possessing the best estimates upon the next important criterion and so on. This rule is based on the assumption that one criterion is more important than all the other criteria together. The evident drawback of such rule is its non-compensatory nature, which makes this rule rather unpopular.

In spite of the fact, that the notion of criterion importance is widely used in decision theory, there almost no attempts to determine what does it mean and to use it in accordance with such definition. For the first time the definition of a relative importance (preferability) of a criterion was given in (Podinovsky, 1978).

According to Podinovsky's definition:

criterion K_1 is more preferable (important) than criterion K_2 , if in all possible pairs of alternatives, with different estimates upon only criteria K_1 and K_2 , an alternative with more preferable estimate upon criterion K_1 is considered to be more preferable.

B (Ladenson & Litvak, 1988) another rule based on ranking of criterion importance was suggested. It is based on the principle of pairwise compensation:

alternative A is not less preferable than alternative B, if for each criterion upon which alternative A is less preferable than alternative B, there exists a not less preferable (important) criterion, upon which alternative A is more preferable than alternative B.

Let us note that this rule is formally correct if we define criterion importance (preferability) as in the previous definition. Let us also mark that in this definition the requirement for preferential independence of the two criteria under consideration from their supplement is presented.

So, the implementation of such decision rule will assume the pairwise preferential independence of criteria.

The evident drawback of such rule is connected with its inability to take into account differences between criterion estimates. So, the introduced notion of criterion importance (preferability) is too strong and not always appropriate for real decision tasks even if it is easy for a DM to rank the criteria.

The notion of criterion importance is also presented in an additive value function (the existence of which is based on the pairwise preferential independence of criteria (see (Keeney & Raiffa, 1976))). But as it has been discussed, the assessment of numerical values for criterion weights (characterizing their importance) and estimation in a numerical form the value of separate estimates upon criterion scales requires too complicated information about DM's preferences (Humphreys, 1977, Humphreys & McFadden, 1980).

What is it possible to do if the requirement of criteria ranking upon importance, is not fulfilled (there is an influence of difference in estimates upon separate criteria on the general preference of an alternative)?

One of the approaches to the solution of such problem is to build a joint ordinal scale (Larichev et al., 1974, Ozernoy & Gaft, 1978) on the base of pairwise comparison of alternatives of special type, differing in estimates by only two criteria.

In such scale estimates upon all criteria used in evaluation of alternatives, are ordered in accordance with DM's preferences. Implementation of such scale for comparison of real alternatives is also based on the idea of pairwise compensation:

alternative A is not less preferable than alternative B, if for each component of alternative A there exists a component of alternative B with not more preferable estimate upon joint ordinal scale.

It is necessary to note that the correctness of such rule is also based on the preferential independence of all pairs of criteria from their supplement (Gnedenko et al., 1986).

Some decision tasks may be driven to a classification ones. In them the information about, how a DM has classified some subset of alternatives, may be used to construct a decision rule (Larichev et al., 1989, Larichev & Moshkovich, 1990).

So, we are able to conclude that there is a set of rules for comparison and evaluation of multiattribute alternatives based on qualitative judgements of a DM and experts. At the same time, almost all these rules are based on the property of preferential independence of criteria (separate ones or their subsets), and does not guarantee the choice of the best alternative or the

ranking of all alternatives (according to the initial decision task) as there may be uncomparable pairs of alternatives.

Certainly, there are decision tasks, in which the partial ordering of alternatives is enough for a DM. But it is necessary to develop the approaches, based on qualitative judgements, which are able to guarantee the pairwise comparison of alternatives without the resort to rather artificial numerical measurements, but within the framework of verbal and logical description of the initial decision task as well, as the process of it's solution.

In conclusion it is necessary to point out that the implementation of qualitative information and logical rules for its processing is not enough to guarantee the absence of inconsistencies and errors in the recieved information. So, it is necessary to have means for detection and correction of errors in such information, and also means to check the implied axioms, but these tests may be also carried out on the base of qualitative estimation.

Further, two methods (decision support systems) are decribed in which we try to solve some of the indicated problems.

DECISION SUPPORT SYSTEMS BASED ON QUALITATIVE INFORMATION ABOUT DECISION-MAKER'S PREFERENCES

SYSTEM ZAPROS

Problem formulation

System ZAPROS is a rather developed and known one for ordering of multiattribute alternatives (Gnedenko et al., 1986, Moshkovich, 1988). It has been in progress for several years and was being updated in accordance with new knowledge about decision making and human beings' capacities in information processing. Here the last version of the system will be described (Larichev & Moshkovich, 1991).

Let us assume that alternatives are estimated upon a set of criteria with verbal formulations of quality grades upon their scales (Larichev, 1979). And let us imagine that the number of alternatives is rather large. In this case it may be considered logical enough to construct some rule for pairwise comparison of alternatives on the base of DM's preferences in the criteria space. Just this problem is under further consideration.

In general, this problem may be formulated as follows:

Granted:

1. Q - number of attributes;
2. $K = \{1, 2, \dots, Q\}$ - a set of attributes numbers;
3. n_q - number of estimates on the scale of the q -th attribute ($q \in K$);
4. $X_q = \{x_{iq}\}$ - a set of estimates' numbers for the q -th attribute, ordered from the best to the worst one (the scale of the q -th attribute);
 $|X_q| = n_q$ ($q \in K$);
5. $Y = X_1 \times X_2 \times \dots \times X_Q$ - a set of vectors $y_i \in Y$ of the following type $y_i = (y_{i1}, y_{i2}, \dots, y_{iQ})$;

$$N = |Y| = \prod_{q=1}^Q n_q$$

6. $A = \{a_i\}$ - a set of vectors, describing the real alternatives.

Required:

to form an ordering of multiattribute alternatives of the set A on the base of decision-maker's preferences.

In the proposed further approach for ordering of the initial set of multiattribute alternatives two rather simple assumptions about the properties of decision-maker's preference system are used, those are : preferential independence of criteria and transitivity of the resulting preference-indifference relation .

Approach to preferences' elicitation

From the initial data it is known that estimates upon attribute (criterion) scales are ordered from the most preferable for a DM to the least preferable ones. This means that the first

estimate upon an attribute q ($q \in K$) is more preferable for a DM than the second estimate upon the same attribute and so on (in spite of the estimates upon other attributes).

This information defines a relation of strict dominance P° on the set Y :

$$P^\circ = \{ (y_i, y_j) \in Y \times Y \mid \forall q \in K \ y_{iq} \leq y_{jq} \text{ and } \exists q^\circ \text{ such that } y_{iq^\circ} < y_{jq^\circ} \}.$$

Further enrichment of this binary relation on the set Y is possible only due to the elicitation of additional information on a DM's preference system.

Definition 1. Let us say that a pair of vectors from the set Y is admissible for comparison by a DM if these vectors differ in components upon only two criteria.

In general a DM is to be presented for comparison with all admissible pairs of vectors from Y . Nevertheless, the number of comparisons may be essentially decreased, if for criteria from K the requirements of preferential independence of criteria are met (Fishborn, 1970, Keeney & Raiffa, 1976).

This property (as it will be shown later) allows to use effectively information on comparison of multiattribute alternatives near the so called, reference points.

Each combination of estimates upon criteria is an image of a certain alternative for a DM. The two most bright "contrasting" images correspond to the combinations of the best and the worst estimates upon all attributes. Such combinations were called reference points (Larichev et al., 1974).

Definition 2. Vectors with all the best or all the worst estimates upon all criteria will be called reference points.

Let us form lists L_1 and L_2 near the first and the second reference point correspondingly:

$$L_1 = \{ y_i \in Y \mid y_i = (1, 1, 1, \dots, 1, y_{is}, 1, \dots, 1), y_{is} \neq 1, \forall s \in K \};$$

$$L_2 = \{ y_i \in Y \mid y_i = (n_1, n_2, \dots, n_{s-1}, y_{is}, n_{s+1}, \dots, n_Q), y_{is} \neq n_s, \forall s \in K \}.$$

$$\text{It is evident that } |L_1| = |L_2| = N_1 = \sum_{q=1}^Q (n_q - 1).$$

We propose to carry out an interview with a DM for each list of vectors. The procedure will be described for the list L_1 as an

example. An interview near the second reference point (list L_2) is carried out in the same way.

The results of pairwise comparisons by a DM may be presented in a form of binary relations as follows:

1) let us say that $(y_i, y_j) \in P_{DM}$, if $(y_i, y_j) \in L_1 \times L_1$ and according to a DM's opinion y_i is more preferable than y_j , or if $(y_j, y_i) \in P^0$.

2) let us say that $(y_i, y_j) \in P_{DM}^{-1}$, if $(y_i, y_j) \in L_1 \times L_1$ and according to a DM's opinion y_i is less preferable than y_j , or if $(y_j, y_i) \in P^0$.

3) let us say that $(y_i, y_j) \in I_{DM}$, if $(y_i, y_j) \in L_1 \times L_1$ and according to a DM's opinion y_i and y_j are equally preferable, or when $i=j$.

If relations P_{DM} and I_{DM} are transitive, then according to (Mirkin, 1979), the relation I_{DM} is the relation of equivalence and the relation P_{DM} is a quasi-series, possessing the property of common transitivity, and the relation $R_1 = P_{DM} \cup I_{DM}$ is a linear quasi-order on the set L_1 . If not all pairs from $L_1 \times L_1$ are compared, then R_1 is a quasi-order.

In any interview with a DM there is a possibility of errors in his (her) answers. These errors may be random or may occur while comparing similar in quality alternatives. So, in the information, elicited from a DM there may appear inconsistencies (contradictions).

So, in (Moshkovich, 1988) the following heuristic procedure to check and correct DM's answers in the process of carrying out pairwise comparisons is suggested.

As the transitivity of DM's preferences is initially assumed and violations of this assumption are considered to be errors in DM's answers, we suggest the following procedure. After each comparison of vectors from L_1 made by a DM, the obtained information (on vectors' comparison) is spread on the base of transitivity (transitive closure of the binary relation defined on the set L_1 is being built).

After that a DM is presented with the next pair of vectors from L_1 , for which the relation has not been defined. Receiving the answer, the transitive closure is being built and so on up to the moment of establishing relation for all pairs from L_1 . It is

necessary to note that during such procedure violations of transitivity can not occur.

On the other hand, while spreading the information on the base of transitivity it is to some extent, assumed that there are no errors in DM's answers and his (her) answer exactly defines relations between other pairs of vectors. As it is necessary to test answers given by human beings, we suggest to present a DM with additional pairs of vectors for comparison on the base of the following principle.

Let us require for relation between each pair of vectors to be defined directly (by a DM's answer) or indirectly (upon transitive closure) not less than two times. This requirement means that if a DM by two of his (her) answers (may be indirectly - upon transitive closure) has equally defined the relation between some pair of vectors from L_1 , then this relation is considered to be verified. If the relation between some pair of vectors from L_1 has been defined only once and only upon transitive closure, then this pair is presented additionally to a DM for comparison.

If the DM's answer coincides with the previous one, then it is considered that the correctness of the information obtained from a DM is proved. If there is some difference, the system finds out among previous DM's answers a triple of vectors, pairwise comparison of which contradicts to the transitivity of the relation built on L_1 . Such triple may always be detected.

After presenting a DM with contradictory answers, he (she) is asked to change them so, that they are not any more in an intransitive relationship. To correct the set of DM's answers in accordance with this last information the following procedure is suggested.

It is supposed that we only start the interview with a DM (that is we have only these three answers for pairwise comparisons of y_i with y_j ; y_j with y_k ; y_i with y_k). And also at this moment we know that these answers do not contradict to each other. Let us include each of these answers to P_{DM} or I_{DM} accordingly and carry out the transitive closure of the obtained relation as in the initial interview with a DM. After that the system carries out further formation of the binary relation on

L_1 .

During this procedure for pairs of vectors from L_1 for which the relation has not been defined, the information is obtained from the previous answers of a DM. After that the transitive closure is built and so on. As a result we obtain new transitive relation on the set L_1 in which the necessary changes are being made, but all previous answers, not contradictory to the new ones, are maintained unchanged.

After that, the condition of "double test" for each pairwise comparison is checked for this new information.

The proposed approach makes it possible to form an effective procedure for DM's interview to build the required relation, as the redundancy of the obtained information is limited to a reasonable condition of minimal necessary test for DM's answers.

Preferential independence of criteria

As a result of an interview with a DM and transforming his (her) answers to a non-contradictory variant, the transitive and reflexive relation $R_1 = P_{DM} \cup I_{DM}$ of a quasi-order on the set L_1 is built (if this is a connected relation then it is a linear quasi-order).

Analogously the relation of quasi-order R_2 on the set L_2 near the second reference point is built.

As a result we have some amount of information about DM's system of preferences.

To use effectively the information obtained from a DM it is necessary to have preferential independence for all pairs of criteria (Gnedenko et al., 1986).

The problem of checking this axiom (as well as checking many other axioms of multiattribute utility theory) has no simple solution. In reality, the necessity to use this axiom results from the desire to construct an effective decision rule on the base of relatively small amount of rather simple information about DM's preferences (the effectiveness of the decision rule means its possibility to guarantee rather high level of compatability for real alternatives on the base of this rule). On the other hand, the full-scale check of DM's preferences implies

the necessity for a DM to carry out rather many pairwise comparisons of vectors (in some task - enomously large number of comparisons). So, the problem is to make not full-scale but representative enough check of DM's preferences for the axiom.

It is evident that list L_2 contains some vectors with all estimates but one equal to the worst ones and with one estimate at the best level. So, there is a possibility to compare relations between pairs of vectors near two reference points of the following type:

$$\begin{aligned} L_1: y_i &= (1, 1, \dots, 1, n_s, 1, \dots, 1) \\ y_j &= (1, 1, \dots, 1, n_t, 1, \dots, 1); \\ L_2: y'_i &= (n_1, n_2, \dots, n_{s-1}, n_s, n_{s+1}, \dots, n_{t-1}, 1, n_{t+1}, \dots, n_Q) \\ y'_j &= (n_1, n_2, \dots, n_{s-1}, 1, n_{s+1}, \dots, n_{t-1}, n_t, n_{t+1}, \dots, n_Q). \end{aligned}$$

It is clear that if criteria s and t are preferentially independent from the other, relations between vectors y_i and y_j must be the same as between vectors y'_i and y'_j . So, there is a possibility to carry out some justification of the axiom on the base of the information, obtained near two reference points.

Let us emphasize that though such justification is a very limited one, the violation of this condition rather evidently proves the violation of independence and the necessity for additional analysis of the situation (see later), as all these relations have been thoroughly checked during comparisons near two reference points.

If the relations between corresponding vectors are the same, then evidence is to the benefit of preferential independence of the corresponding criteria. Some additional checking of the axiom will also be possible, while implementing the obtained information. This possibility will be discussed later.

Comparison of alternatives on the base of the recieved information

Information on comparison of vectors' pairs obtained near the first reference point may be used for construction of the so called, joint ordinal scale (Gnedenko et al., 1986, Larichev & Moshkovich, 1991).

Joint ordinal scale (JOS) in these works was considered to

be the ranking of vectors of the list L_1 . If the binary relation, built on the set L_1 is a connected one, then it means that on this set a linear quasi-order is defined (in some cases - this relation is a linear order). So, to each vector from L_1 we can assess the number of its place in this ordering.

If we recall the peculiarity of vectors near the first reference point (that only one component in them differs from the best (first) one) it will be easy to understand, that this number may be considered as the place obtained by this estimate (the component not equal to 1) in a general ranking of all estimates upon all criteria. That is why this ranking was called joint ordinal scale (joint scale for all estimates of all criteria).

On the base of the JOS the following rule may be formulated: vector $y_i = (y_{i1}, y_{i2}, \dots, y_{iQ}) \in Y$ is not less preferable for a DM than vector $y_j = (y_{j1}, y_{j2}, \dots, y_{jQ}) \in Y$, if for each criterion $s \in K$ there exists the criterion $t(s) \in K$ such that:

$$(1, 1, \dots, 1, y_{is}, 1, \dots, 1) R_1 (1, 1, \dots, 1, y_{jt}, 1, \dots, 1).$$

In (Gnedenko et al.) it is proved that this rule is correct in the case of pairwise preferential independence of criteria. In (Larichev & Moshkovich, 1991) it is also proved that this rule may be modified for implementation while comparing vectors from Y on the base of the relation R_2 , built near the second reference point, and also for implementation for one part of criteria the relation R_1 and for the other - the relation R_2 .

What does this property lead to?

While comparing real alternatives it is possible to use decision rules, built near both reference points. If the results are the same, then this is an indirect confirmation of preferential independence of criteria and justification of rules being used (this means that we carry out additional check of criteria independence just while comparing real alternatives).

If according to one rule alternatives can be compared, and according to the other we are not able to compare them, then this is not a contradictory situation. We just enlarge compatibility of alternatives on the base of additional information from a DM about comparison of vectors.

If the results of comparison are contradictory, they are connected with violation of criteria independence. The

alternatives are to be considered uncomparable.

So, even if there is some evidence about the dependency of of criteria, we are able to use the built rules to compare those pairs of alternatives, which are not influenced by this dependency..

We are able to estimate the number of alternatives' pairs which it will be possible to compare additionally if we analyse the dependency thoroughly. Analysis and elicitation of dependent criteria and also procedures for reformulation of the initial task, proposed further in this case, are rather labour-consuming. So, a DM is able to evaluate if he (she) wants to spend rather large amount of time and effort, knowing the maximum of additional information about comparison of alternatives which it is possible to obtain.

Task modification in the case of dependent criteria

If we get information about possible dependence of preferences, while comparing vectors on the level of reference point, a question arises as what to do in such situation.

We suggest the following approach for manipulation of dependent criteria. It is based on the data of descriptive studies on human capacities in processing multiattribute information and also on the results of field studies of the method.

At first, the important assumption of the proposed approach is the idea that it is necessary to try to change the initial task description so, as to avoid the dependence of criteria (this means to modify the system of criteria). Analogous attempt was made in (Humphreys & Wishudha, 1982) while implementing the theory of expected utility for comparison of a small number of real alternatives.

For the second, the alterations in preferences near two reference points may be caused by the fact, that the least preferable estimate upon one of the criteria in the second reference point, is not admissable for a DM in reality (this means that alternatives possessing such estimate upon this

criterion are not acceptable as a decision option). In such situation if we ask the DM to compare vectors, possessing such component, the process of comparison may be not stable. Our practice shows that near the second reference point a DM more often uses the answer: "vectors are equally preferable". This answer points out mainly that both vectors are not acceptable for a DM. That is why it is not convenient to try to evaluate what the better one is.

For the third, psychological experiments show that as a rule, group of dependent criteria may consist of two, sometimes - three criteria, but not more. This situation is due mostly to the limitations of the short-term memory. These limitations make it impossible for people to take into account simultaneously large number of estimates upon different criteria.

Definition 3. Let us call pair of vectors $(y'_i, y'_j) \in L_2$ analogous to pair of vectors $(y_i, y_j) \in L_1$ if:
 $y_i = (1, 1, \dots, 1, y_{is}, 1, \dots, 1)$, $y_j = (1, 1, \dots, 1, y_{jt}, 1, \dots, 1)$;
 $y'_i = (n_1, n_2, \dots, n_{s-1}, y_{is}, n_{s+1}, \dots, n_Q)$;
 $y'_j = (n_1, n_2, \dots, n_{t-1}, y_{jt}, n_{t+1}, \dots, n_Q)$.

In (Larichev & Moshkovich, 1991) it is proved that if the comparison between all analogous pairs of vectors near two reference points are the same, (that is: if $(1, 1, \dots, y_{is}, 1, \dots, 1) R_1 (1, 1, \dots, y_{jt}, 1, \dots, 1)$, then $(n_1, n_2, \dots, n_{s-1}, y_{is}, n_{s+1}, \dots, n_Q) R_2 (n_1, n_2, \dots, n_{t-1}, y_{jt}, n_{t+1}, \dots, n_Q)$), then there may be no violations of preferential criteria independence within the information at hand.

So, analysing the result of comparison of analogous pairs of vectors near two reference points, it is possible to find out all pairs of dependent criteria in the following way:

1. Find out a pair of dependent criteria.

To do this let us select criteria s and t , for which comparison of analogous pairs differs near two reference points. After that try to ask a DM about what estimate according to his (her) opinion, has influenced the alteration of preferences. If a DM is not able to answer this question, then one of the criteria, not equal to s or t , is selected. Then a DM is presented with a pair of vectors possessing more preferable (on one gradation)

estimate upon the selected (or pointed out by a DM) criterion. If the comparison is the same as at the first reference point, then we have found out the estimate which caused the alteration of preferences. Otherwise, we select the next criterion and the analogous operations are carried out. And so on.

As a result we find out the number of criterion q and the estimate n_q , which has caused the alterations in DM's preferences.

2. Explore what does the dependence of these criteria mean.

Alterations in preferences near two reference points may have two explanations (in terms of violations of initial assumptions).

First of all, there is a possibility that the order of estimates on the scale of criterion s and/or criterion t is changed with changes in estimates upon criterion q . That's why it is necessary to analyse the ranking of estimates upon scales for criteria s and t with estimate n_q upon criterion q . To do it a DM is presented for comparison with pairs of vectors of the following type:

$$\begin{aligned} y_i &= (1, 1, \dots, y_{is}, 1, \dots, 1, n_q, 1, \dots, 1) \\ y_j &= (1, 1, \dots, y_{js}, 1, \dots, 1, n_q, 1, \dots, 1). \end{aligned}$$

If for one criterion (for example, for s) the order of estimates is being changed (in comparison with the initial one) it means that criteria s and q are mutually dependent. A DM is asked to combine these two criteria into one criterion with a scale, containing, in general, all possible combinations of estimates upon criteria q and s .

If the order of estimates upon scales for criteria s and t is not changed, the conclusion is made that all three criteria s, t and q are mutually dependent. In this situation it is necessary to build aggregate scale for all three criteria.

3. Construct the scale for the new criterion.

A DM is asked to order combinations of estimates upon dependent criteria scales. In general some combinations of estimates may be of equal value for a DM. In this case it is possible to reduce the new scale by reformulating the equal combinations of estimates into one gradation.

If the task of ordering of combinations of estimates upon three criteria becomes too complex, it is possible to use special procedure of ordinal classification (Larichev & Moshkovich, 1990) to fulfil the task.

4. Carry out a new interview with a DM.

After formation of a new set of criteria a new interview with a DM near two reference points is carried out (as the description of the task has been changed).

So, step by step, it is possible to find subgroups of independent criteria, to construct aggregate scales for them and to carry out the ranking of real alternatives on the base of the proposed scheme.

Let us note that the time and labour consumption while working with the method will decrease as in the process of such work the number of criteria will be reduced.

SYSTEM ASTRIDA

Problem formulation

Let us consider the same problem as in the paragraph "SYSTEM ZAPROS" but when number of real alternatives is a rather small one (may be 3 or 4, but not more than 10-15), and it is necessary to find the best one of them. Such tasks are rather spread in strategic decision making.

Let us assume that as in the previous case, alternatives are estimated upon a set of criteria with verbal quality gradations. But in this case it is not logical to build up a rule for comparison of alternatives just in the criterion space, as the number of alternatives is rather small. In such case it is necessary to analyse only a small part of possible combinations of estimates upon criterion scales (corresponding to real alternatives), and on the other hand, the rule built in the criterion space, does not guarantee the possibility to compare just these alternatives.

That is why it was necessary to develop a decision method able to carry out the comparison of multiattribute alternatives on the base of qualitative information about DM's preferences.

ASTRIDA (Advanced STRategic Intelligent Decision Aid) is a decision support system for strategic choice, which is to provide problem structuring and the necessary analysis and estimation of possible alternatives to solve the initial problem (Berkeley et al., 1989, Berkeley et al., 1990).

The ASTRIDA design is based on the assumption that a decision-maker in general has enough knowledge about the problem he (she) is facing and also about the possible ways of its solution. The system is used for the purposes: 1)to organize the DM's thoughts about the task and the best alternatives among which to seek the decision; 2)to provide the aid for a DM as to how the best alternative may be determined (not simply selected, but sometimes generated in the process of decision making).

Two main mutually connected functions are combined in the system: 1) problem structuring; 2)aiding the choice of the best alternative (which does not have to be one of the initially considered).

Choice of the best alternative is carried out on the base of pairwise comparison of alternatives. So, one of the main problems the designers faced, was the procedure for comparison of two alternatives, estimated upon a set of criteria, on the base of DM's preferences.

While developping such procedure we tried to take into account analogous principles connected with descriptive data on human capacities to make judgements.

We assume that:

- 1.A DM is able to compare the preferability of alternatives' estimates upon one and the same criterion.
- 2.A DM is able to compare the preferability of one of the two alternatives if they differ in estimates upon only two criteria.
- 3.A DM is able to compare the preferability of one of two alternatives if they differ in estimates upon more than two criteria, but one alternative is more preferable than the other upon all these differing criteria, except one.
- 4.A DM is able to give one of the following answer for pairs of such alternatives:
 - 1)alternative A is more preferable than alternative B;
 - 2)alternative B is more preferable than alternative A;

3) alternatives A and B are equally preferable;

4) it is difficult to give an answer.

In addition, analogously to the previous case, we shall use the assumption about the preferential criteria independence and transitivity of DM's preferences.

Based on such assumptions, we elaborated a special procedure for comparison of two multiattribute alternatives on the base of pairwise compensation principle (when disadvantages of one alternative are tried to be compensated by drawbacks of the other and as a result, one tries to analyse, which of these two alternatives possesses less preferable drawbacks (or more preferable advantages)).

Procedure for comparison of two alternatives

The goal of this stage in the decision process is to compare a pair of real alternatives on the base of thorough analysis of the information about them and in accordance with DM's preferences.

Let there be two alternatives y_i and y_j , estimated upon Q criteria. So, we have two vectors: $y_i = (y_{i1}, y_{i2}, \dots, y_{iQ})$ and $y_j = (y_{j1}, y_{j2}, \dots, y_{jQ})$. Alternative y_i is more preferable than alternative y_j upon several criteria, and upon the other part of criteria alternative y_j is more preferable than alternative y_i .

To make it more clear let us consider that alternative y_i is less preferable upon the first m criteria, and alternative y_j is less preferable than alternative y_i upon the rest n ($n = Q - m$) criteria. So, $K_1 = \{1, 2, \dots, m\}$ and $K_2 = \{m+1, m+2, \dots, Q\}$ are corresponding subsets of criteria numbers upon which y_i is less and more preferable than y_j . It is evident that $K_1 \cap K_2 = \emptyset$.

The procedure for eliciting information about the relative preferability of different estimates upon criteria, is based on comparison of hypothetical objects (alternatives), differing in estimates by only two criteria.

Let us form an alternative y'_i , differing from alternative y_i only by estimates upon criterion 1 and criterion $m+1$. For these criteria estimates are changed by corresponding estimates of alternative y_j . Let us also form the alternative y'_j differing

from alternative y_j by estimates upon the same criteria 1 and $m+1$. Upon these criteria the hypothetical alternative possesses estimates of alternative y_i .

As a result we have two pairs of alternatives (y_i, y'_i) and (y_j, y'_j) , in which alternatives differ in estimates upon only two criteria. At the same time upon one of these criteria one alternative is more preferable than the other, and for the other criterion it is vice verser.

In accordance with initial assumptions, these pairs may be presented to a DM for comparison. Let us receive DM's answers for each of such pairs.

If a DM points out that it is difficult for him (her) to answer this question, we consider that it is the result of one of two reasons:

- 1) alternatives are equally preferable for a DM, but he (she) is not aware of it;
- 2) alternatives' description (in terms of estimates upon the selected criteria) is not complete (or exact) and this makes it impossible for a DM to compare these alternatives.

In this case the system tries to detect the reason of alternatives incomparability. If alternatives are equally preferable, this information is stored in the system. Otherwise, the attempt is made to modify the structure of alternatives' description (decompose those criteria upon which alternatives differ from one another). If even in such case the comparison is too difficult, the comparison of such hypothetical alternatives is postponed and the system forms another pair of hypothetical alternatives.

If a DM is able to compare, we receive one of the first three variants of possible answers. If the relation, given for the pair (y'_i, y_i) is the same as for the pair (y'_j, y_j) , then it is supposed that DM's preferences for estimates' combinations $y_{j1}y_{j(m+1)}$ and $y_{i1}y_{i(m+1)}$ do not depend on the estimates upon the other criteria. Really, in the first pair the estimates upon all other (than 1 and $m+1$) criteria were equal to those of alternative y_i and in the second pair - equal to those of alternative y_j . So, there is a partial test of DM's preferences for preferential independence, and as the level of fixed

estimates is equal to that of the real alternatives being compared, such test may be considered enough representative.

So, if DM's answers for two pairs of hypothetical alternatives are the same, the result of comparison is stored in the system and next two pairs of hypothetical alternatives, differing in estimates upon criteria 2 and $m+2$ are formed.

If DM's answers are different, then it is supposed that there are dependent criteria and a special procedure is carried out to find them. First, a DM is asked to mark the estimate which to his (her) opinion has influenced the change in the answer. If it is too difficult for a DM, then, for example for the pair (y_i', y_i) , estimates upon criteria are sequentially (one by one) replaced by estimates of the corresponding criteria of alternative y_j . This means that the following pairs of alternatives are presented to a DM for comparison:

$(y_{j1}, y_{i2}, \dots, y_{j1}, y_{i(1+1)}, \dots, y_{j(Q_1+1)}, y_{i(Q_1+2)}, \dots, y_{iQ})$ and
 $(y_{i1}, y_{i2}, \dots, y_{j1}, y_{i(1+1)}, \dots, y_{i(Q_1+1)}, y_{i(Q_1+2)}, \dots, y_{iQ})$
 $l=2, 3, \dots, m, (m+2), \dots, Q.$

The procedure is continued up to the moment of preferences' change. Such change marks the criterion l the estimates upon which influence the preferences. In this case estimates upon criteria 1, $m+1$ and l are combined for y_i and y_j into one according to DM's information. The procedure is continued on the base of this new description of these two real alternatives.

As a result we are able to have a set of partial combinations of estimates (upon small number of criteria, mainly 2-3) which it is possible to use for comparison of real alternatives y_i and y_j on the base of pairwise compensation rule, analogous to the introduced earlier.

Let R be the binary preference-indifference relation on the set of estimates of alternatives y_i and y_j , containing subset of estimates' combinations with the determined preference relation.

Then the following may be formulated:

alternative y_i is not less preferable than alternative y_j if there exists such partition of criteria from K , that for each resulting subset of criteria, the combination of estimates of alternative y_i is not less preferable, than the corresponding combination of

estimates for alternative y_j according to the relation R .

If at the current level of information it is not possible to compare y_i and y_j , then the system tries to receive additional information about the comparison of combinations of estimates upon other criteria from K_1 and K_2 . If the list of possible combinations is exhausted, it is resumed that it is not possible to compare alternatives y_i and y_j within the framework of the current criterion description.

A DM is proposed:

- 1) to try to change the description of alternatives by aggregating some criteria into logical groups and try to compare alternatives in such new description;
- 2) to add the alternatives' description by some new characteristics (criteria) and compare alternatives in such new description;
- 3) to give minimal changes (improvements) in estimates of one of the alternatives so, that it would be possible to say that it is more preferable than the other;
- 4) to postpone the comparison of these alternatives and to start comparison of a new pair of alternatives.

To see more details on these four possibilities see (Berkeley et al., 1990).

After all necessary pairwise comparisons of real alternatives have been carried out, the general analysis of the results is done. The information is displayed on the final set of the "best" alternatives. If there is only one best alternative the task is considered to be fulfilled. If there are several alternatives in the final set, then the conclusion is made, that it seems there is no enough satisfactory alternative among their initial set to be the real choice. In this case a DM is proposed to look through the list of hypothetical "improved" alternatives formed in the decision process (by "minimal changes" in incomparable alternatives), and try to think about their actual realisability.

If there is a realisable alternative in this list, then it is suggested to consider it as a real choice. If there are several such alternatives, it is proposed to use the system for comparison of these new set of alternatives.

CONCLUSION

The proposed procedures for comparison and partial ordering of multiattribute alternatives are directed on DM's preferences' elicitation in a qualitative form and on logical transition from such information to construction of the decision rule. Criteria with ordinal scales and verbal description of estimates for them are used. There exists a possibility to develop a rational elicitation procedure to receive the necessary information on DM's preferences, to detect and correct possible inconsistencies and errors in such information.

Special procedures for identification and elimination of criteria preferential dependence are suggested. This allows to use effectively the received information.

The dialogue is carried out through the use of DM's own natural language (i.e., in describing the problem, creating and developing alternatives and criteria, estimating these alternatives upon the criteria). In the process of comparison a DM is presented with the possibility to change the alternatives' description (i.e., through decomposition or aggregation of criteria, introduction of additional information and etc.), and also the generation of new alternatives on the base the carried out analysis.

So, the approach of multiattribute utility theory is not the only possibility in decision aid.

REFERENCES

- Berkeley D., Humphreys P.C., Larichev O.I. and Moshkovich H. (1989), ASTRIDA: Advanced strategic intelligent decision aid: Global design. - Doc. ASTRIDA-GD-001. London School of Economics.
- Berkeley D., Humphreys P., Larichev O.I., Moshkovich H.M. (1990), Aiding strategic decision-making: Derivation and development of ASTRIDA - In Proceedings of the IFIP TC-8.3 conference.
- Clancey W.J. (1984), Classification problem solving. - In proceedings of the National Conference on Artificial

- Intelligence, IAAAAI, University of Texas at Austin.
- Dror Y. (1989), Public Policymaking Reexamined. - Transaction Publishers: New Brunswick (USA) and Oxford (UK).
- Edwards W. (1979), Use of multiattribute utility measurement for social decision making - In: Bell D.E., Keeney R.L. and Raiffa H. (eds), *Conflicting Objectives in Decisions*. - New York: Wiley (Interscience).
- Edwards W. (1983), Human cognitive capabilities, representativeness and ground rules for research in decision making. - In: P.C.Humphreys, O.Svenson and A.Vari (eds), *Analysing and aiding decision processes*. - Amsterdam: North Holland.
- Fishburn P.C. (1970), *Utility theory for Decision Making*. - New York: Wiley, 1970.
- Gallhofer I.N. & Saris W.E. (1989), Decision trees and decision rules in politics: The empirical decision analysis procedure. - In: Svenson O. and Montgomery H. (eds), *Processes and structure in human decision making*. - Chichester: Wiley.
- Gnedenko L.S., Larichev O.I., Moshkovoch H.M., Furems E.M. (1986), Procedure for construction of a quasi-order on the set of multiattribute alternatives on the base of reliable information about the decision maker's preferences. - *Automatica i telemechanica*, 9, p.104-113 (in Russian).
- Hoffman P.J., Slovic P., Rorer L.G. (1968), Analysis-of-variance model for assessment of of configural cue utilization in clinical judgement. - *Psychological Bulletin*, No.69, p.338-349.
- Humphreys P.C. (1977), Application of multiattribute utility theory. - In: H.Jungerman and G. de Zeeuw (eds), *Decision making and change in human affairs*. Dordrecht: Reidel.
- Humphreys P.C. and McFadden W. (1980), Experiences with MAUD: Aiding decision structuring versus bootstrapping the decision maker. - *Acta Psychologica*, 45, p.51-69.
- Humphreys P.C. & Wishudha A. (1982) MAUD4. Technical Report 82-5. London: LSE.
- John R.S., von Winterfeldt D. and Edwards W. (1983), The quality and user acceptance of multiattribute utility analysis performed by computer and analyst. - In: P.C.Humphreys, O.Svenson and A.Vari (eds), *Analysing and aiding decision*

- processes. - Amsterdam: North Holland.
- Johnson L., Keravnov E.T. (1984), Expert Systems Technology: a Guide. - ABACUS PRESS.
- Kahneman D., Slovic P., Tversky A. (eds) (1982), Judgment under uncertainty: heuristics and biases. - Cambridge: Univ. Press.
- Keeney R.L. (1975), Energy policy and value trade offs. - IIASA, Res.Mem., RM-75-76.
- Keeney R., Raiffa H. (1976), Decisions with multiple objectives: preference and value tradeoffs. - New York: Wiley.
- Khaynish S.V., Vlasov A.G. (1983), Interactive systems resulting from a descriptive approach to the solution of complex ill-structured problems. - Theory and practice of multiple criteria decision making. North-Holland.
- Korhonen P. (1986), A Visual Reference Direction Approach to Solving Discrete Multiple Criteria Problems. - Working Paper F-134, Helsinki School of Economics.
- Ladenson A.B., Litvak B.G. (1988), Principle of ordered criteria for multiattribute alternatives. - Izvestia Akademii nauk SSSR. Technicheskaya kibernetika, N6, p.49-54 (in Russian).
- Larichev O.I. (1979), Science and art of decision making. - Moscow: Nauka press, (in Russian).
- Larichev O.I. (1982), A method for evaluating R&D proposals in large research organizations. - Collaborative paper CP-82-75. - Laxenburg, Austria: IIASA.
- Larichev O.I. (1984), Psychological validation of decision methods. - Journal of Applied Systems Analysis, v.11, p.37-46.
- Larichev O.I. (1987), Objective models and subjective decisions. - Moscow: Nauka press, (in Russian).
- Larichev O.I., Mechitov A.I., Moshkovich H.M., Furems E.M. (1989) Expert knowledge elicitation. - Moscow: Nauka press, (in Russian).
- Larichev O.I., Moshkovich H.M. (1986), Task of direct classification in decision making. // Dokladi Akademii nauk SSSR, vol.287, 6, p.567-570 (in Russian).
- Larichev O.I., Moshkovich H.M. (1988a) Limits to decision making ability in direct multiattribute alternative evaluation. - OBHDP, 42, 217-233.
- Larichev O.I., Moshkovich H.M. (1988b), Non-quantitative methods

- for multiattribute alternatives' utility evaluation. -In Proceedeings of the 4-th International conference on the Foudations and Applications of Utility, Risk and Decision Theories (Abstracts), p.116-117.
- Larichev O.I., Moshkovich H.M. (1990), Decision Support System Class for R&D Planning. - In Proceedings of the First International Conference on Expert Planning Systems, (Brighton, United Kingdom, June), p.227-232.
- Larichev O.I., Moshkovich H.M. (1991), ZAPROS: a method and a system for ordering of multiattribute alternatives on the base of decision-maker's preferences. - Preprint, Moscow: VNIISI press, 1991 (in English).
- Larichev O.I., Moshkovich H.M., Rebrik S. (1988), Systematic research into human behavior in multiattribute object classification problems. - Acta Psychologica, 68, p.171-182.
- Larichev O.I., Polyakov O.A., Nikiforov A.D. (1987), Multicriterion linear programming problems (Analytical Survey) - Journal of economical psychology, 8, p.389-407.
- Larichev O.I., Zuev Yu.A., Gnedenko L.S. (1974), Method classification of applied R&D projects. - In: S.V.Emelyanov (ed.), Perspective apllied R&D planning. - Moscow: Nauka press, 1974, p.28-57 (in Russian).
- Luce R.D. and Raiffa H. (1957), Games and decisions: Introduction and crucial survey. New York: Wiley.
- Mechitov A.I., Furems E.M. (1990), New version of an interactive system for knowledge elicitation for diagnostic systems. - VNIISI press, (in press, in Russian).
- Mirkin B.G. (1974), Problems of group choice. - Moscow: Nauka press, (in Russian).
- Montgomery H. (1977), A study of intransitive preferences using a think aloud procedure.- In: H.Jungerman & G.de Zeeuw (Eds), Decision making and change in human affairs. - Dordrecht: Riedel.
- Montgomery H.(1983), Decision Rules and the search for a dominance structure: towards a process model of decision making. - In:P.C.Humphreys, O.Svenson and A.Vari (eds), Analysing and aiding decision processes. - Amsterdam: North Holland.

- Montgomery H., Garling T., Linberg E. and Selart M. (1990), Preference judgement and choice: Is the prominent effect due to information intergration or information evaluation? In: Borcherdig K., Larichev O.I., Messick D.M. (eds), Contemporary issues in decision making. Amsterdam:North Holland.
- Montgomery H. and Svenson O. (1976), On decision rules and information processing strategies for choice among multiattribute alternatives.- Scandinavian Journal of Psychology, 17, p.283-291.
- Moshkovich H.M. (1988) Interactive system ZAPROS (for ordering of multiattribute alternatives on the base of decision maker's preferences). - In: Larichev O.I., Emelyanov S.V. (eds), Man-machine procedures for decision making. - Moscow: VNIISI press, N11, p.13-21 (in Russian).
- von Neumann J.& Morgenstern O. (1947) Theory of games and economic behavior. - Princeton University Press, Princeton.
- Ozernoy V.M., Gaft M.G. (1978), Multicriteria problems of decision making. - In: Emelyanov S.V. (ed.) Multiattribute decision problems. - Moscow: Mashinostroenie press, p.14-47 (in Russian).
- Payne J.W. (1976), Task complexity and contingent processing in decision making: An information search and protocol analysis. - OBHP, 16, p.366-387.
- Podinovsky V.V. (1978), On relative importance of criteria in multiattribute decision problems. In: Emelyanov S.V. (ed.) Multiattribute decision problems. - Moscow: Mashinostroenie Press, p.48-92 (in Russian).
- Raiffa H. (1968), Decision Analysis. - Reading,Mass: Addison Wesley.
- Rivett P. (1977), The dog that did not bark.-Eng.Econ., vol.22, No.4, p.298-300.
- Roy B. (1968), Classement et choix en presence de points de vue multiples (la methode ELECTRE). - Rev. Franc. Inform. et Recher.Operat., v.2, N8.
- Russo J.E. & Rosen L.D. (1975), An eye fixation analysis of multiattribute choice. - Memory and cognition, 3, p.267-276.
- Slovic P., Fishhoff B., Lichtenstein S. (1977), Behavioral

- Decision Theory. - Annual Psychological Review, v.28.
- Svenson O. (1979), Process descriptions of decision making. - Organization Behavior and Human Performance, 23, p.86-112.
- Svenson O. (1983), Decision rules and information processing in decision making. - In: L.Sjöberg, T.Tyszka and J.A.Wise (eds), Human Decision Making. - Bodafor: Doxa.
- Tversky A. (1969), Intransitivity of preferences. - Psychological Rev., No.76, p.31-48.
- von Winterfeldt D. and G.W.Fischer (1975), Multiattribute utility theory: Models and assessment procedures. - In: D.Wendt and C.Vlek (eds), Utility, probability and human decision making. - Amsterdam: Reidel.
- von Winterfeldt D. (1975), An overview, integration and evaluation of utility theory for decision analysis.- Soc.Science Research Institute, USC- Report, 75-9.

**Multiple Strategies in Judgment and Choice:
A Constructive Process View of Decision Making**

John W. Payne, James R. Bettman, Eloise Coupey,
and Eric J. Johnson

Paper prepared for the thirteenth research conference on subjective probability, utility
and decision making, Fribourg, Switzerland, August, 1991

Much recent decision research indicates that preferences for objects of any complexity are often constructed - not merely revealed - in the generation of a response to a judgment or choice task (Slovic, Griffin, & Tversky, 1990; Payne, Bettman, & Johnson, In Press b). The constructiveness of preferences is attributed by March (1978) to the interaction of the limited memory and computational capabilities of the decision maker with the complexity of task environments. That is, the concept of bounded rationality (Simon, 1955) is seen as the basis of the constructed preferences that we observe. In March's words, "Human beings have unstable, inconsistent, incompletely evoked, and imprecise goals at least in part because human abilities limit preference orderliness" (March, 1978, p. 598).

The idea of constructive preferences means more than simply saying that observed preferences are not the result of a reference to a master list of values in memory. Hand in hand with the notion of constructive preferences is the idea that preferences also are not necessarily generated by some consistent and invariant algorithm such as expected value calculation (Tversky, Sattath, and Slovic, 1988). Instead, it appears that decision-makers have a repertoire of methods for identifying their preferences. The multiple methods or strategies result both from experience and training (Kruglanski, 1989; Larrick, Morgan, & Nisbett, 1990).

The strategy used to construct a preference appears to be highly contingent upon and predictable from a variety of task, context, and individual difference factors. Thus, expressed preferences seem to be highly labile (Fischhoff, Slovic, & Lichtenstein, 1980). Task factors are general characteristics of a decision problem, such as the number of alternatives available or the response mode, which do not depend upon the particular values of the alternatives; context factors such as similarity of alternatives, on the other hand, are associated with the particular values of the alternatives. Task and context factors cause different aspects of the problem to be salient and evoke different processes for combining information. Characteristics of the decision problem, therefore, can evoke different strategies which at least partially determine the expressed preferences. Of course, how a solution to a decision problem is constructed will also be a function of individual difference factors such as processing capacities (Bettman, Johnson, & Payne,

1990), prior knowledge or expertise (Shanteau, 1988), and the meta-goals adopted for the decision episode, e.g., maximize accuracy or justifiability, or minimize effort, regret, or conflict (Einhorn & Hogarth, 1981; Tetlock, 1985).

We believe that many current issues in behavioral decision research can be related to the contingent use of multiple approaches for solving decision problems. This paper reports on an ongoing program of research concerned with the multiple strategies people use when making choices and judgments. The paper is organized into two main parts: First, we briefly review a series of studies that assume that a decision maker selects from his or her repertoire of strategies that particular strategy which represents the best accuracy/effort tradeoff for the task at hand. As will be shown, there is much data supporting a top-down or goal-directed process of strategy selection and the primacy of accuracy and effort concerns in strategy selection. As noted above, such contingent strategy usage will lead to variation in preferences across situations.

In the second part of the paper, we argue that assuming the use of single prototypical strategies and the notion of a top-down process of strategy selection is too restrictive. Problem-solvers not only use information extracted from the original problem definition in deciding what strategy to use but also use information they have already explored to identify promising paths for further search (Langley et al., 1987). That is, as people learn more about the problem structure during the course of making a decision, they can change their processing to exploit that structure. Processing, in other words, can be opportunistic (Hayes-Roth & Hayes-Roth, 1979). Such a view implies that individuals can construct choice strategies on the spot during the course of making a decision (Bettman, 1979). We argue that the opportunistic use of decision processes also accounts for labile or constructive preferences.

CONTINGENT DECISION BEHAVIOR

Over the past twenty years there have been numerous studies showing that the processes used to solve decision problems will vary as a function of a number of task and context factors (see Payne, 1982; Payne, Bettman, & Johnson, *In press* for reviews). One of the most striking examples of contingent judgment and choice concerns how people adapt their decision processes to deal with task complexity. For instance, when

faced with decision problems involving just two or three alternatives, people often use decision strategies that process all the relevant information and require one to decide explicitly the extent to which one is willing to trade off less of one valued attribute for more of another valued attribute. When more alternatives are involved, people often use heuristics (e.g., elimination by aspects) to cut down the number of options (Biggs, Bedard, Gaber, & Linsmeier, 1985; Billings & Marcus, 1983; Johnson, Meyer, & Ghose, 1989; Klayman, 1985; Onken, Hastie, & Revelle, 1985; Payne, 1976; Sundstrom, 1987).

As an illustration of the evidence for the use of multiple strategies in decision making, consider Figure 1. That figure provides some excerpts from the verbal protocols (thinking aloud records) obtained by Payne (1976) in a study of choices among apartments. The protocols represent the responses of two different subjects (A and D) faced with two levels of task complexity: (1) two alternative problems (panels a and b), and (2) multialternative choice problems (panels c and d). Panels (a) and (b) in Figure 1 suggest the consideration of tradeoffs among attributes. For example, subject D explicitly asks a tradeoff question dealing with the exchange of a higher rent for a lower level of noise in panel (b). The excerpts in panels (c) and (d), on the other hand, indicate more noncompensatory processing, such as satisficing (Simon, 1955 panel c) and elimination-by-aspects (Tversky, 1972 panel d). Also, note from Figure 1 that strategy differences are shown both within the same subject (e.g., panels a and c) and across subjects (e.g., panels c and d).

.....
Insert Figure 1 about here
.....

Evidence that decision makers sometimes explicitly consider task demands and plan their processing in a top-down fashion is provided in panel (d) of Figure 1 and in panels (a) and (b) of Figure 2. Figure 2 provides excerpts from verbal protocols collected by Payne (1976) and Payne and Braunstein (1978) that illustrate the selection of a decision strategy determined, at least in part, by the number of alternatives available. In panel (a) of Figure 2, the fact that there are many apartments to choose

from appears to lead to a decision to employ an elimination strategy. In contrast, the verbal protocol in panel (b) of Figure 2 suggests that a more comprehensive information processing strategy is to be employed, given that there are "just two" alternatives to choose from.

.....
Insert Figure 2 about here
.....

EFFORT AND ACCURACY IN CHOICE

Given that people appear to use many different strategies in making a decision, a theoretical framework is needed in order to understand and predict when a particular strategy will be used to solve a particular type of decision problem. In this section, we highlight such a framework. We see the use of various decision strategies as an adaptive response by a limited capacity information processor to the demands of complex task environments. In particular, we emphasize the use of multiple strategies as an adaptive way to balance the goals of achieving an accurate decision and limiting the cognitive effort needed to reach a decision.

The idea that decision making is influenced by considerations of cognitive effort as well as by considerations of decision accuracy is an old one (Simon, 1955). Further, the more general idea that strategy selection involves the consideration of the benefits of and costs of different strategies is the most frequently used framework for explaining contingent decision behavior (e.g., Beach & Mitchell, 1978; Klayman, 1983; Klein, 1983; Russo & Doshier, 1983; Shugan, 1980; Thorngate, 1980; Wright, 1977). Our version of this framework, however, focuses on strategy selection at a more detailed information processing level than the work of most other researchers. In addition, we place more stress upon the role that cognitive effort plays in strategy selection. We consider next how to measure cognitive effort and present evidence validating our effort/accuracy approach.

Cognitive Effort and Decision Strategies.

In the late 1970's and early 1980's, a method for comparing the effort required by

different decision strategies was independently proposed by Huber (1980) and Johnson (1979). Based upon the work of Newell and Simon (1972), they suggested that decision strategies could be described by a set of elementary information processes (EIPs). An EIP could include such mental operations as reading a piece of information into short-term memory, comparing the values of two alternatives on an attribute to determine which is larger, and multiplying a weight and attribute value.

A set of EIPs for decision making that we have used in our research is shown in Table 1. A particular decision strategy would be defined in terms of a specific collection and sequence of EIPs. For example, a lexicographic choice strategy would involve a number of reading and comparison EIPs but no adding or multiplying EIPs. In contrast, a weighted additive strategy would have reading EIPs, a number of adding and multiplying EIPs, and some comparisons (but fewer comparisons than the lexicographic strategy).

.....
Insert Table 1 about here
.....

The number of EIPs required to execute a strategy in a particular task environment reflects the cognitive effort required to make a decision in that environment using that specific strategy. Cognitive effort is also a function of the specific mix of EIPs utilized as well as the total number of EIPs used. The latter idea reflects the belief that people will find some EIPs, e.g., multiplications, more effortful than others, e.g., comparisons.

How valid is this EIP approach to measuring decision effort? In Bettman, Johnson, & Payne (1990), we examined the assumption that EIP counts provide a measure of cognitive effort by having decision makers make choices using different prescribed strategies for choice sets varying in size. Both decision latencies and self-reports of decision difficulty were obtained as measures of strategy execution effort (See Bettman, et. al., 1990, for details on the experimental method).

Overall, the results yielded strong support for the EIP approach to measuring

strategy effort. A model of effort based on weighted EIP counts provided good fits for response times ($R^2 = .84$) and self-reports of effort ($R^2 = .59$). In addition, the fit of the weighted EIP model to the data was statistically superior both to the fit of an alternative model of effort based on simply the number of items of information acquired and to the fit of an equal-weighted EIP model. Thus, a model of cognitive effort in choice requires concern not only for the amount of information processed but also for different weights for the particular processes (EIPs) applied to that information. Interestingly, the estimates of the time taken for each EIP were mostly in line with prior cognitive research. Also, the estimated weights for the various EIPs were essentially the same regardless of the decision strategy used. This supports the assumption of independence of EIPs across rules.

Finally, the results showed significant individual differences in the effort associated with particular EIPs (i.e., the fit of a weighted EIP by individual model was significantly better than that of the weighted EIP model). This suggests that individuals may choose different decision strategies in part because certain component EIPs may be relatively more or less effortful across individuals. In fact, the processing patterns used by subjects in an unconstrained choice environment were related to the relative costs of certain EIPs, although the limited number of subjects in that study precluded any strong conclusions. Subjects for whom arithmetic operators were relatively more difficult, as indicated by the coefficients for the various EIPs, showed greater selectivity in processing.

To summarize, strong support for the EIP approach to measuring decision effort was found. Next, we illustrate how a general accuracy/effort framework can be used (1) to generate specific predictions about how the use of strategies will vary across task environments, and (2) to test the extent to which actual decision behavior adapts in ways predicted by a concern for both accuracy and effort.

Adaptive Strategy Selection

In this section, we first present some specific process-level predictions regarding adaptivity in decision processes. These predictions were derived using computer simulation. Details on the simulation studies can be found in Payne, Bettman, and

Johnson (1988) and Payne, Johnson, Bettman, and Coupey (1990). Next, we present some process-level data on the extent to which actual decision behavior involves shifts in strategies of the type predicted by the simulation.

Monte-Carlo Simulation Studies. To determine the effort and accuracy of various heuristics in different environments, we first modeled each of a set of decision strategies (e.g., the weighted additive (WADD), elimination-by-aspects (EBA), lexicographic choice (LEX), satisficing (SAT), majority of confirming dimensions (MCD), and the equal-weight (EQW) rules as production systems (Newell & Simon, 1972). A production system is a collection of IF-THEN rules; a discussion of the value of production systems as representations of human cognitive processes can be found in Newell (1980).

We then implemented these production system models as computer programs and ran Monte-Carlo simulations using these models of each strategy in order to estimate how the effort and accuracy of the various strategies vary with changes in decision environments. Effort was calculated on the basis of counts of EIPs, as discussed above. For each heuristic, accuracy was measured in terms of the relative performance of that heuristic when compared to the optimal choice (given by the weighted additive rule, which uses all the relevant problem information) and the choice that would be expected if a random choice procedure (RAND) was used, which involves no processing of information. Specifically, we measured relative accuracy as follows, in terms of the values of the alternative chosen by each rule indicated: $(\text{heuristic} - \text{random}) / (\text{weighted adding} - \text{random})$.

Based upon a review of factors which might have important effects on either the effort or accuracy of decision strategies (e.g., see Thorngate 1980; Beach, 1983; McClelland, 1978), we varied several factors in the simulations to provide different choice environments: the number of alternatives and number of attributes, time pressure, the presence or absence of dominated alternatives, and the degree of dispersion of weights across attributes. To illustrate the latter variable, a problem with low dispersion might have relative weights on the attributes of .30, .20, .22, and .28, respectively, for a four attribute decision problem. On the other hand, a problem with a high degree of dispersion in weights might have weights of .68, .12, .05, and .15 for the four attributes.

For several of the empirical studies reviewed below, these weights were operationalized in terms of the probabilities of various outcomes that might be experienced, given that a particular alternative was chosen. In other words, the decision maker was asked to select the best gamble from a set of gambles.

The conclusions from the Monte-Carlo simulations can be summarized as follows: First, in some decision environments, the use of a non-normative strategy (e.g., the lexicographic rule) may not only significantly reduce the effort needed to reach a decision but can also provide a level of accuracy comparable to that obtained by the weighted additive rule. Thus, the use of heuristic decision strategies often makes sense when both accuracy of choice and decision effort are considered.

Second, under time pressure the weighted additive rule rapidly degrades in accuracy, while heuristics like elimination-by-aspects and lexicographic choice show much smaller losses in accuracy. In fact, under severe levels of time pressure, elimination-by-aspects is often the most accurate rule. Thus, under time constraints the preferred strategy is to process at least some information about all alternatives as soon as possible rather than to worry about processing each alternative in depth.¹

Third, no single heuristic was the most efficient strategy across all task environments. In the low dispersion, dominance-possible environment, for example, simplifying processing by ignoring weight (probability) information, i.e., using the equal weight strategy, is quite accurate. In contrast, when the dispersion in weights is higher, the lexicographic rule is the most accurate of the heuristics and is substantially better than the equal weight rule. Thus, a decision maker wanting to achieve both a reasonably high level of accuracy and low effort would have to use a repertoire of strategies, with selection contingent upon task demands.

The simulation results reported above highlight what an idealized decision maker might do to shift strategies adaptively as task environments change. In the next section we discuss the extent to which actual decision behavior involves shifts in strategies of the type predicted by the simulation. To assess the match between the simulations'

¹Interestingly, Eisenhardt (1989) reports that firms in the computer industry operating in rapidly changing environments (time pressure) did better if they used a "breadth-not depth" strategy for evaluating options.

predictions and choice behavior, we have conducted a number of experiments designed to validate the results of our simulation studies. Those experiments have used computer program for monitoring information acquisitions called Mouselab to collect process-level data. This involves setting up the decision task so that the subject must use a mouse to view or select information. Data can be obtained on what information the subject seeks, in what order, how much information is examined, and how long the information is examined. A typical Mouselab display is given in Figure 3. Further details on Mouselab's capabilities can be found in Johnson, Payne, Schkade, and Bettman (1991).

.....
Insert Figure 3 about here
.....

Our first experiment examined the sensitivity of decision behavior to variations in the goals for the task (emphasis on accuracy or emphasis on effort savings). The second series of experiments examined the sensitivity of decision processes to variations in time pressure and to variations in the dispersion of the probabilities (weights) associated with the outcomes of the alternatives in a choice set. More complete details on each set of studies can be found in Creyer, Bettman, and Payne (1990) and Payne, Bettman, and Johnson (1988), respectively.

Effects of Accuracy and Effort Goals on Decision Processes. A key assumption underlying any accuracy/effort approach to strategy selection is that processing should be sensitive to the relative emphasis placed on accuracy versus effort. For example, people should utilize strategies that provide greater accuracy (often at the cost of greater effort) when the incentives for accuracy are increased. However, as several authors point out (e.g., Ashton, 1990; Tversky & Kahneman, 1986; Wright & Aboul-Ezz, 1986), incentives sometimes enhance performance, sometimes have no effect, and at times may actually decrease performance.

One concept important for understanding incentive effects is the distinction between working harder versus working smarter (Einhorn & Hogarth, 1986; Tversky & Kahneman, 1986). Working harder denotes devoting more effort to the same strategy;

working smarter, in contrast, refers to changing strategies appropriately to take advantage of a specific situation. We believe that a common response to *general* incentives is simply to work harder at the same strategy. However, we believe that *specific* incentives which explicitly change the relative salience of effort and accuracy considerations in the decision environment can lead to strategy changes.

Subjects used Mouselab to acquire information and make decisions for 32 sets of four alternatives, each defined by six attributes. The subjects' task was to select the alternative in each set that they thought was best overall. The sets varied within subjects with respect to 1) the dispersion of the weights provided for the attributes (high or low), 2) the explicit goal of the decision maker for the set (minimize effort or maximize accuracy), and 3) the presence or absence of effort and accuracy feedback (these feedback factors are not discussed in this paper).

We manipulated effort/accuracy tradeoffs by explicitly emphasizing either a goal of maximizing accuracy relative to effort or a goal of minimizing effort relative to accuracy for each choice set. Subjects were told that an index of overall performance would be developed based both upon the time taken and the accuracy achieved for each trial.² They were told further that for trials when the goal was to minimize effort, time taken would have a weight of one and accuracy a weight of three. Thus, both accuracy and effort (time taken) mattered for all trials; we tried to manipulate the relative importance of those two goals.

Subjects did more processing when the goal was to maximize accuracy rather than to minimize effort; that is, more information was acquired and more time was spent on the information. In addition, information acquisition was less selective under a goal of maximizing accuracy. Subjects spent proportionally less time on the most important attribute, were less selective in processing over attributes, and less selective in processing across alternatives. Finally, processing was more alternative-based when the goal was to maximize accuracy. This more extensive, less selective, and more alternative-based processing is more consistent with normative strategies and also leads to better

²Accuracy was measured relative to the weighted additive rule.

performance, since subjects attain greater relative accuracy levels when the goal is to maximize accuracy.

To summarize, when we emphasized the goal of accuracy more than effort, we found a shift in strategies in the direction predicted by the effort/accuracy framework. These results provide the clearest evidence available to date for the effects of differences in goals on process-tracing measures of decision strategies (see Billings and Scherer, 1988; Ford, Schmitt, Schechtman, Hults, and Doherty, 1989, pp. 101-102).

Effects of Time Pressure and Dispersion in Weights on Decision Processes.

These experiments asked the following questions: (1) to what extent do people vary their choice processes as a function of context factors such as the dispersion of probabilities and task factors such as time pressure?, and (2) are these changes in processing in the directions suggested by our simulations? As outlined above, the simulation results provide a fairly clear picture of how a decision maker might adapt to such decision environments. Specifically, as implied by the good performance of heuristics like the lexicographic and elimination-by-aspects rules under high dispersion, the simulations suggest that an adaptive decision maker should exhibit more attribute-based processing, greater selectivity in processing across attributes and alternatives, and a greater proportion of processing devoted to probabilities and the most important attribute in a high-dispersion environment.

The simulation results also indicate the kind of processing that should be expected under increased time pressure: processing characteristic of strategies like elimination-by-aspects and the lexicographic rule, which performed well in the simulations under time pressure. In particular, there should be more attribute-based processing, greater selectivity in processing, and a greater proportion of processing focused on probabilities and the most important attribute under high levels of time pressure.

We conducted experiments in which subjects were asked to make a series of choices from sets of options where dominated options were possible. Each choice set contained four risky options, with each option offering four possible outcomes (attributes). For any given outcome, the probability was the same for all four options. After completing the series of choices, subjects actually played one gamble and received

the amount of money corresponding to the alternative they had chosen.

The sets varied in terms of two within-subjects factors: (1) presence or absence of time pressure, and (2) high or low dispersion in probabilities. In addition, half the subjects had a 15 second constraint for the problems with time pressure, while the other half had a 25 second time constraint (the average response time for the no time pressure conditions was 44 seconds). Information acquisitions, response times, and choices were monitored using Mouselab. For trials with time pressure, Mouselab ensured that subjects could not collect any additional information once the available time had expired. A clock on the display screen indicated the time left as it counted down (see Figure 3).

Overall, the results for subjects' choice processes validated the patterns predicted by the simulation. Subjects showed a substantial degree of adaptive decision making, although this adaptivity was not perfect. More specifically, subjects processed less information, were more selective, and tended to process more by attribute when dispersion in probabilities was high rather than low. Since accuracy was equivalent in the two dispersion conditions, subjects took advantage of changes in the structure of the available alternatives to reduce processing load while maintaining accuracy.

Individual subjects who were more adaptive in their patterns of processing (i.e., who were relative more selective and more attribute-based processors in high dispersion environments) also attained higher relative accuracy scores. Importantly, this increase in performance was not accompanied by a significant increase in effort. Hence, more adaptive subjects also appeared to be more efficient decision makers.

We also found several effects of time pressure. First, under severe time pressure, people accelerated their processing (e.g., less time was spent per item of information acquired), selectively focused on a subset of the more important information, and changed their pattern of processing in the direction of relatively more attribute-based processing. This general pattern of results is consistent with the simulation results suggesting that an efficient strategy under severe time pressure would involve selective and attribute-based processing.

The effects of time pressure were substantially less for those subjects with a 25 s opposed to a 15 second constraint. Under more moderate time pressure, subjects

showed some acceleration and some selectivity in processing, but provided no evidence for a shift in the pattern of processing. These results suggested a possible hierarchy of responses to time pressure. People may initially try to respond to time pressure simply by working faster. If this is insufficient, they may then focus on a subset of the available information. Finally, if that is still insufficient, people may change processing strategies, e.g., from alternative-based processing to attribute-based processing.

To summarize, the results of the experiments outlined above provide compelling evidence for adaptivity in decision making. While not perfectly adaptive, our subjects were able to change processing strategies in ways that the simulation indicated were appropriate given changes in context and task features of the decision problems. Our conceptual framework thus receives strong support in these empirical studies. Individuals appear to weigh accuracy and effort concerns in selecting decision strategies

Although we are excited and pleased by the power of an accuracy-effort viewpoint to explain contingent decision behavior and lability of preferences, we do view the framework we have developed to date as incomplete. The framework presented so far assumes that preferences are constructed using different strategies selected from an existing repertoire. However, as noted at the beginning of this paper, we believe that assuming both the use of single prototypical strategies and the notion of a top-down process of strategy selection is certainly too restrictive. In the next section we explore the idea that decision behavior often reflects a more bottom-up, data-driven, or opportunistic use of decision processes. This notion implies that preferences also can be constructive because the strategy itself is being developed on the spot. The following section is adapted from chapter five of Payne, Bettman, and Johnson (In press a).

CONSTRUCTIVE PROCESSES IN DECISION MAKING

Several examples may help clarify our view that individuals construct choice strategies on the spot during the course of making a decision. For instance, an individual may intend to evaluate a set of gambles using a weighted adding (expected value) strategy. However, if the individual noted that the probability of one outcome was extremely high (e.g., .8), then that individual might drop plans to carry out an expected

value calculation and simply look for the alternative with the best payoff for the highly probable outcome. In other words, this represents a data-driven shift from a compensatory process to a noncompensatory, lexicographic strategy. We have observed such strategies in our data (Payne, Bettman, & Johnson, 1988). As another example, taken from a consumer choice context, a consumer may start to compare alternatives on what is a priori the most important attribute and discover that the values on that attribute are very similar across alternatives. He or she might then edit or cancel out that attribute (Kahneman & Tversky, 1979). Then he or she may decide to look at another attribute but find that the information available on that attribute is too hard to understand. In both these cases, individuals make spur of the moment shifts in processing direction rather than merely executing some strategy determined beforehand.

Elements of Decision Heuristics

The basic notion behind a constructive choice process is that the heuristic used is developed at the actual time of choice. The individual essentially makes up the strategy as he or she goes along. The individual may begin a particular choice process without any complete rule or heuristic stored in memory that is deemed applicable for that choice. Rather, a heuristic is constructed using fragments or elements of heuristics stored in memory; the overall strategy is built dynamically at the time of choice from elements or subparts. These elements or fragments may be beliefs about alternatives; evaluations; simple rules of thumb involving subsets of beliefs (e.g., "compare these alternatives on attribute A to see if they differ very much"); rules for integrating information (e.g., "count how many attributes alternative X is best on" or "average these ratings"); rules for assigning weights (e.g., "if the values of the alternatives on attribute B are very similar, then assign a low weight to attribute B"); and so on.

Bettman and Park (1980) and Biehal and Chakravarti (1986) have argued that simple processing operations such as those described above may represent the level at which consumers store much of their information processing repertoire in memory. Rules such as "If the values of the alternatives are similar, then assign a low weight to that attribute" are also easily representable as productions or procedural knowledge (Anderson, 1983).

Effects of the Decision Task on the Construction of a Heuristic

An individual may enter many decision situations with only a general plan to guide the construction of a heuristic. Thus, choice heuristics will in general vary from one situation to the next if a constructive method is used, depending upon how the elements available are put together. The elements used for a particular choice and the sequence in which they are used will be a function of such factors as what information is available (e.g., whether the same data are available for all alternatives); the format in which the information is presented (e.g., in a consumer choice, prices may not be compared if unit prices are not provided and different brands have different-sized packages); the salience of various pieces of information; intermediate processing results; and other task specific factors.

The constructive view reinforces the argument of Yates et al. (1978) that events affecting order of attention in the real world are likely to be numerous and powerful; such events should not be dismissed as just experimental nuisance factors. An example of the effect of physical proximity of items in an information display on processing is provided by Russo and Rosen (1975). Based on an analysis of eye movements, they found that 63% of all paired comparisons were between alternatives that were spatially adjacent, even though only 47% of the possible pairs were adjacent. More active ways to influence the sequence of processing include the setting of choice agendas.

If individuals build up heuristics as they go along, and the elements used are sensitive to many task specific factors (salience, format, and so on), the resulting choice "heuristic" may consist of a sequence of elements with a less than coherent overall structure. The following sequence illustrates such a structure: (1) An individual uses the element "compare several attribute levels to standards" for each of several alternatives (a component of a satisficing rule); (2) While doing so, the individual notes an outstanding value for a particular alternative on some attribute and eliminates all alternatives still being considered which are not "close" to that value (like an element of an elimination by aspects heuristic) (3) Next, the individual compares two remaining alternatives to see which is better on more attributes (a procedure which is part of a majority of confirming dimensions rule (Russo and Doshier, 1983)), and so on. Thus, each element or short

sequence of elements may be used to process only a few alternatives. Different sequences of elements may be used for different alternatives. Examples of such fragmented heuristics have been found in several studies (e.g., Bettman and Park, 1980). For a specific example of a choice among microwave ovens from Bettman and Park (1980), see Figure 4. Thus, constructive processes imply that the resulting heuristics are very sensitive to specific features of the situation.

Insert Figure 4 about here

Constructive Processes and Labile Preferences

Another implication of the opportunistic use of heuristics and the task sensitivity of constructive processes is the labile nature of preferences mentioned at the beginning of this paper. That is, the preferences that we observe often may reflect a constructive process in which attention to information and the methods used to combine information vary across tasks.

The compatibility hypothesis of Slovic, Griffin, and Tversky (1990) provides an example of such variability in preferences. According to that hypothesis, the weight given to a stimulus attribute is enhanced by its compatibility with the response mode. A study of the prediction of academic performance reported in Slovic et al. (1990) is a nice illustration of a compatibility effect in choice. In that study, subjects were presented with pairs of students and asked to choose the student in each pair who would achieve either a higher grade in history (half the subjects) or achieve a higher class rank in history (the other half of the subjects). The information to be used to predict relative achievement was the students' prior performance levels in two other courses, English literature and philosophy. The performance in one course was given by a grade (from A+ to D), and the performance in the other course was given in terms of a class rank (from 1 to 100). As expected from the compatibility hypothesis, subjects who had to choose which student would receive a better history grade chose the student in a pair with the higher prior grade in a course significantly more often than subjects who had to decide which student would achieve a higher class rank.

The notion of compatibility also may be the basis for the effect of dimensional commensurability on the use of attribute information (Slovic & MacPhillamy, 1974), i.e., that the weight of an attribute will be enhanced to the extent that values on the attribute can be more easily compared across alternatives. Once again, therefore, how individuals process information is to some extent made up on the spot. In this case it is not the compatibility of an attribute with a response mode that is important, but the compatibility of attribute information across alternatives.

It is important to note that both the effects of compatibility of an attribute with a response mode and the effects of compatibility of attribute information across alternatives can be interpreted in terms of a more general cognitive effort effect. For instance, Slovic et al., (1990, p. 5) suggest that "noncompatibility between the input and the output requires additional mental operations." Thus, a decision maker may adjust his or her processing to take advantage of special features of a problem that will reduce cognitive effort. Our idea is that individuals respond to such special features as they are noticed during the course of solving a problem, not just at the beginning of the decision episode.

When Will Constructive Processes Be Used?

Constructive processes are not always used, of course. In cases where there is a good deal of prior experience with a particular decision, top-down use of complete heuristics is much more likely. It would be too inefficient for individuals to construct heuristics in such situations where the structure of the decision task was well-known. However, when choices are made where there is little prior knowledge, constructive processes are more likely.

Constructive processes may also be more likely as decision problems become more complex or stressful. In such situations, it may be too difficult to attempt to determine an overall strategy a priori (see also Wood and Locke, 1990; Klein and Yadav, 1989). Of course, more difficult and stressful decisions will also tend to be less familiar, so the arguments above about prior experience would also apply. Janis has argued that under severe stress, people evaluate information in a hasty, disorganized, and incomplete fashion (Janis, 1989; Janis & Mann, 1977). Keinan (1987) provides experimental

evidence that people do scan alternatives in a more nonsystematic fashion under stress. If so, this suggests that we would see more data-driven processing patterns under situations of stress.

Bettman and Zins (1977) have done one of the few studies attempting to examine the frequency of constructive choice processes. They used verbal protocol data gathered from two consumers while they shopped over several shopping trips. Judges were given transcripts of the protocols and definitions of constructive choice and were asked to categorize each choice episode. In roughly 25% of the choices, the consumers appeared to be constructing a choice heuristic on the spot. This amount of constructive processing seems reasonable; in most cases, individuals will try to develop simplified rules they can merely apply for repetitive decisions, and most everyday decisions are somewhat repetitive. However, in many experimental settings, individuals will be faced with unfamiliar tasks and may exhibit constructive processes.

Local Accuracy/Effort Assessments

What does the notion of constructive processing imply about our accuracy/effort view of contingent decision making? In our view, the two are perfectly consistent. When individuals construct heuristics on the spot, these constructions are based upon more local, momentary accuracy/effort assessments. When an individual notes that all values on an attribute appear to be similar and shifts to another attribute, that shift in effect reflects a tradeoff of the low benefits from continued processing of that attribute versus the costs. The earlier example of a shift in processing by a consumer due to the noticing of common attribute values illustrates this point. Tversky (1972; Kahneman & Tversky, 1979) has argued that people often disregard components that alternatives share as one method for simplifying the choice among alternatives. Such spur of the moment shifts in processing direction are still based upon accuracy/effort considerations, even though the individual is building or realizing a heuristic on the spot rather than using one selected a priori.

A Dynamic View of Properties of the Choice Task and Contingency

These arguments lead to an extension of our notions about contingent choice processes. We have mainly used examples where contingent processing is based upon

relatively stable task properties of the decision (e.g., number of alternatives, time pressure) or context factors that were relatively easy to perceive quickly (e.g., dispersion in probabilities). However, the constructive view implies a more complex notion of contingency, where many properties of the "choice task" itself change as the individual progresses. Thus the elements of choice heuristics used may not even be the same for all alternatives during a given choice. The task is not the same for all alternatives. The elements used to process a given alternative may depend upon which alternatives have already been processed (e.g., whether a "good" alternative has appeared yet or not); upon the particular sequence of elements already used (e.g., if certain alternatives have been eliminated because of their values on a given attribute, that attribute may have a relatively restricted range when further operations involving that attribute are considered); upon which other alternatives happen to be near a given alternative in the information display (e.g., because this will affect the magnitude of the differences on various attributes); and so on.

The level of intercorrelation among the attribute values defining the choice alternatives provides another example of how the properties of the "choice task" itself may change as an individual progresses. Decision problems often may include dominated alternatives or be characterized more generally by moderately high positive correlations between attribute values (see Einhorn, Kleinmuntz, & Kleinmuntz (1979) for a discussion of the related concept of redundancy in cue values). It has often been suggested that people first eliminate dominated alternatives from consideration when making a choice (e.g., Coombs & Avrunin, 1977); if this is so, then at some point the intercorrelation structure will become negative. Such a negative intercorrelation structure may then trigger the application of new decision procedures. Evidence that changes in the inter-attribute correlation structure can influence strategy usage is provided by Bettman, Johnson, Luce, & Payne (In preparation).

Thus, the constructive view implies a more detailed contingency notion: the elements of choice heuristics used at any given time are contingent upon the properties of the choice task at that particular time. Finally, a constructive viewpoint also suggests that context variables, which reflect specific values of the alternatives rather than more

general structural properties of the choice task, may play an even more important role in determining decision processes.

If we take a constructive perspective, then our conceptual framework still applies, but at a more detailed level of analysis. Hence, constructive processes do not appear to present a conceptual problem for our framework. However, to implement our framework at this level of detail, we would need to extend the set of EIPs we use to include new operators (e.g., "bookkeeping" operators which keep track of possible regularities in the decision environment). We would also need to implement more detailed elements of heuristics for our simulations. However, our conceptual framework appears capable of handling such extensions. Recent work by Klein and Yadav (1989) seems consistent with such a local, bottom-up approach to accuracy/effort tradeoffs.

Throughout the above discussion, one of the most important underlying concepts has been the idea that individuals can notice and exploit regularities in the decision task. That is, individuals learn about the structure of the task as they gather information, and they may then change their processing to take advantage of what they have learned.

Noticing and Exploiting Regularities

These notions of noticing and exploiting regularities are very similar to the idea of interrupts and reactions to interrupts (Simon, 1967; Bettman, 1979). Once they have begun to work on a particular goal or to implement some element of a strategy, individuals do not necessarily blindly follow their original direction to completion. Rather, if conditions warrant, they can interrupt current processing, assess the situation, and switch direction if necessary. Interrupts are generally brought about by departures from expectations; unexpected events are noticed, assessed, and reacted to if necessary. This structure of interrupts and reactions is of course very parallel to that of noticing and exploiting structure in a decision task.

However, there is a cost associated with opportunistic decision processes. As Anderson (1983) has noted with regard to the Hayes-Roth and Hayes-Roth (1979) model of opportunistic problem solving, there is a problem with distractibility. Specifically, if a decision maker's attention can always be captured by an currently interesting piece of information, overload of working memory may lead to a failure to maintain a coherent

decision process. This possibility of overload necessitates a theory for when information will be considered or ignored and a theory for resolving conflicts about which pieces of available information are relevant to the overall goal. Anderson offers production system models as one way to incorporate noticing and conflict resolution mechanisms into theories of human problem solving.

We conjecture that distractibility is a fairly common problem for decisions of any complexity. Important attributes may get ignored, and much too much time may be spent on relatively unimportant attributes. In fact, we would argue that one of the primary benefits of decision analysis techniques for improving decisions is that they mitigate the problem of distractibility by forcing the decision maker to consider information in a more systematic and explicit fashion. The issue of distractibility is also related to the problem of inconsistency in judgment, and why boot-strapping often works.

Two aspects of noticing (or interrupts) are particularly relevant: what gets noticed, and when things are noticed. As we just discussed, departures from expectations are often noticed. In the context of choice situations, this could include extreme values or negative information (Fiske, 1980); missing information (e.g., if one is trying to implement a LEX rule but finds information is missing for some alternatives (Burke, 1990)); or information in a different format from other alternatives. There may also be differences in the ease of noticing different types of information. Some task factors, such as time pressure or problem size, may be relatively easy to ascertain. However, many context factors are more difficult to perceive (e.g., interattribute correlation (Crocker, 1981; Klein and Yadav, 1989)).

The ease of noticing task properties is generally related to the concept of problem "transparency" that is receiving increased attention in the decision making literature (Hammond, 1990; Tversky and Kahneman, 1983, 1986). The idea is that a particular procedure, such as eliminating dominated options, will be used in situations when its application is transparent and will not be used in nontransparent situations.

Hammond (1990) relates the concept of transparency to the difference between surface (immediately apparent in the display of information) and depth (not displayed or not immediately apparent) properties of a task. Nontransparency can be thought of as a

situation where the surface properties of a task are inconsistent with and mask the depth properties of the task. Why would this happen? One explanation offered by Tversky and Kahneman (1990) is that rules of mental economy often result in the acceptance of a problem as presented (i.e., based upon surface properties), without the spontaneous generation of more depth-related problem representations (see also Slovic's (1972), concreteness principle).

The distinction between task and context factors mentioned above also affects when things are noticed. Obviously task factors are much more likely to be noticed a priori or very early in the decision. Context effects, on the other hand, are often noticeable only after some information has already been examined. In general, given the importance of noticing factors to a constructive view of choice, research examining individuals' focus of attention during information gathering would be very important.

In the next few sections, we examine aspects of noticing and exploiting structure in somewhat more detail. In particular, we examine notions of editing and problem restructuring.

Editing Processes in Decision Making

Editing processes have been proposed as an important component of choice (Kahneman and Tversky, 1979; Goldstein and Einhorn, 1987), with individuals supposedly editing choice problems into simpler form before choosing. Editing could involve dropping outcomes which are identical across alternatives, eliminating some alternatives, or eliminating redundant attributes, for example. To the extent that editing can simplify choice, it is potentially a major component of adaptivity to different choice environments.

Whereas Kahneman and Tversky (1979) and Goldstein and Einhorn (1987) argue that editing processes come first, with alternatives edited and then simplified options evaluated, we argue instead that editing is opportunistic. Editing may occur throughout a choice whenever individuals notice some structure in the choice environment that can be exploited. Hence, editing can be a bottom-up process, driven by the data, as well as a priori or top-down. Editing processes may be involved earlier in the decision process the more experience one has in a given choice environment (Johnson and Russo, 1984).

Editing is probably also adaptive, in that the particular editing operation used may be a function of the immediately preceding processing. That is, different types of processing will leave different traces in short-term memory, and these traces will be more or less compatible with different editing operations. For example, processing a pair of alternatives one attribute at a time and noticing how one compared to the other on each attribute would enable the detection of dominance, whereas processing each alternative in its entirety without direct comparison to the other would discourage such detection. Hence, different choice strategies enable different editing operations. Therefore, different choice environment properties will affect editing because they affect processing. This is likely to be particularly true for the effects of information display. Of particular importance is Slovic's (1972) principle of *concreteness*, mentioned earlier: he argues that individuals tend to use information in the form in which it is displayed (see also Tversky and Kahneman (1990) and the earlier discussion of problem transparency). To the extent this is true, display should exert a strong influence on editing processes by encouraging or discouraging various types of processing.

A similar argument has been offered by Kahneman and Tversky (1979) regarding the order in which particular editing operations are carried out. They illustrate the importance of order with the following example: the gamble (\$500, .20; \$101, .49; \$0, .31) will appear to dominate another gamble (\$500, .15; \$99, .51, \$0, .34) if the second components of both gambles are first simplified by rounding to (\$100, .50). Thus, "the final edited prospects could, therefore, depend on the sequence of editing operations, which is likely to vary with the structure of the offered set and with the format of the display" [p.275]. Further, the preference order between gambles need not be invariant across contexts due to differences in the order in which problems are edited.

This view of editing is of course very consistent with the idea of constructive processes. Note that it also implies that individuals can devote effort not only to applying a heuristic to make a choice but also to setting up the problem in such a way that further processing is made easier. In the next section, we discuss work by Coupey (1990) which considers this tradeoff between decision processing and problem restructuring in more detail.

Restructuring Decision Tasks

Coupey (1990) defines problem restructuring as the application of operations to a set of information to yield a new problem representation. Restructuring operations might include transformations of information (e.g., rounding off, standardizing, or performing calculations), rearranging information (e.g., the order of brands or attributes), or simplifying by eliminating information. Figure 5 provides a protocol that illustrates restructuring. While editing and restructuring are clearly related, Coupey's definition of restructuring appears to be somewhat more general than that of editing; hence, editing could be thought of as a subset of restructuring.

Insert Fig. 5 about here

The purpose of restructuring is to help make difficult decision problems more manageable. That is, by transforming, rearranging, or eliminating information, the decision maker may be able to use a processing strategy that will result in a fairly accurate choice with reasonable levels of effort. In general, decision makers can decide to put effort into restructuring in hopes that the restructuring will make later choice processing more efficient. Individuals may trade off restructuring effort and effort devoted to choosing among the alternatives in the restructured problem.

Coupey examined restructuring by providing individuals with five decision problems with five alternatives and from four to six attributes. These decision problems were presented on a computer monitor via the Mouselab system described earlier. Two aspects of the problems were manipulated between subjects: 1) problems were either well-structured or poorly-structured, and 2) information was presented either simultaneously or sequentially. In well-structured problems, all information for a given attribute was expressed in the same units, and information on any given attribute was presented in the same order within each alternative. For poorly-structured problems, the information was presented in different units within the same attribute, and information on any given attribute could appear in a different order within each alternative. Samples of well-structured and poorly-structured problems are given in Figure 6. For

simultaneous presentations, information on all alternatives and attributes was presented on one screen in matrix form; in the sequential conditions, information was presented one alternative at a time. By crossing the structure and presentation conditions, four types of problems were created (well-structured, simultaneous; well-structured-sequential; poorly-structured, simultaneous; and poorly-structured- sequential). Each individual received five instances of one of the four problem types.

Insert Fig. 6 about here

Coupey gathered information about restructuring by allowing half of the subjects to take notes and then coding the restructuring operations evident in those notes. All individuals given the opportunity to take notes did so, and 94 percent of those taking notes used them to restructure the information. Individuals in the poorly-structured and sequential conditions used notes to create alternative by attribute matrices. For individuals receiving poorly-structured input, transformations, calculations, and rearranging were used to arrive at such matrix representations. An example of notes with a coding of restructuring operations is provided in Figure 7. Coupey characterizes this almost universal tendency to develop matrix representations as a top-down type of restructuring, since individuals appear to carry out such restructuring almost without regard for the particular values of the information. After such a matrix is developed, individuals carry out eliminations and additional transformations in a more bottom-up fashion, depending upon the particular values of the information. In cases where the initial presentation was already in alternative by attribute form (the well-structured, simultaneous condition), individuals often developed rankings of the alternatives within each attribute. These transformations were not as evident in the other conditions, perhaps because individuals had already had to exert some restructuring effort simply to develop an alternative by attribute representation.

Insert Fig. 7 about here

Individuals who restructured (i.e., almost all those who made notes) were more likely to ultimately utilize alternative-based strategies when processing the restructured information than those individuals who did not have the opportunity to make notes. Hence, one interpretation of these data is that individuals put effort into restructuring so that later they could carry out a more accurate heuristic with a reasonable amount of effort.

These descriptions of restructuring imply that restructuring occurs at early stages in the decision process. While this may in fact be true, we should observe that the methodology of using notes to study restructuring could bias the results in this direction. That is, individuals would be likely to take notes and then do further processing based upon those notes. Therefore, although restructuring certainly often occurs in early stages of a choice process, we still believe that restructuring can occur at any time that some aspect of the choice that can be exploited is noticed.

CONCLUSION

This paper reviews a program of research aimed at understanding the contingent and constructive use of strategies in decision making. We argue that the observed flexibility in the use of decision strategies often reflects adaptive behavior when both the effort and accuracy of decisions are considered. Evidence in support of an accuracy/effort framework is presented. We also argue that individuals will often build strategies opportunistically, changing their processing on the spot depending upon the information they encounter. Whether this constructive processing is to make a choice or involves editing or problem restructuring, however, such opportunistic processes still involve accuracy/effort tradeoffs. Both contingent strategy usage and opportunistic strategy construction lead to labile or constructive preferences.

References

- Anderson, J. R. (1983). The Architecture of Cognition. Cambridge, MA: Harvard University Press.
- Ashton, R. H. (1990). Pressure and performance in accounting decision settings: Paradoxical effects of incentives, feedback, and justification. Journal of Accounting Research, 28 (Supplement), 148-180.
- Beach, L. R. (1983). Muddling through: A response to Yates and Goldstein. Organizational Behavior and Human Performance, 31, 47-53.
- Beach, L. R., & Mitchell, T. R. (1978). A contingency model for the selection of decision strategies. Academy of Management Review, 3, 439-449.
- Bettman, J. R. (1979). An Information Processing Theory of Consumer Choice. Reading, MA: Addison Wesley.
- Bettman, J. R., Johnson, E. J., & Payne, J. W. (1990). A componential analysis of cognitive effort in choice. Organizational Behavior and Human Decision Processes, 45, 111-139.
- Bettman, J. R., Johnson, E. J., Luce, M. F., & Payne, J. W. (In preparation). Correlation, conflict, and choice.
- Bettman, J. R., & Park, C. W. (1980). Effects of prior knowledge and experience and phase of the choice process on consumer decision processes: A protocol analysis. Journal of Consumer Research, 7, 234-248.
- Bettman, J. R., & Zins, M. A. (1977). Constructive processes in consumer choice. Journal of Consumer Research, 4, 75-85.

- Biehal, G. J., & Chakravarti, D. (1986). Consumers' use of memory and external information in choice: Macro and micro processing perspectives. Journal of Consumer Research, 12, 382-405.
- Biggs, S. F., Berard, J. C., Gaber, B. G., & Linsmeier, T. J. (1985). The effects of task size and similarity on the decision behavior of bank loan officers. Management Science, 31, 970-987.
- Billings, R. S., & Marcus, S. A. (1983). Measures of compensatory and noncompensatory models of decision behavior: Process tracing versus policy capturing. Organizational Behavior and Human Performance, 31, 331-352.
- Billings, R. S., & Scherer, L. M. (1988). The effects of response mode and importance in decision making strategies: Judgment versus choice. Organizational Behavior and Human Decision Processes, 34, 1-19.
- Burke, S. J. (1990). The effects of missing information on decision strategy selection. In M. E. Goldberg, G. Gorn, & R. W. Pollay (Eds.), Advances in Consumer Research, Volume 17, pp. 250-256. Provo, UT: Association for Consumer Research.
- Coombs, C. H., & Avrunin, G. S. (1977). Single-peaked functions and the theory of preference. Psychological Review, 84, 216-230.
- Coupey, E. (1990). Decision restructuring in consumer choice. Unpublished doctoral dissertation, Duke University.
- Creyer, E. H., Bettman, J. R., & Payne, J. W. (1990). The impact of accuracy and effort feedback and goals on adaptive decision behavior. Journal of Behavioral

Decision Making, 3, 1-16.

Crocker, J. (1981). Judgment of covariation by social perceivers. Psychological Bulletin, 90, 272-292.

Einhorn, H. J., & Hogarth, R. M. (1981). Behavioral decision theory: Processes of judgment and choice. Annual Review of Psychology, 32, 53-88.

Einhorn, H. J., & Hogarth, R. M. (1986a). Decision making under ambiguity. Journal of Business, 59, S225-S250.

Einhorn, H. J., Kleinmuntz, P. N., & Kleinmuntz, B. (1979) Linear regression and process-tracing of models of judgment. Psychological Review, 86, 465-485.

Eisenhardt, K. M. (1989). Making fast strategic decisions in high velocity environments. Academy of Management Journal, 32, 543-575.

Fischhoff, B., Slovic, P., & Lichtenstein, S. (1980). Knowing what you want: Measuring labile values. In T. Wallsten (Ed.), Cognitive Processes in Choice and Decision Behavior. Hillsdale, NJ: Erlbaum.

Fiske, S. T. (1980). Attention and weight in person perception: The impact of negative and extreme behavior. Journal of Personality and Social Psychology, 38, 889-906.

Ford, J. K., Schmitt, N., Schechtman, S. L., Hults, B. M., & Doherty, M. L. (1989). Process tracing methods: Contributions, problems, and neglected research questions. Organizational Behavior and Human Decision Processes, 43, 75-117.

Goldstein, W. M., & Einhorn, H.J. (1987). Expression theory and the preference reversal phenomena. Psychological Review, 94, 236-254.

Hammond, K. R. (1990). Functionalism and illusionism: Can integration be usefully

- achieved? In R. M. Hogarth (Ed.), Insights in Decision Making: A Tribute to Hillel J. Einhorn. Chicago: University of Chicago Press.
- Hayes-Roth, B., & Hayes-Roth, F. (1979). A cognitive model of planning. Cognitive Science, 3, 275-310.
- Huber, O. (1980). The influence of some task variables on cognitive operations in an information-processing decision model. Acta Psychologica, 45, 187-196.
- Janis, I. L. (1989). Crucial Decisions. New York: Free Press.
- In H. Selye (Ed.), Selye's Guide to Stress Research (Volume 3). New York: Van Nostrand Reinhold.
- Janis, I. L., & Mann, L. (1977). Decision Making. New York: Free Press.
- Johnson, E. J. (1979). Deciding how to decide: The effort of making a decision. Unpublished manuscript, University of Chicago.
- Johnson, E. J., Meyer, R. M., & Ghose, S. (1989). When choice models fail: Compensatory representations in negatively correlated environments. Journal of Marketing Research, 26, 255-270.
- Johnson, E. J., Payne, J. W., Schkade, D. A., & Bettman, J. R. (1991). Monitoring information processing and decisions: The Mouselab system. Unpublished manuscript, Center for Decision Studies, Fuqua School of Business, Duke University.
- Johnson, E. J., & Russo, J. E. (1984). Product familiarity and learning new information. Journal of Consumer Research, 11, 542-550.
- Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision making

- under risk. Econometrica, 47, 263-291.
- Keinan, G. (1987). Decision making under stress: Scanning of alternatives under controllable and uncontrollable threats. Journal of Personality and Social Psychology, 52, 639-644.
- Klayman, J. (1983). Analysis of predecisional information search patterns. In P. C. Humphreys, O. Svenson, & A. Vari (Eds.), Analyzing and aiding decision processes. Amsterdam: North Holland.
- Klayman, J. (1985). Children's decision strategies and their adaptation to task characteristics. Organizational Behavior and Human Decision Processes 35, 179-201.
- Klein, N. M. (1983). Utility and decision strategies: A second look at the rational decision maker. Organizational Behavior and Human Performance, 31, 1-25.
- Klein, N. M., & Yadav, M. S. (1989). Context effects on effort and accuracy in choice: An inquiry into adaptive decision making. Journal of Consumer Research, 15, 411-421.
- Kruglanski, A. W. (1989). The psychology of being "right": The problem of accuracy in social perception and cognition. Psychological Bulletin, 106, 395-409.
- Langley, P., Simon, H. A., Bradshaw, G. L., & Zytkow, J. M. (1987). Scientific Discovery. Cambridge, MA: MIT Press.
- Larrick, R. P., Morgan, J. N., & Nisbett, R. E. (1990). Teaching the use of cost-benefit reasoning in everyday life. Psychological Science, 1, 362-370.
- March, J. G. (1978). Bounded rationality, ambiguity, and the engineering of choice. Bell

Journal of Economics, 9, 587-608.

- McClelland, G. H. (1978). Equal versus differential weighting for multiattribute decisions. Unpublished working paper, University of Colorado.
- Newell, A. (1980). Harpy, production systems, and human cognition. In R. Cole (Ed.), Perception and Production of Fluent Speech. Hillsdale, NJ: Lawrence Erlbaum.
- Newell, A., & Simon, H. A. (1972). Human problem solving. Englewood Cliffs, NJ: Prentice-Hall.
- Onken, J., Hastie, R., & Revelle, W. (1985). Individual differences in the use of simplification strategies in a complex decision-making task. Journal of Experimental Psychology: Human Perception and Performance, 11, 14-27.
- Payne, J. W. (1976). Task complexity and contingent processing in decision making: An information search and protocol analysis. Organizational Behavior and Human Performance, 16, 366-387.
- Payne, J. W. (1982). Contingent decision behavior. Psychological Bulletin, 92, 382-402.
- Payne, J. W., Bettman, J. R., & Johnson, E. J. (1988). Adaptive strategy selection in decision making. Journal of Experimental Psychology: Learning, Memory, and Cognition, 14, 534-552.
- Payne, J. W., Bettman, J. R., & Johnson, E. J. (In press a). The Adaptive Decision-maker. Cambridge University Press.
- Payne, J. W., Bettman, J. R. & Johnson, E. J. (In press b). Behavioral decision research: A constructive process perspective. Annual Review of Psychology, Vol. 43.
- Payne, J. W., & Braustein, M. L. (1978). Risky choice: An examination of information

- acquisition behavior. Memory & Cognition, 6, 554-561.
- Payne, J. W., Johnson, E. J., Bettman, R. R., & Coupey, E. (1990). Understanding contingent choice: A computer simulation approach. IEEE Transaction on Systems, Man and Cybernetics, 20, 296-309.
- Russo, J. E., & Doshier, B. A. (1983). Strategies for multiattribute binary choice. Journal of Experimental Psychology: Learning, Memory, and Cognition, 9, 676-696.
- Russo, J. E., & Rosen, L. D. (1975). An eye fixation analysis of multialternative choice. Memory and Cognition, 3, 267-276.
- Shanteau, J. (1988). Psychological characteristics and strategies of expert decision makers. Acta Psychologica, 68, 203-215.
- Sugan, S. M. (1980). The cost of thinking. Journal of Consumer Research, 7, 99-111.
- Simon, H. A. (1955). A behavioral model of rational choice. Quarterly Journal of Economics, 69, 99-118.
- Simon, H. A. (1967). Motivational and emotional controls of cognition. Psychological Review, 74, 29-39.
- Slovic, P. (1972). From Shakespeare to Simon: Speculations - and some evidence - about man's ability to process information. Oregon Research Institute Bulletin, 12 (3).
- Slovic, P., Griffin, D., & Tversky, A. (1990). Compatibility effects in judgment and choice. In R. M. Hogarth (Ed.), Insights in Decision Making: A Tribute to Hillel J. Einhorn. Chicago: University of Chicago Press.
- Slovic, P., & MacPhillamy, D. (1974). Dimensional commensurability and cue utilization in comparative judgement. Organizational Behavior and Human Performance, 11,

174-194.

- Sundstrom, G. A. (1987). Information search and decision making: The effects of information displays. Acta Psychologica, 65, 165-179.
- Tetlock, P. E. (1985). Accountability: The neglected social context of judgment and choice. Research in Organizational Behavior, 7, 297-332.
- Thorngate, W. (1980). Efficient decision heuristics. Behavioral Science, 25, 219-225.
- Tversky, A. (1972). Elimination by aspects: A theory of choice. Psychological Review, 79, 281-299.
- Tversky, A., & Kahneman, D. (1983). Extensional vs. intuitive reasoning: The conjunction fallacy in probability judgment. Psychological Review, 90, 293-315.
- Tversky, A., & Kahneman, D. (1986). Rational choice and the framing of decisions. Journal of Business, 59, S251-S278.
- Tversky, A., & Kahneman, D. (1990). Cumulative prospect theory: An analysis of decision under uncertainty. Unpublished working paper, Stanford University.
- Tversky, A., Sattath, S., & Slovic, P. (1988). Contingent weighting in judgment and choice. Psychological Review, 95, 371-384.
- Wood, R. E., & Locke, E. A. (1990). Goal setting and strategy effects on complex tasks. In B. M. Staw and L. L. Cummings (Eds.), Research in Organizational Behavior, Volume 12. Greenwich, CT: JAI Press.
- Wright, P. L. (1977). Decision times and processes on complex problems. Unpublished manuscript, Stanford University.
- Wright, W. F., & Aboul-Ezz, M. E. (1988). Effects of extrinsic incentives on the quality

Figure 1
Verbal Protocols of Choice Strategies

(a) Additive Utility	(b) Additive Difference
A24: O.K., the decision is now between the two rent prices A25: In accordance with the other qualities A26: Now apartment A has the advantage A27: Because the noise level is low A28: And the kitchen facilities are good A29: Even though the rent is \$30 higher than B	D238: O.K., we have an A and a B D239: First, look at the rent for both of them D240: The rent for A is \$170 and D241: The rent for B is \$140 D242: \$170 is a little steep D243: But it might have a low noise level D244: So we'll check A's noise level D245: A's noise level is low D246: We'll go to B's noise level D247: It's high D248: Gee, I can't really very well study with a lot of noise D249: So I'll ask myself the question, is it worth spending that extra \$30 a month, to be able to study in my apartment?
(c) Satisficing	(d) Elimination-by-Aspects
A163: The rent for apartment E is \$140 A164: Which is a good note A165: The noise level for this apartment is high A166: That would almost deter me right there A167: Ah, I don't like a lot of noise A168: And, if it's high, it must be pretty bad A169: Which means, you couldn't sleep A170: I would just put that one aside right there - I wouldn't look any further than that A171: Even though the rent is good	D289: Since we have a whole bunch here D290: I'm going to go across the top and see which noise levels are high D291: If there are any high ones, I'll reject them immediately ... D295: Go to D D296: It has a high noise level D297: So, we'll automatically eliminate D ... D303: So, we have four here D304: That are O.K. in the noise level

Source: Payne and Braunsstein (1978)

Fig. 2 Verbal Protocol Examples of Task Contingent Processing

(a)

C98: Well, with these many apartments (six) to choose from,
C99: I'm not going to work through all the characteristics
C100: Start eliminating them as soon as possible.

(b)

B1: With just two (gamble)s to choose from,
B2: I'm going to go after all the information
B3: It won't be that much trouble.

(a) Excerpt from Payne (1976).

(b) Excerpt from Payne and Braunstein (1978)

of frequency assessments. Organizational Behavior and Human Decision Processes, 41, 143-152.

Yates, J. F., Jagacinski, C. M., & Faber, M. D. (1978) Evaluation of partially described multiattribute options. Organizational Behavior and Human Performance, 21, 240-251.

TABLE 1

ELEMENTARY EIPS USED IN DECISION STRATEGIES

READ	Read an alternative's value on an attribute into STM
COMPARE	Compare two alternatives on an attribute
DIFFERENCE	Calculate the size of the difference of two attributes for an alternative
ADD	Add the values of an attribute in STM
PRODUCT	Weight one value by another (Multiply)
ELIMINATE	Remove an alternative or attribute from consideration
MOVE	Go to next element of the external environment
CHOOSE	Announce preferred alternative and stop the process

Note: STM = short-term memory.

Figure 3

	Outcome 1	Outcome 2	Outcome 3	Outcome 4
Probs.				
Gamble A		\$8.39 ₊		
Gamble B				
Gamble C				
Gamble D				
Choose one:	Gamble A	Gamble B	Gamble C	Gamble D

Verbal Protocol of a Constructive Choice Process for Microwave Ovens

- 1 Montgomery Wards
- 2 Hum, a lot of information here
- 3 That's not so hot. I don't like that
- 4 That makes Amana a plus.
- 5 Is Amana the only one that comes out like that?
- 6 No, everything that stays below 1 would probably be acceptable
- 7 I don't know, after that I would begin to become a little bit leery, maybe even 1.2. I'm not sure
- 8 This one, Sharp
- 9 And maybe this 1.2
- 10 I don't know, after that, I begin to get leery about it
- 11 Okay, after that, it has to have a browner
- 12 So now we're in trouble
- 13 I'm getting particular
- 14 Probably to the cooking levels or timers
- 15 But the Montgomery Wards has 7 and the Sharp number was 4
- 16 I might sacrifice this, but this is so important, at least to me, that I might even sacrifice the other three cooking levels
- 17 Then that would leave me with so much less that I would probably, looking at this, would go with this one
- 18 But, by dollars, that is such a determining factor, I don't think I would go for that over this, but it would probably be between these two
- 19 I was not aware that so many models didn't have [a browner]
- 20 I thought that this was something that was fairly common
- 21 But I was not aware that browners were not that widespread
- 22 I might eliminate the browner which goes on the oven itself
- 23 And then things such as the cooking levels and the timer would become more important factors
- 24 So, in looking them over again, this would be acceptable now
- 25 No, only two cooking levels doesn't do it
- 26 This one would probably be acceptable now
- 27 And, I don't know, probably not this one
- 28 All three of them have expanded timers so where would we go from there
- 29 Probably to the amount of space I could use in cooking
- 30 So this one, this one, so probably between these three at this point
- 31 I still have the same two with the browner except one
- 32 And now I'd go back to using the browner to eliminate
- 33 Which still leaves me with the large Sharp or the large Montgomery Ward
- 34 I'm going to pick the biggest model as my first choice.

An Example of Restructuring*

Okay, let's see. The information isn't in the same order for each brand. What I'm going to do is to make a table to reorganize the information. It'll be easier to decide if I can see everything at a glance, too. (Subject sets up a brand/attribute matrix.) Okay, what I did was to put down all the brands and the categories in order for each brand. Now, let's see. Hmm, number of needles -- A is best with 80, then B. That's the most important category. Warranty, A is 56 months, B, 70 months, C -- oh, gee, that's right--they're not all in the same units. C is weeks, so is D. E is years. Okay, what I'm going to do now is to do some math to convert them so I can compare them better (Subject does calculations in notes). Okay, that's better. Look at the others and see if I can standardize them. Quality rating, 2.2 out of 5, 8.8 out of 10, so A is 4.4 out of 10, 3.3 out of 5 is 7 out of 10, and E is oh, out of 100, so 7.2 out of 10. (Subject converts quality ratings mentally, noting transformations in matrix.) Price, well, this is hard. Price per needle or price overall? 3.39 per needle, how many needles? Eighty, so do the math. There. 5.88 times 68. Okay, now I'm all set. (Subject begins comparing brands.)

* This protocol is taken from Coupey (1990). The task was to recommend a knitting machine to purchase. Restructuring is used to standardize information in order to increase processability.

Figure 6

A SAMPLE DECISION: Storage Buildings

WELL-STRUCTURED PRESENTATION *

Brand	Price	Size (in square feet)	Durability (in months)	Difficulty to build (0=easy, 5=very hard)
Brand A	\$1500	150	72	3
Brand B	\$2352	168	50	2
Brand C	\$1700	100	54	2
Brand D	\$769.50	81	72	4
Brand E	\$3388	121	120	1.5
Brand F	\$1568	128	64	4

Panel A

POORLY STRUCTURED PRESENTATION

Brand A	Total price: \$1500	Size 151 x 10w	Difficulty to build: avg. (5 pt scale)	Durability 6 years
Brand B	Price per square foot \$14	Durability 50 months	Size 121 x 14w	Difficulty to build: 4 out of 10 (0=easy)
Brand C	Total price: \$1700	Size 101x10w	Difficulty to build: easy (5 pt. scale)	Durability 4.5 years
Brand D	Total price: \$769.50	Durability 72 months	Difficulty to build: 8 out of 10 (0=easy)	Size 91 x 9w
Brand E	Price per square foot \$28	Difficulty to build: 29 (100 pt. scale. 0 = easy)	Size 121 sq ft	Durability 10 years
Brand F	Total price: \$1568	Size 161 x 8w	Durability 64 months	Difficulty to build: somewhat difficult (5 pt. scale)

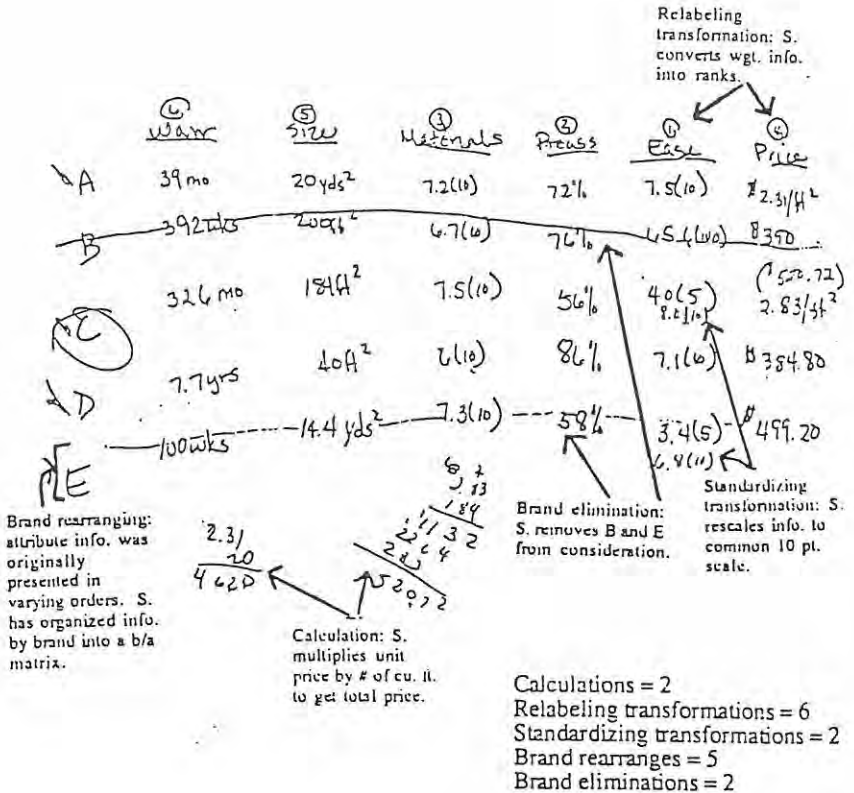
Panel B

Adapted from Coupey (1990)

Figure 7

AN EXAMPLE OF NOTES CODING *

Choice Among Five Storage Buildings: Attributes are Ease to Build, % Preassembled, Quality of Materials, Price, Size, Length of Warranty



* Adapted From Cooper (1990).

84

Behavioral decision theory and environmental risk management:

What have we learned and what has been neglected?

VLEK Charles (University of Groningen) & KEREN Gideon
(University of Utrecht)

Major environmental hazards are the resultant of massive human behavior mediated through needs, habits, attitudes and decisions. By unraveling the determinants of choice at these behavioral levels, by clarifying the use (if any) of future time perspective, and by comprehending the conflict between individual and collective rationality; behavioral decision theorists could effectively provide various support in environmental risk assessment and management. The paper would argue and illustrate the usefulness of behavioral decision theory for environmental risk management and it would offer research questions and proposals derived from both classical and more recent theories of human decision making.

(Full text was not available for inclusion in this booklet.

A copy will be distributed separately.)

The Presumptions of Prescription

Stephen R Watson

Judge Institute of Management Studies
University of Cambridge

1. Introduction

How one ought to make decisions has been an important subject of study for a long time. The normative framework of the rational decision making paradigm was widely accepted many decades ago as an ideal to follow; more recently it has been recognised that there are difficulties in conforming to this ideal. As a result a distinction has been made between normative methods - those that we would ideally like to follow - and prescriptive methods, that is, ways to prescribe how decision makers should approximate to this ideal in practice.

We shall explore several approaches to prescription in this paper, and the presumptions which underly them. Firstly, we discuss the prescriptions associated with various forms of the rational decision making paradigm. In section 2.1 we explore the arguments for prescriptive additions to standard decision theory, criticising the presumption that such additions could be satisfactory. Next, in section 2.2, we look at attempts to modify the axiomatic presumptions of standard decision theory in order to construct a more satisfactory prescriptive theory; we conclude that this effort has not yet been successful. Finally, in section 2.3 we look at other prescriptive approaches which vary from standard decision theory, but which are still exemplars of the rational decision making paradigm; we explore whether the presumptions of such approaches are acceptable.

Secondly, we examine the presumption that prescription based on the rational decision making paradigm is the right way to go. There are other descriptions of how decision making takes place which do not conform to the rational decision making paradigm. Can these be used for prescription? Moreover, there are now several approaches to the analysis of situations which involve decisions, and which, it is argued, help decision makers, but which stop short of explicit prescription. Some of these are referred to as 'soft systems methodologies'. Are these approaches based on fewer presumptions, and are they more acceptable as decision-making aids? We discuss both of these questions in section 3.

Finally, we discuss the desirability of prescription of any form, in the light of the discussions in the previous sections.

2.0 Presumptions of the 'Rational Decision Making' Paradigm.

The notion of establishing a procedure to guide us in choice is far from new. Most religions provide rules for ethical behaviour, and the search for the moral life has always been a strong theme in literature. Although some of these guides for decision-making have been very detailed (look at the book of Leviticus, for example, for an extremely detailed guide to behaviour in a particular context), they tend to be of the form of a

look-up table (in this situation do such-and-such etc), or general ethical principles ("love thine enemy and do good to those who despitefully use you"), rather than procedures for balancing one goal against another in situations which have not previously been encountered. Procedures of this latter kind have been lacking, until relatively recently.

The rational decision making paradigm, which has become very popular over the last 50 years or so, has as its most important aim the provision of explicit prescription. The steps to be taken when following this paradigm may be stated as follows:

When facing a decision, go through the following stages:

1. Identify options
2. Identify objectives
3. Predict the outcomes of options
4. Evaluate the outcomes of each option in terms of the objectives
5. Select the option whose evaluation is best

In pursuit of this approach, a whole new academic subject was born, namely operational research (OR). It can reasonably be argued that much of the successful work in OR has been addressed at stages 3 and 4 (modelling) and stage 5 (optimisation). OR has been criticised for failing, by and large, to address stages 1 and 2. Decision Analysis (see Watson and Buede (1988), or von Winterfeldt and Edwards (1986)) would claim to be addressing at least stage 2 (by showing how to construct utility functions), and to be making an important contribution to stage 5 (by showing that it is expected utility which should be maximised).

Do we now, therefore, have a satisfactory paradigm for prescribing what to do when faced by any decision? Can we not simply use the normative rules of Decision Analysis and the procedures of Operational Research to give us a prescription in any situation? If this were the case, one would expect Decision Analysis by now to have a far wider domain of application than it has. Brown (1970) and Ulvila et al. (1982) both reported the extent of use of Decision Analysis, and found it disappointing. It would be interesting to see more recent reviews on the application of Decision Analysis, to know if there is still a disappointing take up. There are several consulting companies, particularly in the United States which seem to earn their living using Decision Analysis, but it still seems to be the case, particularly in the United Kingdom, that what is in theory a compelling prescriptive framework for decision making is not widely adopted.

There seem to be several explanations for this phenomenon. Firstly, it is not as straightforward as we might first imagine to turn the normative framework of decision theory into a prescription for action. We presume too much if we think that this subject is closed. In section 2.1 we examine the nature of this presumption, and analyse suggestions for getting round the problem.

Secondly, the key principle of decision theory, the maximisation of expected utility, is deduced from axioms of behaviour which are themselves presumptions about how people should behave. Much has been done to challenge these presumptions, particularly in the light of considerable evidence that people do not, typically, make decisions in accord with these axioms. There is a considerable literature now on what happens when these presumptions are challenged; we argue in section 2.2, however, that little

progress has been made in showing that any modified set of presumptions leads to more satisfactory prescriptions.

Thirdly, there are alternative prescriptive frameworks to that of Decision Analysis, still within the rational decision making paradigm, such as the Analytic Hierarchy Process, the Outranking methods, and the Multi-criteria Decision-Making movement. Some of these frameworks are dismissed by some members of the Decision Analysis community; we explore in section 2.3 the presumption in doing so. In particular, we ask whether the difference between these frameworks simply reflects different sets of presumptions about human decision making.

2.1 The presumption that Decision Analysis provides clear prescription

Ever since the invention of the term by Howard (1966), and its clear presentation by Raiffa (1968), the idea that Decision Analysis provides a compelling prescription for decision-making in any situation has become widespread. It all seems so straightforward: all we have to do is to specify the options, construct the utility function, determine the probabilities and then choose the option which maximises expected utility. What is more, there have been notable examples of success for the method, particularly if you measure success by satisfied customers (see, for example, Part II of Watson and Buede (1988) or French (1989) for a description of many satisfactory applications of Decision Analysis).

The presumption that the prescriptions of Decision Analysis can be straightforwardly applied to any decision situation has been challenged in several different ways. For example, Simon (1983) argues that Decision Analysis is impractical because of the limited ability of the human mind to reason. In principle, an application of decision theory entails the need to construct utility functions involving all possible future states for the entire world and the assignation of a joint probability distribution over them. Any Decision Analysis can, therefore, only be an approximation, since no human possesses the reasoning power (or the time) to do this: Decision Analysis operates in a world of bounded, rather than true rationality. This argument is normally countered by the believing Decision Analyst on two grounds: (a) given that we arbitrarily bound our contemplations of what to do anyway, it is surely better to choose consistently within that bounded world than to do otherwise; (b) very often we can feel pretty confident that the approximation to the complete world represented by the Decision Analyst 'captures all that matters' in our decision. Both these points again have their counter-arguments; what is clear, however, is that we are making an unfounded presumption - sometimes called a step of faith - in thinking that normative decision theory can be simply applied.

This presumption is also challenged by those who try to make Decision Analysis work, by making slightly different observations about human cognitive abilities from those identified by Simon. It appears to have been Keeney and Raiffa (1976) who first made the distinction between the normative and the prescriptive, in this context. The decision theory used by Howard and Raiffa to form their Decision Analysis was normative, in that it showed what a decision-maker ought to do, starting from certain plausible assumptions about desirable properties for human preferences. Given that it was found that peoples' preferences often did not possess these properties, so that the measurement of probabilities and utilities was

far from easy, for example, how could one follow the norms? How could one prescribe?

A nice example of the problems created in attempts to apply Decision Analysis is given by Bell et al. (1988, pp 11-14), who analyse a framing phenomenon uncovered in an experiment by McNeil et al. (1988). Subjects were presented with data on the distribution of survival time for cancer patients following two different treatment regimes; one had a higher frequency of early death and a higher frequency of longer life than the other. What was remarkable was that the proportion of subjects saying that they preferred one treatment to the other was highly dependent on the way the situation was framed: in one case statistics were presented in terms of death rates, and in the other in terms of survival rates. The point of this experimental result (and there are many others which show that peoples' expressed preferences depend on the way the problem is framed), is that if a Decision Analysis was done which involved an assessment of the utility for the two treatment alternatives, how should the analyst frame the problem? A prescriptive theory would give an answer to this question; normative theories do not.

The need for adding to the normative theory with prescriptions for how to implement it has perhaps best been expressed by Brown (1989). He points out that there are several areas where a decision analyst often has to make decisions about how to do the analysis without any guidance from the normative theory. For example:

How do you specify options - in terms of the first step,
or in terms of a broad category of actions?

Where in the hierarchy of goals do you focus the analysis-
e.g. long term survival of the company or return on
investment on a particular project?

Other problems facing the prescriber are mentioned by Keller (1989 a and b). One such is as follows:

Which methods should you use to elicit utility
functions, given that people regularly come up with
different utility functions by using different methods;
do you assess certainty equivalents, or do you assess
the probability at which a sure thing is indifferent to
a standard gamble?

It is clear, then, that in applying the normative model of decision theory to real decision problems, there are many ways in which the analyst is stopped short, and has to guess what to do.

Two ways have been suggested for dealing with this problem, both of which seem unsatisfactory. The first is that the answer lies in education. Fischer (1989 p496) states:

"Thus, developing an effective prescriptive science of decision-making requires both research on theories and methods of decision-making and improved education of the decision makers who will use these theories and tools".

Von Winterfeldt (1989) argues that the core axioms of normative decision theory (such as transitivity, the independence of irrelevant alternatives

etc.), are so obviously sensible that "their violation would be an embarrassment to most reasonable people". He goes on to argue that no reasonable person would be prepared to insist on such a violation when threatened by a Dutch Book made out against them. The difficulty with this argument is that people would have to stick to a set of mutually inconsistent preferences in order to lose through the Dutch Book. What is much more likely to happen is that their preferences would change in time, particularly when they became aware that they could lose money. We know that preferences are labile anyway (Fischhoff et al. (1980)), so that it is doubtful whether a person with inconsistent preferences could be pinned down in practice by the Dutch Book argument.

But the argument that education about the normative theory will remove the need for any prescriptions to add to it, can be more strongly criticised for being incomplete. As Lindley et al. (1979), showed when discussing how to cope with a set of inconsistent probability assessments, a highly motivated and educated assessor may want to change his assessments to make them coherent, but not know which of his assessments to change. All education about the nature of decision theory will do, is to point out to the assessor that he has a problem, but not show him how to solve it.

Brown (1989) takes a different view. He sees the problems outlined above as real and important in attempts to use the normative theory for prescription. If we are to be able to prescribe, he argues, we must find ways to determine what to do when confronted with one of the unsignposted forks in the path of analysis mentioned above. If our probability judgements are incoherent, we must have a procedure for making them coherent. If our preferences depend on the way that a problem is framed, we must have a procedure for determining which frame to use. This is the rationale for the work on reconciling the incoherent probability judgments that individual assessors often make, that Brown carried out with Lindley and Tversky (Lindley et al. (1979)), and this forms the motivation for the programme of research in Prescriptive Decision Science that he sets out in Brown (1989).

However, if the research methodology to be used in researching Prescriptive Decision Science is akin to that used in Lindley et al. (1979), we should pause to ask if it will work. In that paper the problem of reconciling incoherent judgments is likened to the problem of triangulation in surveying. Each measurement is presumed to be inaccurate, and so the best estimates of the probabilities involved are obtained by a least squares analysis, taking into account that some probabilities are 'better' than others and giving them an appropriate weight. This procedure has a theoretical elegance, but it appears not to have been used in practice since it was suggested over ten years ago. It may be that the reason for this is that the inputs to such an analysis involve not only elicited probabilities, but also second-order estimates of the 'goodness' of these probabilities; echoing Simon (1983), we wonder whether anybody has the cognitive apparatus to make such estimates satisfactorily. Underlying this methodology is the presumption that a 'correct' set of probabilities exist; but since they can only be presumed to exist because of the existence of a complete preference order on the likelihoods of all possible events, and that patently does not exist because of the observed inconsistencies in the first place, such a presumption is untenable.

It seems more satisfactory to suppose that, in the process of eliciting probabilities and utilities, we are creating preferences and judgments of relative likelihood just as much as we are revealing them. We cannot, then, expect to resolve questions such as what frame to use, or how to reconcile

incoherent probabilities by a deeper introspection. Rather, we must accept that to some extent there are many possible outcomes to a Decision Analysis, depending on the paths we choose at the forks in the road on the analytic process, and that there is no obvious way to choose between them. Rather than supposing that there is a unique correct path, we must accept that there are many possible paths, possibly leading to different prescriptions for action. We choose between them not by reference to the Guide Book to the True Way, but by other criteria, such as whether those for whom we are attempting to prescribe action find the procedure satisfactory. In fact Brown adopts this view himself, when he suggests (Brown (1989), section 2.3), that one of the criteria for determining whether a prescription for decision is satisfactory or not is 'acceptability, both institutional and psychological'.

We conclude therefore that there are significant problems in presuming that we can use standard decision theory as a practical prescriptive device; that these problems cannot be overcome either by education or by deeper analysis; and that, therefore, we must be less presumptuous in what Decision Analysis has to offer.

2.2 The Presumption that Generalized Decision Theories Will Provide Better Prescription

If education is not going to make decision analysis work, and the cognitive demands of additional prescriptions are too great, are there other ways to make decision theory more directly prescriptive? If the problem is that people cannot elicit probabilities and utilities satisfactorily, can we not change the basic axioms of decision theory to discover a decision theory that is immediately prescriptive? This is a line of argument that has been approached by many authors over the last decade. The purpose of this section is to review this work, and to ask whether this approach is likely to provide a more satisfactory and usable kind of decision analysis.

We need not rehearse here the evidence that people often have difficulty conforming to the norms of behaviour which constitute the axioms of standard decision theory. This evidence is set out in many places, for example, chapter 4 of Watson et al. (1988), or Kahneman et al. (1982). As a result of this body of knowledge, numerous alternatives to standard decision theory have been suggested in the last decade or so to describe how decision making takes place; Prospect theory, suggested by Kahneman and Tversky (1979), the theory of ambiguity of Einhorn and Hogarth (1986), or the Image theory of Beach (1990) are notable examples. But these new descriptive theories do not claim to be prescriptive. They aim to explain observed behaviour, rather than to prescribe satisfactory decision making.

The normative theories which do attempt to replace standard decision theory are motivated by a desire to accommodate the evidence that people often violate the axioms of that theory. Accounts of these theories are given in Bell et al. (1986), and Machina (1987). An extensive review is given in Fishburn (1989). As Fishburn notes: "...their variety can be cause for bewilderment". He lists 25 such theories, each of them starting from slightly different axioms from those used in standard decision theory. They all lead to slightly different normative recommendations for action than that of maximising expected utility; but the recommendations are of a similar kind. For example, a theory of Becker and Sarin (1987) recommends that instead of maximising an expected utility, one should maximise the

expectation of a modified utility function which itself depends on the probability distribution. There is a normative recommendation for action, but it is slightly different from that of the standard theory.

The question that interests us here is whether these theories offer more satisfactory prescriptions. Are the elicitation requirements for these new theories any easier than those for the standard theory? Are framing problems avoided? The evidence is, regrettably, thin on this point. For example, Becker and Sarin (1987), although seeing their model as useful for descriptive or predictive applications, also saw potential for its prescriptive use. They hoped to be able to modify the procedures of Decision Analysis to incorporate their extended utility theory, but (Sarin, 1990, private communication), they have not yet done so. It may be that all the generalised utility theories are too new to have been applied, and we must wait for descriptions of attempts at application before we can know whether they are an improvement in prescription. We anticipate, however, that no such advance will be forthcoming, for two reasons.

Firstly, the motivation for developing these new theories seems not to be the need for more satisfactory prescription, but rather the intrinsic interest of exploring the implications of a different axiomatic base. Not many of the authors report a concern to test their theories as prescriptions. Secondly, many of the theories still require the elicitation of elements that have caused trouble with the standard theory. To use the Becker and Sarin theory again as an example, it is necessary to know the probability distributions over outcomes for all the options being compared; if we produce incoherent probability assessments, then we still need some prescriptive addition to resolve this incoherence, such as that suggested by Lindley et al. (1979). As we argued above, this is not a satisfactory prescription, since it seems to be cognitively even more difficult than the first order probability elicitation process.

Our anticipation of failure here may be proved wrong; it may be that a prescriptive theory will emerge whose demands in terms of elicitation, structuring and framing are all within human cognitive abilities. That is to say, the questions asked of subjects about their preferences and beliefs will be easy to answer, and the answers will be independent of how the questions are asked. There is no sign yet of such a theory emerging, however. At present we have to conclude that this route to a satisfactory prescriptive scheme is not likely to lead anywhere. It currently seems to be somewhat presumptuous to hope that a modified decision theory will emerge to solve our problems of prescription.

2.3 The presumptions of competing prescriptive frameworks within the rational decision making paradigm.

The expected utility model, or, as it has been referred to above, the standard decision theory model, is not the only prescriptive approach to decision making that can be said to lie within the general approach of the rational decision making paradigm. There are other distinct approaches, some of which are inconsistent with standard decision theory. For example, the outranking methods developed by Professor Roy and his associates at the University of Paris (see Roy (1985) for example) have gained a widespread following in France and Belgium. The Multi-Criteria Decision-Making movement (Zeleny (1982)) can also be said to espouse a distinct prescriptive approach to that of standard decision theory.

Perhaps the best known, and most controversial alternative theory for prescription, however, is the Analytic Hierarchy Process of T L Saaty (see Saaty (1980)). This is clearly a method that lies within the rational decision-making paradigm, as set out at the beginning of this section. It is also clear that the method has been used widely for prescription, not only by Saaty and his coworkers. But it is also distinct and in conflict with standard decision theory. The debate between Dyer and Saaty and his colleagues which appeared in Management Science in 1990 shows just how strong this conflict is (Winkler (1990), Dyer (1990a), Saaty (1990), Harker et al. (1990) and Dyer (1990b)). The question we must address here is whether the AHP is a better prescriptive theory than standard decision theory. Is it less presumptuous?

The first point to make is that the AHP, although making different presumptions from standard decision theory, nonetheless makes presumptions. When Saaty first introduced the AHP, it was somewhat ad hoc, in the sense that while the steps of the process were plausible, they were not derived from some fundamental axioms. In response to this criticism, Saaty (1986) produced an axiomatic formulation of the AHP, based on, to quote Dyer (1990b) "a 'primitive notion' that essentially assumes the existence of (a) ratio scale". Dyer argues that, in distinction to standard decision theory, where the axioms are plausible statements of how reasonable people might want to behave, Saaty's axiom has no such quasi-empirical basis. Of course, as we have observed previously, the empirical evidence is that people do not make up their minds as though they were obeying the axioms of standard decision theory; this criticism of Saaty's approach somewhat loses its force, therefore. The most that can be said is that some people would like to make up their minds as though they were following the axioms of decision theory; and it is at least plausible that others might want to make up their minds as though they were following Saaty's primitive notion. We conclude that the argument which places standard decision theory above the AHP on the ground of better axioms, is weak.

But a second point can be made about using Saaty's method, which seems to be much more telling. As first pointed out by Belton et al. (1982), it is possible for the most preferred option in an AHP analysis to depend on whether or not another option is included even though the newly included option does not come out best. Belton et al. (1982), give an example of this. They show that if some options A, B and C are compared using the AHP, they are evaluated $A > B > C$, while if an option D is included, the order becomes $B > A > D > C$. The preference between A and B is reversed, by the inclusion in the analysis of a less satisfactory option, D. Saaty and his colleagues response to this was firstly, that it did not matter; and then subsequently, that if several very similar options were to be compared, only one of that group should be included in the analysis. Many observers find this property of rank reversal unsatisfactory. It is a consequence of the presumptions of Saaty's method that rank reversal can occur. A person with a decision problem seeking a formal method of analysis must ask whether this property is acceptable, even if it does not apply to the problem being faced.

Thirdly, however, the presumptions about human cognitive abilities in the AHP seem to be weaker than in standard decision theory. We are not asked to make fine judgements between hypothetical options; we are simply asked to make a response on a verbal scale. People report that the AHP makes cognitive demands on decision-makers which are possible to respond to. Of course, one of the criticisms of the AHP is that these verbal responses are converted into a somewhat arbitrary 1 - 9 scale, which is followed by

some rather precise mathematics. We might well question whether what the AHP does with the verbal responses is the best, or indeed only, thing possible. Nonetheless, the AHP is less presumptuous about human cognitive abilities than standard decision theory.

What, then, are we to make of the AHP? It makes presumptions about how people might want to choose, cast into axioms, as does standard decision theory; it presumes that rank reversal is unimportant, unlike standard decision theory; it makes presumptions about human cognitive abilities, as does standard decision theory. There appears to be no higher argument about which procedure one should adopt, other than which set of presumptions seem more appealing.

This presents a problem for the person or organisation who feels the need for prescriptive help in decision problems; which framework should they adopt? It is possible that the answer to this meta-question is likely to be determined by cultural issues - much as the choice of a religion (or none) seems to be. It can be argued that the reason people adopt a religion, particularly if they convert to it in later life, is that they may be confronted by a convincing and passionate evangelist, and also come to know a community of satisfied believers. A similar phenomenon might be operating in the choice of a decision making framework; if confronted with an enthusiastic evangelist for a particular framework, and a community of satisfied practitioners, the new user may well accept the plausible arguments that are put forward. Manager X uses framework A because a proponent of framework A makes a presentation of framework A just at the moment that manager X has an important decision to make. Similarly a manager in company Y uses framework B, not because it is intrinsically better than framework A, but merely because the company has used framework B satisfactorily before.

Many of us who are educated in the rational decision making tradition might find this conclusion distasteful. Can it be that there is no one correct way to good decision-making? Can it just be a matter of adopting the most attractive set of presumptions? Once one accepts that this is indeed the case, all sorts of consequences follow, in particular that, as aids to decision, there may even be sensible alternatives to the rational decision making model outlined at the beginning of this section. We now go on to look at some of these in the next section.

3.0 Prescription outside the rational decision making paradigm

To many people, including most decision analysts, the rational decision making model seems self-evidently sensible, and yet resolute adherence to it has led to what has been described as a 'crisis' in Management Science (Dando and Bennett (1981)). There is a widespread intuition that there is something wrong with the standard model. For example, Deber et al (1990) have recently suggested that there are occasions when we should approach medical decision making using rules other than the maximising of expected utility. This is not the place to answer Deber et al. in detail; the important point is that there are people attempting to apply rational decision making procedures in practical contexts, and finding difficulty in so doing. As another example, an interesting recent contribution by Zhang et al. (1989) comments that attempts to apply the rational decision making paradigm in China often results in great difficulties, in that Chinese people are uneasy with the notion of methodically following the steps of the

paradigm; they suggest that the essentially Western notions that underlie rational decision making fit ill with Chinese culture.

It would not be correct to argue that concern over the rational decision making paradigm is new. For example, both Simon (1976) and March (1978) argue that there is something not quite satisfactory about a strenuous search for rational decision-making - as March points out, 'sin' is doing what we don't want to do, and this is rife! We seem to need to do what we don't want to do, clearly not following the rational decision making paradigm. As another example, the work of the RAND Corporation in attempting to help New York City with its decision-making problems was only partially successful, and was spectacularly unsuccessful in dealing with the organisation of public health. RAND were renowned for their adherence to the rational decision making paradigm, and yet it did not work. As Greenberger et al. (1976) show, this failure could be explained by the conflicting goals of the people involved in this area of work, by the difficulty involved in forecasting the consequences of action, and by the collegial nature of decision-making.

Recently, however, some interesting alternatives to the rational decision making paradigm have emerged. In this section we will describe some of these alternatives and analyse the presumptions they make - and in particular whether they presume to prescribe what decision to make. In section 3.1 we summarise alternatives to the rational decision making paradigm as descriptions of decision making and explore their potential for prescription. Then, in section 3.2, we investigate the Soft Systems movement which has recently emerged, principally in Britain, to see whether this alternative to the rational decision making paradigm leads to satisfactory prescription.

3.1 Prescription through alternative descriptions of decision making

We start from a recent exposition by Lipshitz (1990), who shows that of the many frameworks that have been put forward to describe decision making, only some of them follow the paradigm of rational decision making which we set out at the beginning of section 2. Moreover, he goes on to argue that there are circumstances in which it would be proper to use some of the methods which do not fall into the rational decision making framework. He is, thus, recommending prescriptive methods which differ markedly from those discussed in section 2 of this paper. In the next few paragraphs we shall summarise Lipshitz's argument.

Lipshitz analyses 11 different descriptive explanations of decision making, and concludes that they may be divided into three classes. The first class he refers to as Consequential Choice. In a Consequential Choice theory of decision making, the decision maker makes a careful comparison between a set of options at a particular time, and evaluates these options in the light of their possible future consequences. What we have referred to as standard decision theory is a Consequential Choice theory; so is The Analytic Hierarchy Process. In fact all versions of the rational decision making paradigm, as defined by the five stages outlined in section 2 above, are Consequential Choice theories. What is of interest to us here are the other two classes, which contain theories which differ from the rational decision making paradigm.

The second of Lipshitz's classes contains theories referred to as Matching theories. Here a decision maker only considers one option; that option is arrived at by assessing the situation, and determining the appropriate action. This idea needs further explication. The theories included in this class are the Obligatory Action theory of March (1981), the script processing theory of Abelson (1976) and the Story-based Decision Making theory of Pennington et al. (1986). Lipshitz's classification scheme is not quite exclusive and exhaustive, since two theories have characteristics of more than one class. One of these, the Recognition-Primed Decisions theory of Klein (1989) is in most respects a Matching theory, however. In fact, a description of Klein's theory may help to explain the characteristics of Lipshitz's class of Matching theories.

Klein carried out an extensive study of people actually making decisions, particularly under time pressure. It included fire chiefs, making decisions at the site of a fire, tank platoon leaders and design engineers. To quote from Klein (1989):

"In conducting 150 interviews we found that experienced decision makers can usually recognise ways that situations are typical, including typical responses. Relatively few decisions are made using analytical processes such as generating a variety of options and contrasting their strengths and weaknesses."

An important element of Klein's model, then, is the presumption that when facing a decision, his studied decision-makers first recognised, and appropriately classified the situation, and then deduced the appropriate action. Very rarely did they deliberate about the advantages and disadvantages of different actions. Rather, once they knew what type of problem they had encountered, they also knew what the 'correct' response should be.

Klein's theory is an interesting one, although we should stress that he has not yet proved that his theory actually describes what goes on in the minds of his decision-makers. One would like to see further rigorous experimentation to confirm that his theory does indeed describe what happens, before adopting it as a good model of the world. We include it, nonetheless, because of its radical distinction from rational decision-making theories.

Before describing Lipshitz's third class of theories, we return to the theme of this paper by asking whether Matching theories, developed as descriptions, are candidates for prescription. Both Klein (1989) and Lipshitz (1990) believe they are. Klein suggests that if efficient decision makers acting under time constraints do behave according to his theory, and the result is satisfactory decision making, then perhaps decision aids which help people to recognise situations, and to prompt with the appropriate response would be better than those which try to do a very quick version of the rational decision making approach. There may be a lesson here for those who construct decision support systems for decision makers under time pressure, such as those in the financial services sector, or the military. Lipshitz argues that theories in each of his three classes might be suitable for prescription, depending on the circumstances.

Let us move on to describe Lipshitz's third class, that of Reassessment. Theories in this class are categorised by the presumption that many people approach decision making with at least a partial commitment to a particular action, and that the cognitive processes involved are mainly

the reassessment of this action to construct an argument for its validity, rather than a comparison of the action with any other. The theories put in this class by Lipshitz are the Escalation of Commitment (Staw et al. (1987)), the Search for Dominance Structures (Montgomery (1989)) and Theories of Action (Argyris et al. (1985)). It is clear from many empirical examples that Reassessment is a plausible description of what goes on in many choice situations. Lipshitz (1990) gives the following, which was written by one of his subjects.

"Once I planned a one-day trip for the coming weekend. On Saturday morning, as I was about to start, I discovered that there was some problem with the car. I could not figure the exact nature of the problem but it was obvious that a long drive might be risky. The weather was outstanding, and I felt that it would be a pity to miss it. I decided to trust to 'lady luck', take my chances and go as planned. I could of course, change my plans and avoid the risk of getting stuck in the middle of the trip".

Although Lipshitz shows how this decision could be described by using all three kinds of theory, it is the Reassessment interpretation which seems most satisfactory. In approaching the decision on how to spend the day, the decision maker finds an argument to support what he had already planned to do. He does not seem to evaluate going against not going as two potentially possible actions; nor does he seem to evaluate the situation (good day, but potentially dodgy car) to reach into his memory for the appropriate response. That is not to say that he spends no time in analysis; he does think about not going, and it is a possible course of action, but his cognitive processes seem to be spent more on justifying a precommitment than on a dispassionate search for the 'best' solution.

Now we must ask whether we should share Lipshitz's view that the Reassessment class of descriptions of decision making could be used as a prescription. It has long been recognised, of course, that one of the perceived values of decision analysis, and indeed, of operations research more generally, is its ability to justify a precommitment to a particular course of action, and to persuade others of its validity. Perhaps a sensible prescription could be that, if one comes to a decision situation with a fairly strong instinct as to the correct course of action, a satisfactory prescription would be to construct an argument to support that course of action, only moving from it to some other if the strongest argument is frankly implausible. Against this, the disciple of rational decision making would argue, of course, that such a prescriptive approach would not attack, and seek to rectify the prejudices of the decision maker, and his potential failure to see the logical implications of the possible courses of action open to him. Yet, if we accept one measure for the value of a prescriptive theory of decision making to be its acceptability to decision makers (as Brown (1989) does in his section 2.3), then it is clear that many people do find a Reassessment approach to decision making satisfactory. We conclude by agreeing with Lipshitz:

"In conclusion, the best answer to the question of (which class, or mode, of decision making is superior prescriptively)...is that different modes may serve better in different circumstances. Decision analysts

should therefore entertain all three in their analyses." (Lipshitz (1990) p23)

Quite what a prescriptive decision aid, based on a Reassessment theory of decision making would look like has yet to be determined, but it deserves serious investigation.

In summary, then, Lipshitz's analysis shows that there are numbers of descriptive theories of decision making which do not conform to the rational decision making paradigm; and there are good arguments that some of these could be turned into satisfactory prescriptive approaches. Whether the presumptions of such approaches are more or less acceptable to decision makers than those of rational decision making prescriptions, has yet to be determined.

3.2 The Presumptions of Prescription in Soft Systems Theories

In the last twenty years several distinct approaches to the analysis of decision situations have emerged in the United Kingdom. They are well described in Rosenhead (1989). These approaches have come to be referred to as 'soft systems' approaches, following the terminology originally due to Checkland (1981). In this section we will discuss the nature of prescription within these approaches, mentioning in particular the Soft Systems Methodology (SSM) of Checkland, and the Strategic Options and Development Analysis (SODA) approach of Eden. Both are procedures for helping an organisation, rather than an individual, to move toward action. It could, thus, be argued that we are changing the subject here, moving from individual to group decision-making. It is a curious fact however, that most of the reported applications of the rational decision-making paradigm are to organisational rather than individual decision-making. Prescription for organisations would thus seem to be an important part of prescriptive decision-making, and we make no apology for describing these new approaches to prescription here.

3.2.1 Checkland's Soft Systems Methodology

Checkland's Soft Systems Methodology is perhaps the best known of these approaches. We have not the space to provide a detailed statement of SSM here; good accounts are given in Checkland (1981), Wilson (1984), or Checkland (1989). It is necessary, however, to describe the broad structure of Checkland's approach. SSM emerged from a dissatisfaction with the rational decision making paradigm, particularly as practiced in solving complex problems in organisations. Unlike the other methods discussed above, therefore, it is concerned not with an individual taking a single decision, but with organisational decision making, where many interlocking decisions might be involved. Its importance for us however, is that it is a procedure for leading to purposeful action. To quote Checkland (1989) p78:

"...SSM is a learning system. The learning is about a complex problematical human situation, and leads to taking purposeful action in the situation aimed at improvement, action which seems sensible to those concerned".

In an SSM analysis the decision makers construct alternative representations of their situation (recognising that different people will have different perceptions of the problem). These representations are usually expressed as diagrams on paper, full of arrows connecting bubbles in which the words describe concepts. They then compare those representations with reality, to define differences, which then lead to the representations being changed. After iterating this procedure, a desirable course of action emerges. There is no formal evaluation of alternatives, nor of identifying the 'correct' solution for the particular situation being considered, nor even of reassessing a solution for which prior commitment already existed. No mathematical model is constructed, and there is no mathematical analysis. The method is useful for creating consensus and commitment within an organisation, rather than for 'prescribing what to do'. It can be criticised for being inherently conservative, reinforcing existing asymmetries in the power distribution. Nonetheless, if the purpose of prescription is to lead to action which is accepted by those involved, SSM clearly does this; there are many satisfied users of the Soft Systems Methodology (Checkland, 1981).

Perhaps the aspirations of SSM are best captured by quoting from Checkland's conclusions of a case study describing an application of the methodology: (Checkland (1989), p117-118).

"The study was 'successful' in that it made a significant contribution to bringing about change in a particular real situation. But the account of this 'successful' study is completely defenceless against assertions that it should have been done better, or quicker, or that some other approach would have been even more successful. Since the same human situation cannot be investigated twice, methodology is undecidable..... Ultimately the only powerful argument in favour of SSM is that it has been found to be transferable to people beyond those who developed it, has been used in several hundred studies and is still developing."

SSM makes modest presumptions. It is not presumed that an application of the method will lead to a clear, and unambiguous recommendation for action. What it does often lead to, however, although this is not guaranteed, is a better, and more widely shared, understanding of the problem situation, and a consequential commitment to action. This is a different kind of prescription from that issuing from the rational decision making paradigm. It is a lesser kind, but perhaps a more acceptable kind.

3.2.2 Eden's Strategic Options Development and Analysis Approach

Eden's SODA approach has several characteristics in common with SSM, but is distinct in several important ways, as both Eden and Checkland are keen to point out. Good accounts of SODA are given in Eden (1989a and b). Eden's development of his approach seems to have been born out of similar experiences to those which led Checkland to develop SSM. Both Eden and Checkland, after a technical university education, worked in industry on the analysis of industrial decision making problems, and both seem to have become dissatisfied with the methodology of Operational Research, based as it was on the rational decision making paradigm. Both joined new universities founded in the UK in the 60's, and developed their new

approaches from their universities while working in a consulting capacity on practical problems. The approaches have been designed, therefore, to meet unsatisfactory aspects of current procedure, rather than by theorists seeking to improve the intellectual structure of decision making approaches.

The heart of the SODA methodology is the construction of cognitive maps which represent the perceptions of the people involved in the decision concerning the nature of the situation that they face. These maps are based on Kelly's theory of personal constructs. A map is created from each member of the decision making team, and the consultant works with the client to aggregate the maps. Then, to quote Eden (1989a p30):

"There are two principal ways of working on a map with the client. The first is to explore the goal system further, and then gradually work down the map towards increasingly detailed options to achieve goals. Alternatively, one can start from the detailed options and gradually work up the map towards goals by exploring each concept in turn as a potential option."

Once these working sessions have led to a map which the participants agree is a satisfactory representation of the problem, the way is open for more traditional forms of analysis - or alternatively, as in SSM, action will have become obvious.

Eden stresses the importance of understanding that SODA is designed as an approach for a consultant involved in an organisational decision problem. It does not, therefore, offer any prescriptions for individuals facing choice situations. SODA also stresses that managers' understanding of the world in which they are acting may well be created during the process of analysis. If SODA is at all prescriptive, therefore, it prescribes an approach to the process of analysis, rather than what 'the right answer' should be.

As with SSM, the presumptions of prescription are small. There is no desire to establish a procedure which will provide a prescription of what to do. Rather, all that is provided is a prescription on how to approach decision problems in organisations. Once again, however, Eden reports many satisfied users of his approach (personal communication). It is a working methodology for approaching organisational decision making, which appears to lead to satisfactory conclusions.

4.0 Conclusions

This paper has been concerned with the search for good prescriptive methods for decision-making. That search seems to have been for an answer to the question: "What is the correct procedure to adopt when making a decision?" We have argued that attempts to modify the standard normative decision theory model either by making prescriptive additions to it (2.1), or by modifying its axioms (2.2), do not seem to be likely to answer this question. Other prescriptive schemes within the rational decision making paradigm (2.3) cannot be dismissed as wrong, although they do have some undesirable properties. Yet more potential prescriptive schemes may emerge from other descriptive models of decision-making (3.1), or from soft systems methodology (3.2).

We conclude, therefore, that it is an unsupportable presumption that the search for a unique correct prescriptive procedure for decision-making can be satisfactorily concluded. This means that the choice of a prescriptive scheme is a matter of taste. Is this all that can be said? Are there no guides to how one ought to go about decision-making?

This would be far too negative a conclusion. The dominance of the human species in the world has been due to the human ability to analyse and anticipate the consequences of action. The rational decision making paradigm is perhaps the most abstract statement of this ability which has yet been devised. The fact that it cannot be seen as the 'One True Way' to decision making, does not mean that it is not a useful procedure on certain problems and in certain contexts. It is the acontextual nature of the paradigm that is one of its weakest points. Real decision making takes place in the context of the perceptions of the people involved, their ways of understanding those perceptions (which differ from person to person and culture to culture) and the power structure of the human organisation in which the decision is made. It follows that the search for a prescriptive scheme must take this context into account.

The appropriate research question should not be what the correct way is to prescribe how to make a decision; it should be, rather, what a good prescriptive procedure should be in the particular context of the decision. It may be that for fire captains facing instant decisions, prescription should be based on a matching model of decision-making (see 3.1); for managers impressed by friends who have used the AHP before, and have it explained to them clearly, that they should use the AHP themselves (see 2.3); and for Chinese managers who have a culture based on very different understandings of the world from those produced by Western society, to use prescriptions related to their culture.

Before concluding our case for pluralism in prescription, we ought to address one potential criticism of our position. We have not, so far, mentioned the argument that the question of what is the best prescriptive procedure might be answered by experiment. Could we not try different prescriptions to see which give the best results? It does not take long to see the difficulties with this. Firstly there is the impossibility of repetition, so common in the social sciences; we cannot repeat a human decision making process, to test the effects of different approaches. Secondly, the valuation of whether a decision is good is determined by the prescriptive procedure used. If we adopt the rational decision making paradigm then we can either judge the goodness of a decision in the light of whether the goals, as defined by using the paradigm itself, have been fulfilled, or whether the process has been properly adhered to. Alternatively, if we adopt another paradigm, such as one of the matching models described in section 3.1, then again a good decision will be one where the process has been adhered to. The goodness of results is internal to the choice of paradigm, and so different paradigms cannot be compared in terms of the goodness of their results.

The argument in this paper is that the presumption that there is one true way to prescriptive decision-making is false. The research agenda must be that of seeking to relate good prescriptive approaches to different contexts. It may not prove possible to do this, but it is likely to be a more fruitful line of research and less presumptuous than the search for the imaginary Grail of a universally correct prescription.

References

- Abelson, R P, "Script Processing in Attitude Formation and Decision Making"; in Carrol, J S and Payne, J W, (Eds.), "Cognition and Behavior", Erlbaum, 1976.
- Argyris, C, Putnam, R and Smith, D M, "Action Science", Jossey Bass, 1985.
- Beach, L R, "Image Theory: Decision Making in Personal and Organizational Contexts", Wiley, 1990.
- Becker, J L and Sarin, R K, "Lottery-dependent Utility", Management Science, 33, 1367, 1987.
- Bell, D E and Farquhar, P H, "Perspectives on Utility Theory", Operations Research, 34, 179-183, 1986.
- Bell, D E, Raiffa, H and Tversky, A, "Decision Making: Descriptive, Normative and Prescriptive Interpretations", Cambridge University Press, 1988.
- Belton, V and Gear, T, "On a Shortcoming of Saaty's Method of Analytic Hierarchies", Omega, 13, 143-144, (1985).
- Brown, R V, "Do Managers Find Decision Theory Useful?", Harvard Business Review, 48, 78-89, 1970.
- Brown, R V, "Toward a Prescriptive Science and Technology of Decision-aiding", Annals of Operational Research, 19, 467-483, 1989.
- Checkland, P B, "Systems Thinking, Systems Practice", 1981.
- Checkland, P B, "Soft systems methodology", pp72-100 in Rosenhead, J (Ed), "Rational Analysis for a Problematic World", Wiley, 1989.
- Dando, M R and Bennett, P G, "A Kuhnian Crisis in Management Science?", Journal of the Operational Research Society, 32, 91-103, 1981.
- Deber, R and Goel, V, "Using Explicit Decision Rules to Manage Issues of Justice, Risk and Ethics in Decision Analysis: When is it not Rational to Maximise Expected Utility", Medical Decision Making, 10, 181-194, 1990.
- Dyer, J S, "Remarks on the Analytic Hierarchy Process", Management Science, 30, 249-258, 1990a.
- Dyer, J S, "A Clarification of 'Remarks on the Analytic Hierarchy Process'", Management Science, 30, 274-275, 1990b.
- Eden, C, "Using Cognitive Mapping for Strategic Options Development and Analysis", pp 21-42, in Rosenhead, J, (Ed), "Rational Analysis for a Problematic World", Wiley, 1989a.
- Eden, C, "SODA and Cognitive Mapping in Practice", pp 43-70, in Rosenhead, J, (Ed), "Rational Analysis for a Problematic World", Wiley, 1989b.
- Einhorn, H J and Hogarth, R M, "Decision Making under Ambiguity", Journal of Business, 59, S225-S250, 1986.

Fischer, G W, "Prescriptive Decision Science: Problems and Opportunities", Annals of Operational Research, 19, 489-497, 1989.

Fischhoff, B, Slovic, P and Lichtenstein, S, "Knowing what you want: Measuring Labile Values", in Wallsten, T S, (Ed), Cognitive Processes in Choice and Decision Behavior, Erlbaum, 1980.

Fishburn, P C, "Generalization of Expected Utility Theories: A Survey of Recent Proposals", Annals of Operational Research, 19, 3-28, 1989.

French, S, "Readings in Decision Analysis", Chapman and Hall, 1989.

Greenberger, M, Crenson, M A and Crissey, B L, "Models in the Policy Process", Sage, 1976

Harker, P T and Vargas, L G, "Reply to 'Remarks on the Analytic Hierarchy Process' by J S Dyer'", Management Science, 30, 269-273, 1990.

Howard, R A, "Decision Analysis: Applied Decision Theory", in Hertz, D B and Melese, J, (Eds), Proceedings of the Fourth International Conference on Operational Research, New York, 1966.

Kahneman, D, Slovic, P and Tversky, A, "Judgement under Uncertainty: Heuristics and Biases", Cambridge University Press, 1982.

Kahneman, D and Tversky, A, "Prospect Theory: An Analysis of Decision under Risk", Econometrica, 47, 263-291, 1979.

Keeney, R L and Raiffa, H, "Decisions with Multiple Objectives: Preferences and Value Trade-offs", Wiley, 1976.

Keller L R, "The Role of Generalized Utility Theories in Descriptive, Prescriptive and Normative Decision Analysis", Information and Decision Technologies, 15, 259-271, 1989.

Klein, G A, "Recognition-primed Decisions", Advances in Man-Machine Systems Research, 5, 47-92, 1989.

Lindley, D V, Tversky, A and Brown, R V, "On the Reconciliation of Probability Assessments", Journal of the Royal Statistical Society, A142, 146-180, 1979.

Lipshitz, R, "Making Sense of Decision Making: Towards an Integrative Approach to Understanding and Improving Decisions", Working paper, Department of Psychology, Haifa University, Israel, 1990.

Machina, M J, "Choice under Uncertainty: Problems Solved and Unsolved", Economic Perspectives, 1, 121-154, 1987.

March, J G, "Rationality, Ambiguity and the Engineering of Choice", The Bell Journal of Economics, 9, 587-608, 1978.

March, J G, "Decisions in Organisations and Theories of Choice"; in Van de Ven, A and Joyce, W F, (eds.), "Perspectives on Organisation Design and Behaviour", Wiley, 1981.

McNeil, B J, Pauker, S G and Tversky, A, "On the Framing of Medical Decisions", chapter 26 of Bell, D E, Raiffa, H and Tversky, A, (Eds),

"Decision Making: Descriptive, Normative and Prescriptive Interpretations", Cambridge University Press, 1988.

Montgomery, H, "From Cognition to Action: The Search for Dominance in Decision Making", in Montgomery, H and Svenson, O, (Eds.), "Process and Structure in Human Decision Making", Wiley, 1989.

Pennington, N and Hastie, R, "Evidence Evaluation in Complex Decision Making", Journal of Personality and Social Psychology, 51, 242-258, (1986).

Raiffa, H, "Decision Analysis", Addison-Wesley, 1968.

Rosenhead, J, (Ed), "Rational Analysis for a Problematic World", Wiley, 1989.

Roy, B, "Methodologie Multicritere d'Aide a la Decision", Economica, 1985.

Saaty, T L, "The Analytic Hierarchy Process", McGraw-Hill, 1980.

Saaty, T L, "Axiomatic Foundations of the Analytic Hierarchy Process", Management Science, 32, 841-855, 1986.

Saaty, T L, "An Exposition of the AHP in reply to the paper 'Remarks on the Analytic Hierarchy Process'", Management Science, 30, 259-268, 1990.

Simon, H A, "Rationality as Process and as Product of Thought", Journal of the American Economic Association, 68, 1-16, 1978.

Simon, H A, "Reason in Human Affairs", Blackwell, 1983.

Staw, B M, and Ross, J, "Behavior in Escalation Situations", Research in Organisational Behaviour, 9, 39-78, 1987.

Ulvila, J W and Brown, R V, "Decision Analysis Comes of Age", Harvard Business Review, 60, 130-141, 1982.

von Winterfeldt, D, "A Re-examination of the Normative-Descriptive Distinction in Decision Analysis", Annals of Operational Research, 19, 499-502, 1989.

von Winterfeldt, D and Edwards, W, "Decision Analysis and Behavioral Research", Cambridge University Press, 1986.

Watson, S R and Buede, D M, "Decision Synthesis", Cambridge University Press, 1988.

Wilson, B, "Systems: Concepts, Methodologies and Applications", Wiley, 1984.

Winkler, R L, "Decision Modelling and Rational Choice: AHP and Utility Theory", Management Science, 30, 247-248, 1990.

Zeleny, M, "Multiple Criteria Decision Making", McGraw Hill, New York, 1982.

Zhang, B and Angell, I O, "Decision Support Systems in China: a Clash of Cultures", Working Paper #6, Department of Information Systems, London School of Economics, 1989.

WORKSHOP

PRESCRIPTIVE DECISION SCIENCE

Anna Vari (Hungary), Stephen Watson (U.K.),
and Rex Brown (USA), organizers

Unresolved problems standing in the way of effective use of decision aiding technology will be identified. These may include problems related to the technology itself, the facilitation process, the decision making environment, and the fit among these components of the decision making process. Special attention will be given to problems for which behavioral decision research has the potential to provide support. Relevant research to date will be reviewed; key topics for future research will be identified.

20 August, 1991

Session 1: Deficiencies in Decision-Aiding Technology and Promising Directions for Remedy

Topic Introductions:

Helmut Jungermann: Personal Decision Counseling: Counselors without clients?

Rakesh Sarin: Aiding Organizational Decision Making in the Era of Rapid Change: The Problems of Instability and Ambiguity.

Zur Shapira: Problems with Organizational Decision Aids.

Lawrence D. Phillips: Alternative Paradigms of Decision Aiding.

Discussion and Topic listing.

21 August, 1991

Session 2: Validating and Evaluating Decision Processes and Aiding Interventions

Topic Introductions:

Anna Vari: Multiple Attributes of Appropriate Decision Support Technology.

Oleg I. Larichev: Psychological Validation of Prescriptive Decision Aiding Methods.

Peter E. Polister: Necessity - The Mother of Indirect Validation: Necessary Utility Difference Orders to Test Analyses.

John W. Rohrbaugh: Competing Values Approach to Group Decision Support Effectiveness.

Discussion and topic listing.

22 August, 1991

Session 3: Wrap-up

Anna Vari: Unfinished topics.

Stephen Watson: Closure on a research agenda.

Discussion.

WORKSHOP

THE EFFECTS OF TIME PRESSURE AND STRESS ON JUDGMENT/DECISION MAKING

John A. MAULE (UK), organizer

One of the most common complaints of professional decision makers is that they have to make decisions under time constraints. Despite this, there is comparatively little theory or research in this area. This workshop explores how the imposition of time pressure and other stressors affect the nature of judgment and decision making.

Invited participants:

- L. Benson III (Sweden)
- A. Edland (Sweden)
- U. Lundberg (Sweden)
- H. Mano (USA)
- O. Svenson (Sweden)
- B. Verplanken (Netherlands)
- A. Wearing (Australia)

WORKSHOP

RISK COMMUNICATION

Bernd Rohrmann (Germany), organizer

Risk communication, related to both man-made risk sources and natural hazards, is a rapidly growing field of scientific interest. Relevant tasks include presentation and explanation of risk information; influencing risk-related behavior; and designing cooperative and participative decision making processes. Topics to be discussed at the workshop include: significant substantive areas for risk communication programs; the role of relevant actors; the relevance of decision theory; social and political contexts; shortcomings of present approaches; and evaluating the effectiveness of communication strategies.

Invited participants:

A. Bostrom (USA)
W. Brun (Norway)
K. Burgemeister (Germany)
S. Femers (Germany)
S. Levchenkov (USSR)
N. Pidgeon (UK)
O. Renn (USA)
G. Ruhrmann (Germany)
H. Schütz (Germany)
O. Svenson (Sweden)
C. Vlek (Netherlands)
P. Wiedemann (Germany)

WORKSHOP

ADAPTIVE AND CONSTRUCTIVE DECISION MAKING

J. Edward Russo (USA), organizer

Adaptive decision making refers to how decision makers adapt their decision processes to relevant characteristics of a specific decision situation, such as the number of alternatives and attributes, decision importance, and so on. Constructive decision making refers to the process by which decision makers dynamically link or combine various decision rules and select appropriate inputs in response to evolving decision circumstances. The focus of the workshop will be primarily descriptive, but will also consider notions of optimality and techniques for aiding decisions.

Invited participants:

O. Huber (Switzerland)
E. Johnson (USA)
F. Leclerc (USA)
M. Lewicka (Poland)
D. Mitchell (USA)
H. Montgomery (Sweden)
R. Ranyard (UK)
I. Yaniv (USA)

WORKSHOP

KNOWLEDGE ACQUISITION FOR EXPERT SYSTEMS

Franz Schmalhofer (Germany), organizer

The goal of the workshop is to build a bridge between expert systems research and behavioral investigations of human judgment and decision making expertise. The engineering task of developing an expert system has significant commonalities with modeling human information processing behavior. The workshop will consist of three sections: (1) a discussion of basic objectives and principles for modeling the behavior of humans and machines; (2) an evaluation of current knowledge acquisition methodologies in terms of these objectives and principles; and (3) an evaluation of knowledge acquisition techniques for practical applications.

(On Thursday August 22nd this workshop will meet jointly with the workshop Decision Making Perspectives on Experts).

Invited participants:

D. Janetzko (Germany)
W. Karbach (Germany)
O. Larichev (USSR)
M. Manago (France)
S. Rebric (USSR)
J. Shanteau (USA)
G. Strube (Germany)

WORKSHOP

DECISION MAKING PERSPECTIVES ON EXPERTS

James Shanteau (USA), organizer

Results of decision research have ranged from finding little performance gain in experts to finding consistently superior performance. The goal of the workshop is to explore the sources of these differences and to attempt some resolution. Issues to be addressed will include definitions of expertise; criteria for evaluating expert performance; the adequacy of traditional J/DM approaches for studying experts; implications of research on experts for broader decision making concerns; differences in experts' abilities across domains; implications of research on experts for the development of expert systems; and key issues for future research.

(On Thursday August 22nd this workshop will meet jointly with the workshop Knowledge Acquisition for Expert Systems).

Invited participants:

M. Bar-Hillel (Israel)
H. Barron (USA)
F. Bolger (UK)
E.E.J. de Bruyn (Netherlands)
T. Englander (Hungary)
J. Mumpower (USA)
H. Purkitt (USA)
J.H.L. van den Bercken (Netherlands)
W. Wagenaar (Netherlands)
W. Waller (USA)
A.Y. Terekhina (USSR)
G. Wright (UK)
A. Wright (USA)

EFFECTS OF DYADIC COLLUSION ON CONFIDENCE IN CONTROL JUDGEMENTS

Peter Ayton

Nigel Harvey

City of London Polytechnic

University College London

This paper reports a series of experiments designed to investigate the influence of social interaction between pairs of individuals on their ability to control a dynamic system and on their confidence in their control judgements. In particular we are concerned with whether consensus decisions made after discussion are better than the average of individual decisions made before it.

We hypothesised that consensus decisions, made after discussion, might be better than the average of individual decisions made beforehand for two reasons.

First if knowledge used for control is verbalisable then better controllers should be able to provide convincing reasons for why the consensus should be closer to their individual decisions. In fact, previous work with dynamic systems suggests that control knowledge may not be verbalisable (Berry and Broadbent, 1984).

Second, if confidence in decisions provides some reflection of their accuracy then, even if reasons for decisions are not verbalisable, stronger arguments based on greater confidence would move the consensus towards the better individual decision.

We found that judgements of response efficacy were overconfident and made more so by the act of reaching a consensus. Where individuals initially disagree, the consensus decision is usually closer to the initial individual decision that is more correct. Thus, collusion between subjects can result in the expression and mutual appreciation of knowledge concerning the control task. The consensus decision is also usually closer to the initial individual response that is more confident. However the influence of confidence and correctness on the consensus are independent. Thus when two individuals disagree, differences in their levels of confidence is no guide as to which is the better response. But, as the consensus is more influenced by the more confident response, it will be more overconfident.

THE DISJUNCTION FALLACY AND THE REPRESENTATIVENESS HYPOTHESIS

Maya Bar-Hillel and Efrat Neter

The Hebrew University, Jerusalem

Formally, a conjunction fallacy and a disjunction fallacy cannot be distinguished. Intuitively, they are different. Furthermore, the pragmatics of disjunctions differ from those of conjunctions. This study demonstrates a disjunction fallacy, analogous to Kahneman and Tversky's conjunction fallacy. Subjects received brief case descriptions, and ordered seven categories according to one of four criteria: probability of membership, willingness to bet on membership, readiness to predict membership, and resemblance to the category.

Regarding the first two criteria, ranking a category as higher than its superordinate category (i.e., ranking a category as more probable than its superordinate category, or indicating a greater willingness to bet on it than on its superordinate category) is fallacious. But ranking a category more typical of the description than its superordinate, or indicating a greater readiness to predict membership in a category than in its superordinate is not necessarily fallacious. The relationship between these four ranking criteria is discussed. The findings are explained by reference to basic level versus superordinate categories and to principles of similarity judgment.

PARTIAL INFORMATION ABOUT ATTRIBUTE WEIGHTS IN SELECTING BEST
MULTIATTRIBUTE ALTERNATIVES

F. Hutton Barron
University of Alabama, USA

Weighted additive value models are widely used to rank alternatives in multiple attribute decision making under certainty. Some researchers have suggested that approximate attribute weights may provide sufficient accuracy for particular decisions (e.g. selecting the "best" alternative, or ranking all alternatives). Use of approximate weights would greatly simplify decision analysis since detailed weight elicitation could be avoided.

This paper examines the degree to which ranked weights can be used to identify a best alternative, or failing uniqueness, prescribes an easily implemented rule for selecting a "best" alternative. The study compares the ranking procedure of Kirkwood and Sarin (1985) and a linear programming procedure. These procedures are motivated using three-dimensional graphical analyses - corresponding to three attributes, and are subsequently used to re-analyze five decision analysis applications having four to nine alternatives, and twelve to twenty-eight attributes. Case data have been provided by Lawrence Phillips.

The linear programming procedure is best able to incorporate partial information and thus most compatible behaviorally. Consequently it is most efficient in identifying a single best alternative, or the smallest set of "admissible" alternatives. The prescribed rule uses as weights the centroid of the feasible region.

In the absence of a unique best alternative consistent with the available partial information, the value of the partial information can be determined using an "expected gain from weight precision" (EGWP) measure, analogous to "expected value of perfect information" in decision analysis under uncertainty.

INFORMATION SEEKING AND COMPLEXITY OF DECISION STRATEGY
AS A FUNCTION OF DECISION IMPORTANCE

GIZELA BARTOZAK

(Institute of Sport, Department of Psychology,
ul. Trylogii 2/16, Warsaw, Poland)

This study examines what factors favor a complex or a simple information utilization and decision strategy.

It is known that certain situational factors can influence the information seeking and the choice of a decision strategy. Here it is proposed that perceived decision importance could be such a factor:

- the more important a decision is, the more information is used in order to reach it,
- the more important a decision is, the more complex decision strategy is used.

The specific meaning of the concept 'decision importance' is involved. It has been assumed that in the situation, in which a decision maker awards a reward and/or punishment to subordinates, a decision importance is connected with a value (positive vs. negative) and strength of reinforcement.

Thus, it was hypothesized that:

- 1) While punishing hard or rewarding highly, a decision maker uses many information and uses complex decision strategy.
- 2) While punishing, he/she uses more information and uses more complex strategies than he/she awards a reward.

124 subjects who distributes rewards and punishments to others professionally (like medium level managers and coaches) were selected. They were presented with 24 descriptions of subordinates who were characterized by 5 attributes. They had to choose 10 out of 24 persons for 2 rewards (high vs. low) x 2 punishment (high vs. low). A process tracing technique was used.

The data supported the first hypothesis about information seeking and did not support the second one. It was found that all subjects made their decision mostly with simple strategies. These results are discussed in terms of reduction of cognitive strain and of bounded rationality.

THE PSYCHOLOGY OF PRE-CHOICE SCREENING OF OPTIONS

Lee Roy Beach and Richard E. Potter

University of Arizona, Tucson

Recent experiments cast light upon pre-choice screening of decision options. They show that exclusion of an option from the final set from which a choice will be made is governed by a noncompensatory process that depends solely upon the weighted number of criteria that the option fails to meet, untempered by the number that it successfully meets. Moreover, the information used in screening appears to play no role in subsequent pre-choice evaluation of the options in the choice set: it is as though having been used for screening, the information has been 'used up.' In some circumstances this means that the chosen option is not the one that would have been chosen had all information been considered. In addition, when options in the choice set turn out to be unavailable (e.g., the job candidates on one's final list turn out to have taken other jobs), decision makers strongly prefer to start over with a new pool of options. If no new options are forthcoming and a choice must be made from among the previously rejected (screened out) options, decision makers both raise their rejection thresholds and reduce the weight the place on the various screening criteria in order to create a new choice set. The practical and theoretical implications are discussed.

PREDICTING DECISION DIFFICULTY FROM ATTRIBUTE CLASSIFICATION

Jane Beattie and Sema Barlas

Center for Decision Research, University of Chicago, USA

Undoubtedly some decisions are more difficult than others. Some causes of decision difficulty have received considerable attention (e.g., number of alternatives, time pressure, distance from indifference point). We examine a cause of decision difficulty that has been largely unexplored: the finding that subjects perceive some attributes as more difficult to trade off than others. Generalized, decompositional utility models (e.g., MAUT, SEU, PT) assume that type of attribute has no effect on the decision process or outcome. Yet introspectively some attributes (e.g., "time" and "money") seem easier to trade off than others (e.g., "lives" and "money"). We present two experiments that investigate whether the difficulty of the decision can be predicted from a classification of attributes, and from other features of the decision.

In Experiment 1, 64 subjects considered tradeoffs between alternatives from different proposed categories: commodities (e.g., cameras); non-commodities (e.g., health); and currencies (e.g., \$). Pairs of alternatives were presented as tradeoffs, each embedded in a story. The level on the alternatives (e.g., number of \$) was unspecified, as we were interested in the difficulty of comparing the attributes *per se*, rather than the difficulty of comparing alternatives close to one's indifference point. Subjects first rated each tradeoff for its difficulty, then rated it on seven features that might influence tradeoff difficulty (e.g., morality, attribute similarity, attribute importance). Multi-dimensional scaling revealed that the hypothesized categories were psychologically valid. These categories also predicted the difficulty of the decision: decisions involving commodities were rated significantly easier than those involving other goods. Tradeoffs within a category were significantly harder than those involving items drawn from different categories. The feature scales also were useful in predicting difficulty, as revealed by a principal components analysis.

Experiment 2 (currently in progress) replicates Experiment 1, with the difference that subjects are asked to make choices between the alternatives, in addition to rating the difficulty of the decision. Using a method of Slovic (1975), subjects are asked to choose between pairs of equally valued alternatives (thus controlling for possible utility differences between alternatives). We will investigate whether reaction times are longer for the more difficult decisions, and whether subjects appear to have rules for choosing between alternatives from different categories (e.g., "never choose a commodity over a non-commodity"). Such rules would ease the decision maker's task in deciding between categories, but not within categories. We believe this research has important implications for a psychological theory of the representation and comparison of attributes, and possible application to design of decision aids for "difficult" tradeoffs.

Framing and Mental Accounting in Multiple Play Gambles

Shlomo Benartzi and Richard Thaler

Cornell University, Ithaca NY 14853, USA

Since Paul Samuelson's (1963) early article, a question that has intrigued decision researchers is the role of repeated plays in risky choice (e.g., Lopes, 1981; Keren and Wagenaar, 1987). Are prospects considered individually, or integrated with the rest of an individual's portfolio? This paper takes a new look at this problem.

Subjects are presented with a series of repeated gambles of the following type: Choose between a certain amount z , or the gamble: $(x, p; y, 1-p)$ repeated N times. This is called a segregated presentation of the gamble, since the gamble is described as a single event to be repeated many times. The gambles are constructed so that they all produce virtually identical distributions of expected outcomes. Subjects are then presented with another choice between a certain amount and one of the previous gambles, however in this case the gamble is described in terms of the distribution of possible outcomes (either graphically, or via a table). This is an integrated presentation format. We find that for some classes of gambles, there is a significant discrepancy between the choices in the two frames. Specifically, when the individual gamble is mixed, that is one of the outcomes is positive and one is negative, and the gamble has positive expected value, then risky choice is much more common in the integrated frame than in the segregated frame. This result is explained using three concepts from behavioral decision research: the law of small numbers, loss aversion, and mental accounting.

The law of small numbers suggests that people underestimate the power of repeated play to reduce the probability of a loss after a large number of repetitions of a positive expected value gamble. We confirm the existence of this effect by asking other subjects to estimate the probability of losing money after N plays of the gamble. However, this effect is not sufficient to explain the results, since we only obtain the choice discrepancy for mixed gambles, although the estimated distributions of outcomes are independent of the sign of the outcomes of the gambles. (The effect is weaker or non-existent for strictly positive gambles.) This suggests that loss aversion, the heightened sensitivity to losses suggested by Kahneman and Tversky's prospect theory, is also important to understanding the phenomenon. The role of the framing effects is also related to the mental accounting treatment of gambling suggested by Thaler and Johnson (1990).

EVALUATING OPTIONS BY COMBINED QUALITATIVE AND QUANTITATIVE METHODS

Marko Bohanec⁽¹⁾, Vladislav Rajkovič^(2,1), Božo Urh⁽¹⁾

(1) "Jožef Stefan" Institute, Jamova 39, YU-61111 Ljubljana, Yugoslavia

(2) University of Maribor, Faculty of Organizational Sciences, Kranj, Yugoslavia

The traditional approach to multi-attribute decision making involves *quantitative* concepts such as probabilities, utilities and weights. Recently, several *qualitative* knowledge-based approaches have been developed that use quite different components: rules, examples, qualitative attributes and qualitative models. They can provide some advantageous features such as an explicit articulation of knowledge, understandable knowledge representation, qualitative reasoning and explanation of the suggested solutions. On the other hand, there are some useful features of quantitative decision-making methods that are difficult to obtain when approached qualitatively. These include option ranking and option evaluation that is sensitive to small variations of attribute values. It seems that a reasonable *combination* of the two approaches is needed to provide this type of functionality.

This paper presents an *extension* of a qualitative multi-attribute decision-making method named DECMAX with some quantitative elements that are used for the *evaluation* of options. In addition to the qualitative evaluation that was used so far, each option is separately (but consistently) evaluated by a quantitative method. Therefore, two different evaluation results are obtained for each option: a qualitative *class* and quantitative *utility*. The utility extends the functionality of DECMAX with ranking of options within the class and extended sensitivity to small variations of attribute values. The quantitative evaluation method is derived directly from the (qualitative) knowledge base, so no additional effort is required for the extended functionality.

In the paper, two methods for the construction of quantitative evaluation methods from rules in the knowledge base are presented and discussed. Their performance is illustrated by an application in the field of personnel management.

Context effects in probability estimates:
the effect of the presence of related categories

Nicolao Bonini
University of Trieste, Italy

Els C.M. van Schie
University of Amsterdam, the Netherlands

Dipartimento di Psicologia
Via dell'Università 7
I-34100 TRIESTE
Italy

Two experiments will be presented on context effects in probability estimates. More specifically we will investigate the effects of the presence vs. absence of related causes on probability estimates. Subjects (students from the University of Amsterdam, $n=61$) were presented with a series of causes of death and a catch-all category. Results show a general overestimation of the presented causes and an underestimation of the catch-all category. Moreover, if a specific cause is presented alongside related causes, estimates for the cause are reduced as compared to a situation in which this cause is presented in the context of dissimilar other causes. These findings were obtained with two different probability elicitation techniques and suggest that people adopt relatively flexible definitions of possible causes, i.e. redefine each causal category on the basis of its immediate environment. The first elicitation technique is based on Fischhoff (1978) and simply asks subjects to give a probability estimate between 0 and 100 for each cause. Subjects were told that their estimates should add up to 100. The second technique is based on Gettys et al. (1986) and is a magnitude estimation technique in which the most likely cause receives a value of a 100, next subjects are asked to assign lesser numbers to the remaining alternatives.

The second experiment showed that providing subcategories of a general causal category which also have to be rated, tends to result in higher estimates summed over these causes as compared to the estimate for the corresponding more general causal category. A total of 101 students from the University of Trieste participated in this study. These findings are discussed in the context of availability and the anchoring and adjustment heuristic. In this second experiment findings were again obtained with two different probability elicitation techniques. Overall results indicate relatively unstable probability estimates due to judgmental context.

ASSESSING RISK COMMUNICATION DOCUMENTS:
COMPLETING AND CORRECTING MENTAL MODELS OF
HAZARDOUS PROCESSES

Ann Bostrom, Cynthia Atman, Baruch Fischhoff and M. Granger Morgan

Department of Engineering and Public Policy

Carnegie Mellon University

Pittsburgh, PA 15213-3890 USA

In the public domain, risk communication is often intended to inform individual decision-makers so that they can mitigate risks, or accept them knowingly and not unnecessarily (e.g., patient package inserts, warning labels on alcohol and cigarettes). For risk communication to educate the public effectively, its impact on people's *mental models* of the underlying hazardous processes must be evaluated. We propose a decision-analytic framework, called a mental models approach, for assessing risk communication effectiveness at educating the public to enable them to make decisions about that risk. Multiple evaluation methods are incorporated, including talk-aloud protocol analysis, problem solving, and the development of a true-false test that allows the expression of uncertainty. The true-false test is based on previous investigations of lay people's mental models of indoor radon and designed to evaluate subjects' expert knowledge and to diagnose their misconceptions. This mental models approach to evaluating documents is illustrated in empirical comparisons of three risk communication brochures on indoor radon.

THE INFORMATION ON WHICH A GENERAL PRACTITIONER OR A
PSYCHIATRIST DECIDES THAT PSYCHOLOGICAL DISORDERS ARE AT ISSU

Jozé Braspenning

GG&GD, subdivision of social and psychiatric epidemiology,
Valckenierstraat 2, Postbus 20244, 1000 HE Amsterdam

Studies on judgements within the framework of psychological disorders are often limited to circumstantial factors (marital status, social class, education) which influence the diagnosis of the physician. So far, little has been done to obtain the explicit criteria on which the physician basis his or her judgement. In order to acquire such explicit criteria, there have been performed two judgemental studies. In the first one several physicians were interviewed about the cues, which they use to recognize psychological disorders.

In the second study the question of the weight given to the different cues in the diagnosis (the judgement) was explored. To answer this question a so called, Judgement Analysis (JA) study was conducted. The purpose of a JA study is to describe (quantitatively) the relation between someone's judgement and the judgemental stimulus set. This shows how explicit cues (pieces of information) are used by judges (in our case the general practitioners and the psychiatrists) in the judgemental process.

In the current study "paper patients" were used for the judgement of psychological-somatic disorders. These "paper patients" or cues were judged on the degree with which a somatic or psychological disorder was present. By using a sufficient number of "paper patients", it is possible to described by means of a regression analysis the judgemental policy.

Preliminary results will be reported on the judgemental policies of general practitioners and psychiatrists using this method.

RESOURCE DILEMMAS WITH ENVIRONMENTAL UNCERTAINTY:
SIMULTANEOUS AND SEQUENTIAL PROTOCOLS OF PLAY

David V. Budescu

Amnon Rapoport

University of Haifa, Israel

Ramzi Suleiman

University of Arizona, USA

We consider a class of resource dilemmas with the following structure. Members of a group of size n are asked to share a common resource pool whose exact size, x , is not known. Rather, x is sampled randomly from a probability distribution which is common knowledge. Each member j ($j=1, \dots, n$) requests r_j from the resource pool. Requests are made either simultaneously or sequentially with perfect knowledge of the previous requests. If $(r_1 + r_2 + \dots + r_n) \leq x$, each member is granted his/her request; otherwise, group members get nothing.

We derive Nash equilibrium solutions for the games, examine their properties, and state several qualitative and quantitative implications. We then report the results of two experiments designed to (i) compare the two protocols of play, and (ii) compare the equilibrium models to psychological models incorporating the notion of focal points.

COGNITION AND BEHAVIOR IN ALTERNATING OFFER SEQUENTIAL BARGAINING

Colin Camerer

Eric J. Johnson

The Wharton School, University of Pennsylvania, Philadelphia, PA, 19104-
6371, USA; e-mail: Johnson 00@wharton.upenn.edu

Most game-theoretic applications in economics assume players are rational, mutually rational, and purely self-interested. In experimental studies of games one or more of these assumptions often appear to be violated because people do not play Nash or more refined equilibria. (Learning sometimes reduces violations, though not always.) It is usually difficult to tell from strategy choices alone which assumptions a subject is violating. Since game-theoretic solution concepts can be interpreted as algorithms for choosing a strategy, we can test the algorithms directly using a computer system that records what information in a game (e.g., which payoffs) a subject is looking at, and for how long, along with their choices. The information processing data are revealed preferences for information, which can be used to infer an unobservable thinking process.

We examine several games with this methodology, concentrating, for the purposes of this talk, on an alternating offer sequential bargaining game. Prior research clearly indicates that people do not conform to the game-theoretic prediction. Our data suggests that this is because people do not employ backward induction as a means of solving this problem. Our conclusions based on these and other studies suggest systematic departures from the game theory assumption of mutual and complete rationality. Further, the resulting view of behavior in games comes much closer in character to decision-making under risk.

BIASES AS AN EXPERTISE ELICITING TOOL:
ANCHORING EFFECTS IN KNOWLEDGE ASSESSEMENT

Jean-Paul CAVERNI & Jean-Luc PERIS

Centre de Recherche en Psychologie Cognitive
Université de Provence et CNRS, Aix-en-Provence (France)

The psychological reality of cognitive biases - such as anchoring-adjustment - has often been questioned, authors pointing out that these biases might be induced by tasks specificities or by experimental laboratory settings.

This paper presents an overview of just-completed experiments that were designed to explore the cognitive processes involved in knowledge assessment by experts (in this case, French language teachers grading texts written by pupils). Specific goals were to 1) check for the existence of the anchoring adjustment bias in a real world task of expertise, 2) find out the exact nature of this bias effect (over-evaluation versus under-evaluation), and 3) probe its underlying processes.

Results give evidence of an anchoring effect in a real world assessment task. In the absence of an external anchor, evaluators anchor their assessments on the information available in the beginning of the text. This result provides strong support for the idea that anchoring plays a functional role in the assessment process. When an external anchor is given, the anchoring effect stems from an under-evaluation induced by a negative anchor: positive anchors and no anchor conditions do not lead to different grades.

Negative anchors also reduce the processing time for information extracted from the beginning of texts, which leads to conclude that the taking into account of information is different when a negative anchor is present.

The results are discussed in light of a model of the cognitive functioning involved in knowledge assessment.

The interpretation given is that the effects stem from the evaluator's expertise in and of itself. Past experience with evaluation have taught the expert that poor students seldom produce work that is inconsistent with their usual level of performance, while good students are more likely to produce counter-examples of their usual achievement. This would account for the fact that evaluators do not take positive anchors into consideration, given that these anchors do not provide a sufficient guarantee of the quality of the text.

Evidences of the validity of this interpretation are given.

ANTICIPATION AND MEMORY FOR RISK

P.R. Chapman and J.A. Groeger

MRC Applied Psychology Unit, Cambridge, U.K.

There is considerable evidence that when confronted with risk individuals adjust their behaviour, or where possible avoid risk altogether. Such anticipatory behaviour must necessarily be based upon some prior expectations of potential levels of danger. This paper reports results which suggest that drivers' memories of actual events are closely linked with their expectations of risk, though in a rather complicated way.

After driving a car in traffic subjects were asked to recall some of the situations they had driven through. By far the best predictor of the situations they recalled turned out to be ratings of subjective risk given during the drive. To provide a more sensitive measure of memory this result was explored using recognition tests of memory for videos of driving situations viewed in a simulator. In this study two apparently conflicting results were obtained. For junctions which were as a whole regarded as dangerous, individual films of the junction which were themselves judged as risky showed best recognition performance. However, for junctions previously judged less dangerous this result reverses. For such situations the most risky individual films were in fact those for which worst recognition performance was observed. Whilst the first result is relatively easy to interpret in terms of the effects of arousal on memory, the second result is more difficult to explain within this framework. These findings appear to suggest that prior expectations about risk affect subsequent memory for both types of situation.

Other studies to be reported here show that such prior expectations can be manipulated in tasks involving both memory and judgement. Despite such manipulations, subjective risk remains an important determinant of how individual drivers interact with their environment.

POST-DECISION JUSTIFICATION AND THE STRUCTURE OF ARGUMENTS

W.Ray Crozier

University of Wales, Cardiff, UK

and

Rob Ranyard

Bolton Institute of Higher Education, UK

The post-decision phase of decision making is of significance for attempts to provide a systematic account of decision making. Our approach to the study of post-decisional processes stresses the role of argumentation, guided by Montgomery's (1983) maxim that "decision making is a search for good arguments". The question of what counts as a good argument has been neglected within psychology, and whenever researchers have attempted to generate arguments that vary in quality they tend to do so on pragmatic and empirical rather than on theoretical grounds. The paper outlines a scheme for generating different forms of arguments, and tests predictions about judgments of the quality of arguments used to justify a decision. The scheme draws upon concepts from cognitive dissonance theory, theories of argumentation, and cognitive decision research. It incorporates an index of argument strength together with structural features reflecting the role of the rejected decision alternative, attributes that are inconsistent with the decision, and different forms of comparative evaluations of alternatives and attributes. It was used to construct scenarios composed of justifications of hypothetical decisions. These were rated by a sample of subjects on a number of response scales, and principal component analysis of ratings identified two dimensions of argument quality: simplicity and convincingness. Further studies tested hypotheses about the relationships between perceived simplicity and convincingness of the justifications and their underlying structural characteristics.

DESCRIPTION AND PROCEDURE INVARIANCE IN MULTIATTRIBUTE UTILITY MEASUREMENT

Thomas Eppel

School of Management, Purdue University, West Lafayette, IN 47907, USA

Multiaattribute utility theory imposes certain requirements on the scaling parameters (weights). In particular, weights should be invariant towards equivalent descriptions of attributes in a value tree and towards the order in which attributes are given (description invariance). On the other hand, changes in the range of consequences should lead to changes in the weights. These requirements were used as criteria to compare four of the most common weight elicitation procedures in a sequence of experiments: Ratio Method, Swing Method, Tradeoff Method, and Pricing Out.

With respect to manipulations of the value tree, all four methods were affected by splitting up attributes into more specific subattributes, provided that the weights were elicited in a non-hierarchical way (i.e., at the bottom of the tree). Such a splitting bias occurred in two very different decision contexts (evaluating nuclear waste repositories, evaluating apartments). The order of attributes did not significantly influence the weights. Manipulations of the range of consequences were not reflected in corresponding changes in the weights. Splitting biases and the range insensitivity of weights raise some serious questions about framing effects in multiaattribute utility measurement.

Procedure invariance requires that different methods to elicit multiaattribute utility weights lead to the same weights. The reported results show significant interactions between methods and weights. Thus, different methods can lead to very different weights over the considered set of attributes. All experiments used an interactive computer program to elicit weights, which also allowed for immediate feedback about inconsistent weights between different methods. Information about such inconsistencies was provided using different feedback formats. The reconciliation of inconsistencies was influenced by the compatibility of stimulus (feedback) and response (reconciliation) scales, confirming similar observations of some recent research on this topic.

DIFFERENT LEVELS OF REASONING AS ASPECTS OF DECISION
MAKING IN A DYNAMIC ENVIRONMENTAL SITUATION

Andreas M. Ernst

University of Freiburg, Germany

In this paper, the cognitive and motivational processes leading to competent decision making in the context of an intricate socio-ecological situation (a commons dilemma) are explored.

In response to a dynamically changing environment, people acquire, modify and apply specific knowledge structures. We assume that the main sources of knowledge acquisition are (a) learning from experience, (b) observational learning, and (c) learning from the extrapolation of current knowledge into the future. The latter means mentally running one's perception of the state of the world, including one's own as well as the others' behavior, its consequences, and possible long-term side effects.

We propose a knowledge representation on three levels: the basic level of quantitative knowledge about world states (e.g. the state of the resource and its exploitation), the level of relational knowledge (i.e. linking two observations to provide evidence about trends), and the level of qualitative knowledge, in the sense of judgements about the world's states. On each level, certain computational rules are applicable. The levels have each their characteristic result accuracy/processing effort trade-off. Novices are believed to have their qualitative judgements anchored in a relatively inaccurate way, resulting in characteristic, empirically observable biases. The more precise experts' anchoring permits them not to overlook important side effects of their actions.

The knowledge structures and processes sketched above are part of a symbolic computer model of human behavior in socio-ecological dilemmas. The model's implications for theory validation, diagnosis, and decision making training with a micro-world including artificial agents are presented.

THE ROLE OF LANGUAGE CATEGORIES IN DECISION MAKING PROCESS

Frumkina R.M., Institute fo Linguistic Studies of the USST Academy of
Sciences, USSR, Moscow, Semashko Aven., 1/12

Empirical studies of human decision performance often imply scaling procedures, e.g. rating and estimating alternatives as preferred, sound, useful, risky, etc. That is, subjects (both experts and laymen) are regularly suggested to express their opinions (choices) via verbal labels.

According to our experimental evidence, scale labelling presents a problem in itself. Most subjects do not usually interpret unambiguously words used as labels even for comparatively non-opaque scales. Such labels for presumably bipolar and symmetrical scales as often - seldom; risky - safe; most probable - hardly ever probable; most necessary - unnecessary; reliable - unreliable are far from being antonyms, let alone stand for symmetrically opposed concepts.

This tenet is supported by experimental data based on (1) experiments in scaling word frequency and pronounceability estimates; (2) data from a comprehensive public opinion survey questionnaires in USSR. It is claimed that most labels are context-dependent and culturally non-free to an extent that deprive them of any reliability.

Some anchoring procedures moderating various types of response bias are proposed.

The path from individual preferences to a group decision
I.N. Gallhofer W.E.Saris Robert Voogt
Sociometric Research Foundation, Amsterdam

This study deals with the process by which the individual preferences of ministers are transformed into national decisions during cabinet sessions. We assume a process, whereby a collective choice could be taken immediately after a discussion of the arguments for or against, either unanimously or by a majority. If decision makers fail to arrive at a choice based on the argumentation, they will postpone the decision and try to find an acceptable option that is not necessarily based on argumentation, in order to arrive at a collective choice later. This process might be repeated several times and end either by postponement, when no acceptable option is found, or by a majority or unanimous decision, when one is found.

Factors which contribute to a high likelihood that choices will be made unanimously or by a majority, or that they will be postponed, are also dealt with. The literature suggests that factors such as disagreement among the decision makers, a crisis situation and the behavior of the key decision makers will, in general, influence the group process.

The results of our study show that the likelihood of an unanimous decision is 100 % when the group is not divided. When the group is divided, but the key decision makers agree upon an option, the likelihood of postponement is 0. If the key decision makers are also divided, the steps in the choice process will depend on whether or not a crisis situation exists. In both situations (crisis or non-crisis) postponements will initially occur in order that an acceptable alternative might be found. In crisis situations, time is limited and after a sequence of sessions with repeated postponements following at short time intervals, it is most likely that an alternative will be chosen that is supported by a majority. In non-crisis situations there are more postponements but unanimous decisions are more likely. These unanimous choices mainly consist of compromise alternatives. Arguments, i.e. decision trees, play a very limited role in this process.

THE PREDICTION OF CONFLICT OUTCOME IN THE CHAIN OF INTERPERSONAL CONFLICTS

Ilya E. Garber

SF NPO "Agropribor", Saratov, USSR, 410600, Saratov, Pervomaiskaia
st., 154.

The paper deals with the chains of dyadic conflicts with three possible outcomes (win, loss, draw) in which course and outcome of the previous conflict essentially influences course and outcome of the one to follow.

The analysis of inertia tendencies in chains not conditioned by achievement of common goal leads to the functional equation

$$F(G(p_1, p_2), F(p_1, p_2)) = G(G(p_2, p_1), F(p_2, p_1))$$

(p_1, p_2 - initial (a priori) probabilities of win and loss of subject; $F(p_1, p_2)$, $G(p_1, p_2)$ - recalculated (a posteriori) probabilities of his win and loss on condition that event had happened to him which consisted for example in favourable or unfavourable outcome of the preceding conflict).

The rest of the article is devoted to chains conditioned by achievement of common goal.

The analysis of a sequence of conflicts outcomes is based on the psychological theory of motivation and is supported by the data of chess competitions. The "absorption" rules for tendencies are formulated.

The overall outcome of the chain is treated in terms of "expected place" instead of generally accepted "expected score". Our estimations and predictions seem to be more precise than Elo's.

IN SEARCH OF EVIDENCE FOR DOMINANCE STRUCTURING IN DECISION MAKING

Jörgen Garvill¹ Tommy Gärling¹ Erik Lindberg² Henry Montgomery³

¹University of Umeå, Sweden ²Swedish Road and Traffic Research
Institute, Linköping, Sweden ³Gothenburg University, Sweden

Whether subjects, who have to choose between alternatives violating dominance at the level of attributes, activate higher level values in order to obtain a dominance structure was studied in two experiments. In the first experiment 32 subjects made choices from non-dominated and dominated choice sets comprised of two hypothetical alternatives each. After each choice the subjects were required to justify their decision in their own words, or by choosing from a list of attributes and values. Although subjects used values to justify their choices in both conditions the expected difference in frequency of values between the non-dominated and dominated choice sets was not obtained. In the second experiment 36 subjects made choices from the same choice sets as in the first experiment, or made preference ratings of the alternatives presented singly. Before, in connection with, and after making choices or preference ratings the subjects evaluated the attributes comprising the alternatives as well as rated the alternatives' perceived effects on value fulfillment. The subjects also rated the influence of attributes and values on each choice. According to the dominance structuring hypothesis one should expect the subjects to restructure their evaluations in such a way that the chosen alternatives in the non-dominated choice sets should come to dominate the non-chosen more often in connection with choices than before choices and more often than in connection with preference ratings. The expected differences were not found. However, it was found that although the choice sets violated dominance at the level of attributes, the chosen alternative dominated the non-chosen at the level of values in about 60% of the choices. Further evidence of subjects' activation of values and support for the dominance structuring hypothesis was found in that the subjects tended to bolster the chosen alternative at the level of values in connection with choices and in that the subjects rated the influence of values on choices higher for non-dominated than for dominated choice sets.

Component processes of phased decision strategies

Heiner Gertzen

University of Heidelberg

While decision makers tend to use one single strategy in binary choice, it is often assumed that mixtures of strategies including easy heuristic processes are applied in more complex choice tasks. The assumption of phased strategies is mainly based on think-aloud studies, partly combined with global measures of information retrieval strategies. Additional evidence in support of this assumption is therefore needed. Further, the component processes involved in phased strategies remain to be investigated (Stevenson et al., 1990).

In order to generate expectations for an experimental investigation of phased strategies, simulations were run within the framework of effort-quality analysis (Payne et al., 1988). According to this approach, the strategy leading to the best trade-off between a low cognitive effort and a high quality should be selected for a specific choice task. Effort-quality trade-off functions obtained in the simulations for tasks of different complexity favor compensatory dimension-based strategies for binary choice. For more complex tasks a combination of initial dimension-based elimination procedures followed by an alternative-based final processing stage would be expected.

For an examination of these expectations, retrieval behaviour was studied in two experiments. For two-, four- and eight-alternative choice tasks, subjects could successively retrieve parts of the available information. Depending on a subject's menu selection, in each retrieval step one of two events could occur. Either all features of one alternative were displayed together (alternative-based retrieval), or all features on one dimension were displayed together (dimension-based retrieval).

The recorded retrieval sequences are considered as sequences of Bernoulli trials in order to apply techniques known from mathematical learning theory (Suppes & Ginsberg, 1963). The most important indicator of phased versus single strategies consists in a sequence being stationary or not. This property will be analyzed together with Vincent curves of alternative- or dimension-based retrievals. While stationary sequences are compatible with single processing strategies, non-stationarity may indicate the use of phased strategies. In particular, non-stationarity due to decreasing dimension-based retrievals over time would be expected, if subjects first applied dimensional elimination and later derived their choices from more thorough, alternative-based processing of the remaining alternatives.

References: Payne, J., Bettman, J.R. & Johnson, E.J. (1988). JEP:LMC, 14, 534-552. Stevenson, M.K., Busemeyer, J.R. & Naylor, J.C. (1990). In M.D. Dunnett & L.M. Hough (eds.). Handbook of Industrial/Organizational Psychology (2nd ed.), vol.1. Suppes, P. & Ginsberg, R. (1963). PsychRev, 70, 139-161.

Judgmental Forecasting within and across correlated time series

Nigel Harvey, Fergus Bolger and Alastair McClelland

University College London.

Pairs of cross-correlated time series were generated by using the Lotka-Volterra equations that have been employed by biologists to model predator-prey relationships. They were used in an experimental simulation of a forecasting task.

Subjects were told that they were to regard themselves as managers of a section of the London Underground train system. Data on the number of passengers (ie prey) and on the number offences committed by criminals (ie. predators) are collected each week. Subjects were told that they had to use data from the previous five weeks to make forecasts of crime and passenger numbers for the next week - so that the appropriate level of policing and passenger support could be planned for the section.

There were 35 trials in the first part of the experiment. On each one, subjects made forecasts for both series on the basis of a display of the previous five items in the two series. After each forecast, they estimated the probability that it would be correct. In the second part, they still made forecasts based on five week data periods but these periods did not follow on from one another. Furthermore, they had to make forecasts for both series on the basis of previous data from only one series.

Data were analysed to determine how well the within-series forecasting was done in the two halves of the experiment and how well the cross-series forecasting was done in the second half. Time-series regression was used to model forecasts on the basis of previous forecasts and previous actual items in the series. In the first half of the experiment, within-series forecasting was consistent with double exponential smoothing (cf Andreassen, 1990). Subjects took trends into account but had difficulty at turning points of the cyclic series.

Cross-series forecasting in the second half of the experiment was poor. People did not appear to internalise the relationship between the two series adequately.

Effects of risk information and personal control on risk taking
in blind curves

Laurie Hendrickx & Charles Vlek

University of Groningen, The Netherlands

Uncertainty (or risk) may be categorized along two dimensions: (1) the nature of risk information available (i.e., frequency information, about the outcomes of similar situations in the past, versus scenario information, about the way(s) in which a future accident might occur), and (2) the degree of personal control over the task (i.e., the extent to which outcomes depend on one's own behaviour, or on external, uncontrollable factors). In a field experiment, the effects of variations in both uncertainty dimensions on people's risk-taking tendency in an actual traffic task were investigated.

Thirty-two novice and 32 experienced drivers made a test drive on a rural road with many blind curves. Prior to the drive, all subjects had read a 2-page brochure which could contain risk information about blind curves. The latter was systematically varied across subgroups of subjects (no risk information, only frequency information, only scenario information, both). Degree of personal control over the task was also varied: each subject made the same trip twice, once as the car's driver and once as a front-seat passenger. Actual (or driver) speeds were registered at 20 blind curve locations; passengers continuously indicated their 'preferred' speed choice.

Important findings were: (a) presenting subjects with frequency and/or scenario information resulted in substantial (about 6 km/h) curve speed reductions; (b) 'degree of personal control' was also found to significantly affect the subjects' risk-taking tendency, but the effect size was much smaller: actual (driver) speeds were about 1 km/h higher than preferred (passenger) speeds; (c) amount of driving experience was positively related to risk taking, but - surprisingly - only for female drivers. Implications, e.g., for our taxonomy of uncertainty and for the designers of (traffic) safety campaigns, will be discussed.

MENTAL MODELS AND CAUSAL EXPLANATION :
JUDGMENTS OF PROBABLE CAUSE AND EXPLANATORY RELEVANCE

Denis J. HILTON

E.S.C. TOULOUSE - 20 Bd Lascrosses - 31068 TOULOUSE CEDEX

and

Hans-Peter ERB

Universitaet MANNHEIM

ABSTRACT.

Good explanations are not only true or probably true, but are also relevant to a causal question. Current models of causal explanation either only address the question of the truth of an explanation, or do not distinguish the probability of an explanation from its relevance. A diagnosis-explanation model is outlined which shows how the mental models constructed to represent causal scenarios interact with conversational principles to determine the truth and relevance of explanations.

The proposed model makes discriminating predictions about the differential effects of contextual information about alternative explanations on a) the kind of mental models constructed, b) belief revision about probable cause, and c) the perceived quality of a focal explanation. Four experiments are reported that test these predictions. The significance of the notion of explanatory relevance for research on causal explanation is then discussed.

PROBABILISTIC MENTAL MODELS: A BRUNSWIKIAN THEORY OF CONFIDENCE

Ulrich Hoffrage & Gerd Gigerenzer
Universität Salzburg, Austria

Research on people's confidence in their general knowledge has to date produced two fairly stable effects, many inconsistent results, and no comprehensive theory. We propose such a comprehensive framework, the "theory of probabilistic mental models" (PMM theory). The theory (i) explains both the "overconfidence effect" (mean confidence is higher than percentage of answers correct) and the "hard-easy effect" (overconfidence increases with item difficulty) reported in the literature, and (ii) predicts conditions under which both effects appear, disappear, or invert. In addition, (iii) it predicts a new phenomenon, the "confidence-frequency effect," a systematic difference between a judgment of confidence in a single event (i.e., that any given answer is correct) and a judgment of the frequency of correct answers in the long run. Two experiments are reported that support PMM theory by confirming these predictions. Unlike earlier explanatory attempts that explain overconfidence by general biases in information processing (e.g., confirmation bias), the present theory explains the known effects and predicts new ones by assuming mental models that are well-adapted to people's environment, but that are suboptimal if that environment changes or if the experimental task in effect creates such a change.

Gigerenzer, G., Hoffrage, U., & Kleinbölting, K.: Probabilistic Mental Models: A Brunswikian View of Confidence. Psychological Review (in press).

RISKY CHOICES MADE BY CHILDREN AND ADOLESCENTS

David R. Holtgrave and Barbara Tinsley

Univ. of Oklahoma and Univ. of California - Riverside

Although children and adolescents must make important decisions everyday (about matters like drug use and sexual behavior), the processes by which they make risky choices have received relatively little empirical study. Recently, we conducted two large-scale, cross-sectional, survey studies in this area (including nearly 2,000 subjects). One study was aimed at understanding how 4th through 10th grade students make choices about health risk behaviors (such as drug use, tobacco use, alcohol use, "unsafe" sexual behavior, and failure to engage in adequate exercise and usage of automobile seatbelts). The other was a study of the choices 5th through 11th graders make with regard to environmental health risks -- especially those risks associated with the failure to recycle solid waste. Although many hypotheses were tested in both studies, a primary hypothesis was that older children give different weightings to information sources (like mother, father, teacher, and television) when deciding whether to engage in health risk behavior. A second important hypothesis was that risk attitudes (measured in a multi-dimensional, multi-method way) are predictive of risky behaviors.

Both major hypotheses were supported. As children get older they appear to use fewer and fewer sources of information as input to their health risk decision-making processes. As children age, their mothers, fathers and teachers are less influential while TV and radio personalities remain very important to the decision-making process. Risk attitudes are related to health-risk behaviors, but in a complex way (best explained by developmental theory). For example, as children age, risk attitudes play a lesser role in determining the frequency of cigarette use. Cigarette and alcohol use in young children can be predicted from teacher and peer ratings of individual student's social and health risk taking.

SCENARIOS IN PERSONAL DECISION MAKING

Huys, J., Evers-Kiebooms, G., & d'Ydewalle, G.

K.U.Leuven, Leuven, Belgium

With Beach (1990), we suggest that the ability to construct possible futures is a central feature in decision making. Forecasting the future usually consists of constructing a plausible story, in which the decision maker is an active participant. "Constructing a plausible story" means that people unfold the future in a coherent way.

We direct our attention to personal decisions which result in an important change in the life of a person. Having to decide about a next pregnancy in the context of an increased genetic risk is such a decision. We hypothesize that the development of a particular scenario will influence the subsequent decision.

Ninety subjects had to imagine that they were carrier of Duchenne Muscular Dystrophy. They all received the same information about the disease. Two elements were varied. First, half of the women had to imagine that they had decided to have a child and that the child was healthy, and half of them had to imagine that the child did have the disease. The second manipulation consisted of three levels: one third received an enumeration of consequences of having a healthy/ill child, one third read an elaborated scenario, and one third had to write how they saw their future should they have a healthy/ill child. Thereafter they had to decide if they would want a child or not, imagining to be a carrier of Duchenne Muscular Dystrophy.

The results show that people who had to write about having a healthy child are more in favour of a next pregnancy than people who had to write about having a child with the disease. In the reading and enumeration conditions such a difference did not come out. We did expect this effect since we assume that a scenario is a good cognitive tool for structuring the evidence by integrating the given information with the image one has of one's own future. As a consequence of this finding, we directed our attention to the written scenarios. The propositional analysis revealed that people develop only a limited number (between three and eight) of ideas. A further analysis revealed that for each scenario a degree of coherence could be calculated. The existence of interdependencies between the ideas is crucial because it is a diagnostic feature of the very concept of a scenario.

COMPUTER-AIDED NEGOTIATION PROCEDURES FOR MULTI-PERSON
DECISION MAKING - SOME CONCEPTS AND EXAMPLES

Andrzej Jakubowski

Systems Research Institute, Polish Academy of Sciences

Computer-aided multistage interactive-negotiation procedures for collective decision making are presented. The procedures make it possible to solve some typical decision making problems; it is assumed that members of the collective are directly - sometimes financially - interested in the final result of decision made. Two main tasks of collective decision making are analysed:

Problem I. A collective has to choose among discrete alternatives. It is assumed that the preferences of individuals for particular alternatives are significantly different; e.g. such a situation may occur when a decision is to be made on the choice among several investments under restricted budget or - on the choice among a few enterprise development scenarios. To solve the considered decision problems, the so called simultaneous and sequential auction procedures are presented.

Problem II. A collective has to choose the quantity and finance a public divisible good. It is assumed that the individuals differ in their opinions on the amount of this good to be produced (or purchased) as well as on the share of total cost. For instance, such a situation takes place when a decision on the purchase of a plot necessary for some joint venture is to be made - or a joint construction is to be financed.

Taking into account that, in general, the negotiation procedures used to solve Problems I-II can be considered as realizations of cooperative or noncooperative multistage games, different types of equilibrium for such games are discussed. The formalized approach is applied and some specific types of the collective members' utility functions are expressed in mathematical terms. So called Free-Rider (manipulation) problem is then analysed. It is related to false reporting - by some of the collective members - their preferences announced at the successive steps of negotiations; such a behaviour is aimed at misleading other individuals. The examples, illustrating - the Free-Rider versus "fair" behaviour of particular negotiators are presented.

WELL CALIBRATED GENERAL KNOWLEDGE: AN ECOLOGICAL-
INDUCTIVE APPROACH TO REALISM OF CONFIDENCE

Peter Juslin

Department of Psychology, Uppsala University, Sweden

When the experimenter is selecting almanac items to be used in a calibration study he is implicitly forming a reference class in terms of which he evaluates his subjects. The present paper will argue that the overconfidence phenomenon, a frequent finding of earlier studies of realism of confidence, can be seen as an artefact of the experimenters informal selection of almanac items. The reference class assumed in the inferences made by the subjects reflect adaptation to the subjects everyday environment. The selection procedures used by the experimenter involve a number of specific strategies, e. g. the avoidance of items expected to be correctly solved by all subjects, over-representation of items where the modal cues used by the subjects lead to the wrong answer, and so forth, that serve to create a systematically altered reference class. An experiment is reported showing that when items are informally selected by human selectors we observe poor calibration and the traditional overconfidence phenomenon. However, when the selection procedure is "debiased" and the item sample made more representative of the reference class assumed by the subjects, overconfidence disappears and we observe quite accurate calibration. The problems connected with an ecological-inductive approach to realism of confidence are briefly discussed.

IS CHOICE ACCURACY RELATED TO INFORMATION SEARCH PATTERNS?

José H. Kerstholt

TNO Institute for Perception, The Netherlands

A major assumption underlying the information board paradigm is that decision strategies can be inferred from information search patterns. This paper proposes, however, that this assumption ignores the potential influences of top-down effects, such as the use of prior knowledge, and over-emphasizes the data-driven aspects of task performance.

Support for this notion comes from two experiments in which subjects were required to make choices of varying complexity on computerized information boards. Results from the first experiment, in which a "conventional" multidimensional task was used, replicate previous research findings: a high level of task complexity reduces the amount of information requested and increases the variability of search. Choice accuracy however, remained constant over task complexity conditions. This result suggests that in high complexity conditions top-down influences can compensate for a reduced amount of information, allowing subjects to maintain high accuracy levels.

In the second experiment we used a unidimensional task that typically suggests the employment of a linear rule. The results showed that task complexity neither affected search patterns nor accuracy. Conceivably, subjects only switch to simplifying search behaviour when the task allows for the employment of such a heuristic.

Together, these results suggest that people try to reduce cognitive strain within the boundaries of goal attainment. The specific goal, task structure and strategic knowledge are important factors that mediate the relationship between information search patterns and choice accuracy.

BASE RATES AND SELECTION RATIOS

Pieter Koele & Oscar Henneken

Faculty of Psychology
University of Amsterdam
Roetersstraat 15
1018 WB Amsterdam, The Netherlands.

One of the extensively studied forms of incoherence in dealing with probability concerns the clinician's estimation of the predictive accuracy of a clinical test result. If a patient belongs either to a category C+ or C-, and a relevant clinical test yields either a positive (D+) or a negative (D-) diagnosis, this predictive accuracy is given by

$$(1) \quad P(C+|D+) = P(D+|C+) \frac{P(C+)}{P(D+)},$$

in which $P(D+|C+)$ is the sensitivity of the test, $P(C+)$ is the prevalence or base rate of category C+ in the population, and $P(D+)$ is the selection ratio of the test, the proportion of the population receiving a positive diagnosis. This selection ratio is given by

$$(2) \quad P(D+) = P(C+)P(D+|C+) + P(C-)P(D+|C-),$$

in which $P(D+|C-)$ is the complement of the specificity of the test. At least three explanations for the apparent incapability of clinicians to estimate $P(C+|D+)$ accurately have been proposed:

1. Clinicians confound $P(C+|D+)$ with $P(D+|C+)$ (Eddy, 1982).
2. Subjects underestimate the influence of the base rate $P(C+)$, especially when this base rate is extremely low (Bar-Hillel, 1980).
3. Clinicians do not consider the population base rate to be relevant in a specific diagnostic situation (Gigerenzer & Murray, 1987).

And still, "the full base rate story is not yet told" (Bar-Hillel, 1990, p.215).

In this presentation we take as a starting point a heuristic for estimating the predictive accuracy of a test suggested by the structure of formula (1): Use the sensitivity of a test as an anchor and adjust by taking the value of the ratio $P(C+)/P(D+)$ into account. As a first step in testing the applicability of this heuristic we investigated subjects' ability to estimate the selection ratio of a test. Our experimental results show that subject estimates of this selection ratio are remarkably accurate. Consequences for the further testing of this heuristic, and its possible use in statistical training programmes will be discussed, against the background of the three explanations mentioned above.

ON ADAPTIVE METHODS FOR MULTICRITERIAL OPTIMIZATION

Alexander S. Krasnenker, Dr. Sc. (Techn.)

Scientific Research Tractor Institute (NATI), Head of department,
Moscow Region, 142322, Chekhov, Novy Byt, NATI, USSR

Solving the multicriterial or another DM-problems with the formal probabilistic algorithms led to certain difficulties, some of them appearing significant. At first, the information coming from the DM in many cases contradicted the assumption on value function convexity or transitivity, which became clear only after many iterations. Analysis of this decision showed that during the algorithm functioning the improvement direction was chosen, which either contradicted the convexity or transitivity assumption, or the assumption itself was wrong. At second, these algorithms were very slow due to step length going to zero. It led to investigating very similar decisions, and the DM started losing the interest in the search processes. These difficulties showed that the adaptive algorithms have to be applied carefully to the problems with the DM participation. In this report some ideas for adaptive algorithms directed to removing these difficulties are proposed. For the such algorithms the formal definition of DM's errors and method for estimation its influence on the final result are proposed. The estimate of the number of questions for DM, necessary to obtain the approximate optimal solution is in report obtained. Such estimate for 20 men was put to the test. It was a choice of the optimal regime for testing of tractors. The solutions have been obtained after 20-40 iterations, and the probability didn't exceed 30%.

Clinical decision making: comparing experts with novices in a
psychodiagnostic classification task.

Nicole P. Krol, Eric E. De Bruyn and John H. Van den Bercken
N.I.C.I. University, Nijmegen

Classification is an important component of clinical decision making. It focuses on or guides, perhaps unintentionally, the generation of hypotheses concerning the conditions which either elicit or sustain the identified problem. Classification systems that have been proposed to systemize and support this diagnostic task, such as the DSM-III-R (APA, 1980) or the empirical classification system developed by Achenbach (1983), can never give an exhaustive description of the full domain of problem behaviour (Krol & De Bruyn, 1990). This implies that the practitioner regularly is confronted with individual problem behaviour that is not covered by the classification system. Our research aimed to detect heuristic rules an experienced diagnostician might use when he or she has to decide whether such problem behaviour is part of a syndrome of the classification system.

Two groups of experienced diagnosticians were given a problem behaviour classification task. One group of diagnosticians (n=12) was given a structured classification task and the other group (n=10) received an unstructured classification task. The structured task entailed sorting of descriptions of problem behaviour into Achenbach's empirical classification system while thinking or talking aloud. In the unstructured task the diagnostician had to classify problem behaviour into categories based on his or her own clinical experience. To evaluate the expertise of the diagnostician's classification, the structured classification task was also performed by 12 novices.

The thinking aloud protocols were analyzed on the appearance of cognitive operations using a category system based on the inferences Clancey (1985) described for heuristic classification and on the KADS interpretation library (Breuker, 1986). Results show an effect of the task structure on the use of the cognitive operations. Furthermore, association and labelling turn out to be important ingredients of the cognitive process involved in psychodiagnostic classification. This supports the case-based reasoning model Kolodner (1984, 1986) has developed in order to elucidate the difference between novices and experts.

THE FRAMING OF DECISIONS: A NEW LOOK ON OLD
PROBLEMS

Anton Kühberger

University of Salzburg, Austria

Previous research has shown the influence of the wording of decision problems on human choice behavior. If the same problem is presented in different ways, one highlighting gains, the other highlighting losses, the resulting choices are inconsistent. In the domain of gains choices have been found to be generally risk averse and in the domain of losses choices are generally risk seeking. This finding is labelled framing effect. A frequently cited decision problem to demonstrate framing effects is the Asian Disease Problem (Tversky & Kahneman, 1981). This problem is not fully described, that information is missing, however, has never been mentioned. It is implicitly - though it seems unwittingly - assumed that subjects insert the experimenters' default value.

A study that controlled for the missing information showed the following results: (i) Using the classical incomplete version of the Asian Disease Problem we could replicate earlier findings for the domain of losses; (ii) in the completely described version, however, no framing effects could be found; (iii) when we substituted the incomplete classical version by a reversed incomplete version, we could reverse the framing effect in the positive framing condition.

We argue that it is necessary to distinguish between different interpretations of the framing concept and to define the concept independently from results obtained. Results are discussed with respect to Prospect Theory (Kahneman & Tversky, 1979) and in terms of probabilistic mental models (Gigerenzer, Hoffrage, & Kleinbölting, in press). Probabilistic mental models specify cues invoked by the wording of problems, which can change perceived credibility and accuracy of information.

Risk, Ambiguity, and Insurer Decision Processes
March 1991
Abstract

Howard Kunreuther
Jacqueline Meszaros
Robin Hogarth
Mark Spranca

It has long been thought that, although individuals tend to exhibit risk aversion, firm-level decisions should be risk neutral. Market forces and organizational arrangements were expected to overcome individual risk aversion. Some recent theories in economics and in organizational behavior, however, suggest that managers in firms may, for rational reasons, behave as if they are risk averse. It is relatively straightforward to extend these theories to predict that managers will also behave as if they are ambiguity averse.

To empirically test these theories, we surveyed practicing underwriters, asking them to make the same kinds of pricing decisions they make daily as their firms' agents in the marketplace. A large, national sample of underwriters provided prices for a set of hypothetical insurance policies that were systematically varied by levels of uncertainty and ambiguity regarding the size and probability of losses. Results indicate that even among these trained, market-experienced, company-conditioned managers, risk and ambiguity aversion are strong and pertinent to market behavior.

NEW FINDINGS ON SP/A THEORY

Orfelio G. León and Hilda Gambara

Facultad de Psicología, Universidad Autónoma Madrid

The SP/A theory (Lopes, 1987) establishes the existence of two factors: a dispositional and a situational factor. The first factor (SP) reflects a basic attitude toward risk and is represented by two extremes, security as opposed to potentiality. At the 'security' extreme are risk averse ss, and at the 'potentiality' extreme are risk seekers. The situational factor is called 'aspiration level' (A) and is determined by the immediate needs and opportunities in the obtention of objectives marked by the ss. It is a factor determined by external demands. In this study we have proposed a simply linear model (covariance structured) wich expresses the influence of both factors in behaviour under risk.

$$R_{ij} = \mu + \alpha_j + \beta SP_i + e_{ij}$$

R_{ij} : observed risk of a subject i under treatment j ;

μ : overall mean under treatments j ; α_j : effect of treatment j ; β : slope of the overall regression of risk on SP ; SP_i : score of subject i in SP ; e_{ij} : random error.

Using as stimuli lotteries similar to those used by Lopes, the experiment consisted of: 1) Obtaining from 54 ss scores for the SP factor for their preferences for pair of lot. under conditions considered to be base line. 2) Creating two different risk external conditions from the basic situation, we observed the subjects' risk response again (from wich we obtained the estimates for factor A).

The results we obtained were the following: The scores for SP maintained a linear correlation on the risk scores under treatments ($r_{xy}=.67$). The goodness of fit of the model was .48 .

Up until now, we could only affirm that there was some experimental data congruent with the existence of the two proposed factors (León y Dueñas, 1988; Lopes y Schneider, 1987), but we did not know to what extent these exhausted the explanation of subjects' behaviour. The formalization of the SP/A theory manages, from our viewpoint, to obtain a measurement of the degree of explanation for simple tasks involving risk.

BASE RATE USE IN PROBABILISTIC REASONING

Laura Macchi

Università degli Studi di Milano, Italia

Studied how the base rate fallacy in probabilistic reasoning might be due to the verbal structure of the problem, in contrast with some heuristics proposed by other authors (i.e. causality, representativeness). In our hypothesis some elements of the text led to an overestimation of specific information, to the detriment of base rate consideration. Four texts, two from Tversky and Kahneman (1980), used as controls, and two obtained modifying some elements of their verbal form, have been submitted to four different groups of Ss ($N=40$); verbal protocols have been utilized. We used the Suicide and the Books problems that in the hypothesis of these authors, give different results, the first one having information that is causally relevant and the second one not. We hypothesized that the question structure was responsible for the neglect of the base rate. In the Books problem the two data (base rate and specific information) are quoted and related in the question. On the contrary, in the Suicide problem, the question concerns only the specific information (the suicide group). This excludes the consideration of the population (base rate) by the subjects. To verify our hypothesis we exchanged the Suicide text with the corresponding verbal structure of the Books text, and vice versa. The results indicate that a similar verbal structure produces very similar results. A chi-square test evidenced a significant ($p<0.05$) increase of the base rate consideration with the modified Suicide problem (0.67 are intermediate answers, compared to 0.38 with the original text of Tversky and Kahneman). On the other hand, the Books modified version increased the base rate fallacy, as foreseen in our hypothesis (the intermediate answers are 0.34, while in the original version they were 0.70). In this sense it is a problem of the link between information that can be produced by the question formulation or by the way in which the information are given. This suggests a reconsideration of the explicative role of the heuristics, and a consideration of the direct link between the biases and the rules of natural language.

EVIDENCE FOR AN EQUALITY HEURISTIC IN SOCIAL DECISION MAKING.

David M. Messick & Terry Schell

Department of Psychology, University of California, Santa Barbara

Many proposals have been made that complex human performance can be constructed from simple underlying processes, or heuristics. Messick (1991) has proposed that much social decision making can be construed in a similar light. Specifically, the proposal is that decisions involving the fair allocation of "goods" or "bads" to individual members of a group often entail the use of an equality heuristic, that is to say equal divisions among members often made with little deliberation.

In the current study, subjects read a story in which five business partners needed to allocate the profits and expenses of the partnership among themselves in a fair manner. Each of the partners worked independently and they all produced different gross incomes, between \$140 and \$285. The total gross income needed to be divided into collective expenses and profits. Subjects were then asked to fill in a fair allocation in an accounting ledger. In the story and instructions three factors were manipulated: the target of the allocation task, either allocating the expenses column among the partners or the profits column; the columns of the ledger to be filled in, either to complete both columns or not specified; and the causal attribution for the differences in individual gross incomes, either a situational, personal, or ambiguous attribution.

The results supported the heuristic proposal. Subjects allocated their target columns more equally than the non-target columns, even though the allocation of one column necessarily implies an allocation of the other. Furthermore, subjects who were asked to fill in both columns made more variable allocations on their target columns than those were not asked to fill in both. The attribution manipulation also produced the expected result, personal attributions produced less variable allocations of expenses than did the ambiguous or situational conditions.

- 190 -
Evaluative Processes in Choice
and Preference Judgments:
Prominence effects and/or dominance structuring?

Henry Montgomery Marcus Selart
University of Göteborg

Tommy Gärling Erik Lindberg
University of Umeå Swedish Road and
Traffic Institute

Previous research suggests that choices between options, on the one hand, and judgments of the attractiveness of options (preference judgments), on the other hand, may differ in at least two respects: (1) In choice, the more prominent attribute looms larger than in preference judgments (Montgomery, Gärling, Lindberg, & Selart, 1990; for support of a related hypothesis see Tversky, Slovic, & Sattah, 1989). (2) In choice, the chosen alternative under some conditions is more often backed up by dominance structuring (attempts to reduce dominance violations) than is true for the most preferred alternative in preference judgments (Lindberg, Gärling, & Montgomery, 1990; Garvill, Lindberg, Gärling, & Montgomery, 1990). However, the empirical support for the two hypotheses is not unequivocal. In the present study, the validity of the two hypotheses were examined under conditions, which in some respects allowed a more detailed and more controlled comparison between evaluative processes underlying choice and preference judgments than has been possible in earlier research. In a pilot study, a matching task was used to create pairs of options that were equally attractive. In the main experiment, 72 subjects were assigned to conditions in a 2 X 3 choice/preference judgment vs. data elicitation method design. In all conditions, subjects were presented with eight sets of four options, two of which were the matched options in the pilot study. The two other options were constructed to be less attractive than the matched options. Three data elicitation methods were used, viz (1) attractiveness ratings of aspects plus choice or preference judgments of options, (2) think aloud data plus choice or preference judgments of options, and (3) only choice or preference judgments.

The results include the following findings: The so called prominence effect observed by Tversky, Slovic, & Sattah (1989) was replicated inasmuch as among the two matched options subjects tended to choose the one that was superior on the more important attribute. However, the same option was also favored by preference judgment subjects implying that choice and preference judgments did not differ with respect to the prominence effect. This finding contradicts Tversky's et al suggestion that the prominence effect for choice vs. matching is due to the fact that choice requires a more qualitative response than is true in matching (the response compatibility hypothesis). The finding also indicates that a greater prevalence of dominance structuring in choice than in preference judgments, if any, cannot (fully) explain the prominence effect for choice vs. matching. The aspect ratings did not clearly indicate that there in fact was more dominance structuring in choice than in preference judgments. By contrast, the think aloud data showed that choices more often than preference judgments were backed up by dominance structuring. A closer scrutiny of the arguments used in dominance structuring showed that subjects often did not accept the given information, but tried to modify it in order to de-emphasize disadvantages with the chosen alternative (e. g., to read a more entertaining book twice to de-emphasize the fact that the book was shorter than another book). Presumably, the aspect ratings were more clearly linked to the given information, which may explain why these ratings did not clearly support the dominance structuring hypothesis.

It may be concluded that the prominence effect and dominance structuring are two different phenomena, of which the former may be rooted in possibilities of making comparative judgments between given options (which was possible in all conditions of our main experiment), whereas dominance structuring may reflect commitments resulting from the requirement to make a choice. It may also be concluded that the dominance search model should be extended to include notions of experienced possibilities of controlling and modifying given options.

SELF-REFLECTION AND MENTAL SIMULATION: CONTENT, PROCESS, AND OUTCOME EFFECTS ON CHOICE

Professor Deborah J. Mitchell
The Wharton School, University of Pennsylvania, USA

If someone asked you to describe your dream home, or your ideal car, how would you answer? What if they asked you to describe the most important features to look for, if searching for the ideal car? In contemplating and answering these types of questions, factual domain knowledge, as well as knowledge of one's tastes and preferences, is retrieved from memory. In addition, tastes may be constructed on the spot as part of forming an answer. The processes used to retrieve and manipulate this knowledge might vary, ranging from self-analysis based on reasons, to picturing and mentally simulating an ideal.

This paper examines the effects of thinking about one's preferences on later choice processing. In general, elaborating on such preferences should make them more accessible as inputs to choice. Additionally, however, how we think about our preferences can affect choice processing. I will present results from an experiment that addressed the following question: to what extent might self-reflection or mental simulation during an earlier, unrelated task affect later choice processing? Using a priming task, subjects either (1) imagined an ideal product, (2) wrote down the most important features to consider in choosing an ideal product, as well as reasons for their importance, or (3) performed a simple recall task unrelated to the products of interest in the study. Later, under the guise of evaluating a new consumer information system, subjects made choices among alternatives from two product classes (one being the focus of the priming task, one not), using a computer-based information display program. Results will be discussed in terms of content effects (i.e., differences in attributes and their weights), process effects (i.e., choice strategies), and actual choices.

Cognitive Commitment and Affect in a Real-Time
Dynamic Decision-Making Task.

Mary Omodei & Alexander Wearing
University of Melbourne, Australia

Despite a general theoretical interest in the relation between cognition and emotion, there has been little investigation of this relationship in the sorts of complex decision making tasks which occur in everyday life. Much everyday decision making is both sequentially dependent (dynamic) and time dependent (occurs in real-time). A forest fire-fighting simulation was programmed to create a realistic real-time dynamic decision making (RT-DDM) environment. Realism was achieved by involving both fire-fighting personnel and undergraduate students in all phases of the development of the simulation. The present paper reports an investigation of the inter-relationships among a number of personality, cognitive and emotional factors, and the joint effects of these factors on performance on the fire-fighting task. Several causal models of these relationships were assessed. Cognitive commitment to the task was found to be associated with increased performance, approximately half of this effect being mediated by affective involvement, arousal, and their interaction. Some evidence was found to suggest that involvement has several qualitatively different effects on cognitive performance: improving performance by increasing arousal, but impairing performance both directly and through increasing stress levels. Personality factors (extraversion, neuroticism, mastery and optimism) and attentional factors (self-awareness, environmental-awareness and distractibility) were not found to have any overall relationship with task performance. However, extraversion was found to be related to the specific decision strategies adopted, and causal interactions which nullified one another were observed for some of these variables. From the relationships obtained among commitment, involvement and efficient cognitive performance, it is concluded that persons choose the level of emotional involvement they experience in any specific task by controlling the level of cognitive resources they allocate (i.e., commit) to the task.

THE USE OF MULTIPLE CHOICE EXPERIMENTS AS A DEVICE TO COLLECT AND SCALE EXPERT JUDGEMENTS ON THE LIKELIHOOD OF OCCURRENCE OF FUTURE DISCRETE EVENTS: AN APPLICATION INTO RETAILING

Harmen Oppewal (Faculty of Architecture, Building and Planning,
Eindhoven University of Technology, The Netherlands),

Jordan J. Louviere (Faculty of Business,
University of Alberta, Canada) and

Harry J.P. Timmermans (Eindhoven University of Technology & University of Alberta)

This paper focuses on the combined use of multiple choice experiments and choice models as a device to collect and scale (expert) judgements on the likelihood of occurrence of future discrete events. Since the integration of functional measurement, or numerical conjoint analysis, with discrete choice modeling (Louviere and Woodworth, 1983, Journal of Marketing Research), the resulting decompositional approach has increasingly been used in consumer studies to derive utility functions and to predict choice. The present paper explores an extension of this approach into the study of subjective probabilities. The suggested indirect method to the derivation of likelihood estimates is easy to administer and leads to frequency data that can be analysed assuming a Multinomial Logit type of response model to derive scale values for subjective probabilities. The method is demonstrated with an application in which store managers' expectations about future store investments were measured.

PERCEIVED PROBABILITY OF POSITIVE AND NEGATIVE EVENTS:
BEHAVIORAL ANTECEDENTS AND CONSEQUENCES

Wilma Otten and Joop van der Pligt
Faculty of Psychology, University of Amsterdam, The Netherlands

This paper deals with individual probability estimates of negative and positive future life events. Risk perception studies show that absolute risk appraisals tend to be biased. Also, comparative risk appraisals, i.e. appraisals of own probability as compared to the probability of a similar other, seem to be optimistically biased. In the literature this phenomenon is referred to as "unrealistic optimism" (Weinstein, 1980) or "illusion of unique invulnerability" (Perloff, 1982). The present paper will discuss absolute and comparative estimates of events, which are under personal control.

Most of the risk literature presumes a relationship between perceived risk or vulnerability and protective behavior. Empirical evidence for this sequence is relatively scarce. This paper addresses the behavioral consequences of risk appraisals in terms of protective behavior for negative events and 'fostering' behavior for positive events.

In addition to behavior as a consequence of perceived probability, behavior can also be viewed as an antecedent. The hypothesis that risky antecedent behavior heightens probability estimates for negative events and lowers the estimates for positive events will be examined in this paper.

Summarizing, the present study deals with (1) absolute and comparative estimates of positive and negative events, and (2) relations with antecedent behavior, and behavioral consequences. Four groups of Ss (N=100 in each group) were each presented three positive and negative events. The twelve events concerned health, relational, study and financial issues. Ss assessed the probability that the event would happen to them. Furthermore, they rated the extent to which four antecedent behaviors applied to them, and judged their future intentions regarding these behaviors. The same ratings were collected for a similar other.

Preliminary analyses provide some evidence for the expected relations between probability estimates and antecedent behavior and/or intended behavior. The main analyses try to shed some light on the pre- or absence of the expected relations by focussing on attributes such as perceived control of the behaviors and personal experience with the event.

THE FUNCTION OF CONCRETE EMOTIONS IN RATIONAL DECISION MAKING

Hans-Rüdiger Pfister and Gisela Böhm

Technische Universität Berlin

Descriptive models of behavioral decision making are often criticized that they neglect the impact of emotions. This is particularly true for multiattribute decisions under certainty, when alternatives are exhaustively defined by a set of objective attributes. We propose a model based on Tulving's (1983) differentiation between episodic and semantic memory: attributes describing the alternatives pertain to semantic and emotions to episodic memory. Usually, decompositional approaches only consider attribute representations in semantic memory, where options are decomposed into discrete features inducing partial utility judgments. It is assumed that these judgments are modified by holistic evaluations resulting from emotional associations in episodic memory. Matching the available options to similar episodic events generates specific emotional reactions, which are then combined with utility judgments of attributes.

In an experimental study subjects had to give ratings of 'objective' features of different decision options (semantic judgments), as well as to indicate the intensity of several concrete emotions elicited by these options (emotional-episodic judgments). Additionally, type of options (persons or cities), similarity of options, and response mode (rating, ranking) were varied. The following results were obtained: Choices can be better predicted with a regression model by semantic judgments than by emotions. However, emotions significantly increase the portion of variance explained. Concerning response mode, the impact of emotions is larger on rankings than on ratings. Factor analyses show that different emotions can be described as two independent components: positive-pleasant and negative-unpleasant emotion. Surprisingly, only positive emotions seem to be relevant in decision making. When applying a weighted additive model, contrary to attributes, the aggregation of positive and negative emotions is not compensatory. We conclude that emotional impact on decision making primarily consists of a non-compensatory positive and negative component.

**Facilitated work groups:
theory and practice**

Lawrence D. Phillips¹

London School of Economics & Political Science

Maryann C. Phillips

ABSTRACT

This paper concerns the facilitation of a particular kind of working group: one whose general aims are to achieve a shared understanding of issues, a sense of common purpose and a mutual commitment to action. We see the main role of the facilitator in such a group as contributing to process and structure, not content. This view of facilitation is strongly coloured by our assumptions about the nature of groups: that groups have a character or personality as well as an emotional life which affects the way tasks get done, that the group life influences and is influenced by each participant, that the group life is characterised by strong feelings, that groups have the potential for great creativity but also for destructiveness, that the group life can shift suddenly and unpredictably, that participants experience a tension between what is desirable for the group and what is personally desired by each participant, that the size of the group affects the balance between individuality and group life, and that the group is considerably influenced by its environment.

We consider that the facilitator's main tasks are to see and understand the group life, and to intervene, when appropriate, to help the group stay in the present and maintain a task orientation to its work. The basis for understanding the group lies in observing verbal and non-verbal behaviour, and relationships between participants. Inferences about roles and underlying group processes can be drawn from both the overt and symbolic content of the group's discussions, and sometimes from what is not said in the group. Maintaining awareness of his or her own feelings allows the facilitator to gain access to the group life; attempting the role of detached scientific observer can make observation more difficult. Various forms of intervention are available to the facilitator: pacing the task, directing, 'handing back in changed form', reflecting, questioning and summarising.

In performing these functions for some work groups, the facilitator can be helped by computers, which can provide an effective means of externalising many aspects of group work. In particular, decision conferencing software allows individual judgments to be modelled, then shows their combined effect, with the result that new perspectives on the issues are often created. Subsequent sensitivity analyses may show that overall results are unaffected by substantial differences of opinion about details. By assigning to the computer the tasks of storing, retrieving, manipulating and communicating information, group members can concentrate their attention on the judgmental tasks, and the facilitator can attend better to group processes. Effectively used, computers can provide means to help a group maximise creative and minimise destructive aspects of its life.

1. Decision Analysis Unit, London School of Economics & Political Science, Houghton Street, London WC2A 2AE, England.

Decision making in Africa: Study of awele players' strategies

Jean Retschitzki

University of Fribourg (Switzerland)

Paper proposed at the 13th research conference on Subjective probability,
utility and decision making
Fribourg, 18-23 august 1991

For several years, cognitive aspects of the "national game of Africa", the game called "awele" in Ivory Coast were studied in order to find out the kind and level of intellectual processes underlying this activity. The game of awele constitutes a complex conceptual field, with its own concepts, schemas, etc. and it is partly independent from other knowledge domains.

Béart (1955) has reported that the best players are the old unschooled men living in the villages. What kind of cognitive activity allows them to play so efficiently ?

The main question addressed by our study was the nature of the cognitive activity underlying the playing behavior or the strategies of the players. How do they choose the next move? What are the criteria used to make a decision about the next step?

Several techniques were used to address this question: Observations and analyses of games (73 games recorded with 19 boys; 68 games with adults); interviews about game situations; thinking-aloud and didactic games; awele "problems" (a paper and pencil task).

The subjects were encouraged to express as many arguments as possible, in order to establish an inventory of tactics and strategies, on the one hand, and on the other hand to determine to what extent players base their choices upon an anticipation of the next moves, and if it is the case, how long-range these predictions are.

The main results concerning the problems outlined above are the following:

1. The analysis of the "awele problems" task conducted according to the "rule assessment approach" (Siegler, 1976) has shown that simple models are unable to account for the results.

On the data gathered with the boys' group we conducted a comprehensive analysis by building models of rules that could have been used by the subjects to choose their answers. The models differed in several variables like the priority given to attack or defense, the direction of scanning, the depth of anticipation, etc.

2. Verbal reports by adult players indicate a similarity with reasoning exhibited by players of other board games. According to their accounts, players never randomly choose the next move.

3. Analyses of game situations by the illiterate players of Kpouebo illustrate that anticipation and hypothetico-deductive reasoning occurs when considering the choice of a move.

ANTICIPATED REGRET IN PERSONAL DECISIONS UNDER UNCERTAINTY

René Richard

University of Amsterdam

Both behavioral decision theory and attitude-behavior models presume that personal decisions are based on a trade-off between the positive and negative consequences of the various behavioral alternatives. Models of personal decision-making usually add variables such as 'self-efficacy' (i.e. the ability to actually perform the behavioral which maximizes utility) and 'social-norms' (i.e. the norms in one's immediate social environment about the various behavioral consequence). In this presentation we will focus on these factors, but also investigate the role of anticipated affective reactions to the outcomes of behavioral alternatives. This factor is related to "anticipated regret" as a guiding principle in decision making.

The results of two studies will be reported. Both studies concerned decisions about risky behavioral practices (in this case sexual behavior). Both studies were large scale (approx. 800 and 400 Ss. respectively) and attempted to assess the relative importance of anticipated affective reactions as a determinant of behavioral choice. The first study is based on a representative sample of adolescents of approx. 800 subjects. This study attempted to investigate the major predictors of risky sexual practices. Results indicated that anticipated affective reactions are a major determinant of risky behavioral practices. In the second study we attempted to assess whether this factor should be incorporated as a separate predictor of behavioral choice or whether it should be included in the expectancy value component of attitude-behavior models. In both studies the model predicting behavioral preferences was tested with LISREL-analyses. Implications for SEU- and related models of human decisionmaking will be discussed.

THE DELPHI TECHNIQUE: AN EXAMINATION OF FACTORS RESPONSIBLE
FOR IMPROVEMENTS IN FORECASTING ACCURACY

Gene Rowe
Bristol Business School

Fergus Bolger
University College London

George Wright
Strathclyde Graduate Business School

The Delphi technique is a widely used means for obtaining a consensual judgement from a group of experts using questionnaires and anonymous feedback. However, the relatively few empirical studies which have been done in order to assess its validity as a technique for improving accuracy have produced variable results. One possible problem with the laboratory studies is the simplified nature of feedback given (often only summary statistics such as means and medians) which may not adequately convey the knowledge of others. This is the issue this paper addresses.

Using short-range forecast questionnaires on newsworthy national and international events, student subjects were required to give probability ratings as to how sure they were that each of 60 events would or would not occur, allowing calibration of their responses and analysis of changes in calibration over a second round of study. The three conditions used differed in terms of the nature of feedback given on the second round (either none, ie.. iteration; statistical feedback of medians; or written feedback of subjects' rationales for choice). Analysis shows that subjects given written feedback showed better forecasting performance on the second round than the other two conditions and, further, that the amount of opinion change for individuals over rounds depended upon the proportion of disconfirming arguments presented, with subsequent improvements in calibration (supporting Koriati, Lichtenstein and Fischhoff 1980). These results have major implication for the experimental examination of Delphi in the laboratory and for its utilization in the "real world" environment.

ON THE EFFECTS OF REFERENCE POINTS IN MAKING TRADEOFFS AMONG
MULTIATTRIBUTE ALTERNATIVES

Zur Shapira
Department of Management
New York University

In evaluating dimension weights in constructing multiattribute utility (MAU) functions, pairs of alternatives are compared. The two alternatives are equal on all but one dimension which is in its "best" level in one alternative and in its "worst" on the other. If the level at which the other dimensions are kept leads to differential judgments (i.e. reference effects) some difficulties may arise for the determination of linear and multiplicative MAU functions. This paper reports the results of three studies in which MBA students who were actively searching for jobs made bilateral tradeoffs between job attributes. Those tradeoffs were made from the "best," "worst," and "realistic" job descriptions. Reference effects were found and are discussed in terms of what is the "best" reference alternative for eliciting preference.

HYPOTHETICAL MECHANISMS OF PREFERENCES INCONSISTENCY

Joanna Sokolowska & Tadeusz Tyszk
Polish Academy of Sciences, Institute of Psychology, Warsaw.

It is proposed that preferences inconsistency is due to the way of preferences formation (direct experience, inferences, social pressures and fads) and the way of information processing (automatic vs "reflective").

Considering the first relation, it is argued that "general" judgments are mostly based on inferences and social fads while "specific" ones are based on direct experience. Thus, it is hypothesized that there is inconsistency between "general" and "specific" judgments on the same issue.

This hypothesis was supported in a questionnaire survey on socioeconomic preferences of Poles. There were found remarkable inconsistencies in "general" and "specific" judgments about socioeconomic issues like privatization or free market economy. It was also found that "general" judgments were usually consistent with social fads.

The relation between inconsistency of preference judgments and the way of information processing is also investigated. When an individual is made to shape preferences ad hoc without personal involvement (e.g. social survey), subject's responses are made in the automatic way due to chronically accessible "evaluative" perspectives. Contrary, when an individual's involvement is needed (e.g. choice between conflicted values, goals), subject's responses are due to an "evaluative" perspective makes information processing consistent with it. Thus, it is hypothesized that:

1. automatically made preference judgments may be inconsistent with reflective judgments;
2. reflective judgments are consistent among themselves.

These hypotheses were also supported in the mentioned survey.

A NEW PROCEDURE TO VERIFY THE PHASED DECISION STRATEGY

M. Stokmans

Tilburg University

In process studies of consumer decision-making, usually a phased decision strategy is assumed, which consists of two phases. The conventional procedures to verify a phased decision strategy are Payne's index (Payne, 1976) and Van Raaij's indices (van Raaij, 1977). In both procedures two successive information items searched for are categorized on the basis of the product the information items are about (the same or a different one) and the attribute the information items are about (the same or a different one). This categorization results in four categories of the sequence variable: type I, type II, type III or type IV. In both indices, the relative frequency of type II and type III are compared.

Some comments can be made regarding Payne's and Van Raaij's indices. Regarding Payne's index, it can be concluded on statistical grounds that it is a worse procedure to verify the phased decision strategy. Van Raaij's indices are acceptable on statistical grounds, but its usefulness is almost restricted to the situation in which only two phases are distinguished in the decision-making process. The type I error of Van Raaij's indices grows exponentially, if the number of phases distinguished increases. In this study, the single Markov model, is introduced to verify a phased decision strategy. This procedure does not have the limitations of Payne's index and Van Raaij's indices. Each phased distinguished in the decision-making process is modeled in a different Markov chain. By setting some chains equal to one another (and thereby assuming that the corresponding phases of the decision-making process are equal), one can test the phased decision strategy in one test, irrespective the number of phases distinguished.

The Markov procedure is compared with the conventional procedures on the basis of data gathered in choice problems which differ in complexity (3, 5 or 8 products and 4, 6 or 8 attributes). Also the empirical findings suggest that Payne's index is a worse procedure to verify whether the consumer uses a phased decision strategy or not. In comparing the results of Van Raaij's indices with those of the Markov procedure, it turns out that, in all choice problems for which Van Raaij's indices indicate a significant difference, the Markov procedure indicates a significant difference also. Furthermore, in the choice problems for which the Markov procedure indicated a non-significant difference, both indices of Van Raaij indicate a non-significant difference also. We can conclude from the theoretical considerations as well as the empirical findings show that the Markov procedure is a valid and accurate way to verify whether the consumer uses a phased decision strategy or not.

A COMPARISON OF
PROBABILITY ELICITATION TECHNIQUES
FOR AN AUDIT BUDGETING TASK

Melinda S. Stouppe
University of Cambridge

A large literature exists describing comparative studies of many different probability elicitation techniques. Several techniques receive a great deal of support from these studies, including recommendation for their use in practical applications. It seems logical to assume that individuals requiring probability assessments would use these recommended techniques in their work. It continues to be the case, however, that lesser known techniques are used in real world applications such as acid rain studies and medical research. This paper describes a comparison of six different probability elicitation techniques drawn from the recommended techniques and the lesser known techniques using an audit budgeting task as the real world context. The purpose of this study is to determine an elicitation procedure that is quick, efficient, and accurate for use in the auditing environment.

DIFFERENTIATION AND
CONSOLIDATION THEORY
OF HUMAN DECISION MAKING

Ola Svenson
Lund University

Abstract

Differentiation and Consolidation Theory of decision making models human decision making as an active process, in which one alternative is gradually differentiated from other available alternatives. When a selected alternative has become sufficiently differentiated from the others a decision may follow. After a decision there are post decision differentiation processes during what is called consolidation.

Differentiation processes are of two basic types, structurally related and process related. In structurally related differentiation the representation of a decision problem undergoes a change related to the choice alternatives and some decision rule(s). Both facts and value differentiation processes belong to structural differentiation and consolidation processes. In process related differentiation, one or several decision rules, which are applied in a choice are used as differentiation subprocesses. Differentiation and Consolidation Theory assumes that it is not sufficient to make a choice of the best alternative but to make a choice that is so much better that it can withstand future threats, such as, unknown facts surfacing after the decision, uncertain events turning out bad, and changes in the decision maker's own value system.

THE DECISION MAKER'S PREFERENCE SYSTEM MODEL IN MULTICRITERIA CHOICE PROBLEMS

V.G. Tanaevsky

Minsk Radio engineering Institute, USSR, 220600, Minsk, Brovkiy st. 6

The choice of the rational solution in multicriteria problems depends on the Decision Maker's (DM) preferences on the set of alternative solutions. It is clear that direct analysis of multiattribute alternatives is too difficult for DM and the results of psychological investigations testify the same.

The paper deals with an adaptive interactive procedure which is intended to help the DM to construct his preference system model and to choose on its basis the rational solution. The model under consideration is constructed with taking into account the restricted human abilities on information analysis and processing.

The DM's preferences on a set of alternative solutions are determined after the preferences on the sets of criteria and criteria estimates have been elicited. The latter is possible with using permissible elementary operations. One of them is pairwise comparison. DM estimates a relative importance of the criteria in qualitative form and his answers are checked for consistency. The DM's answers are considered consistent if the preference relations are transitive on a set of any three criteria. The method of formal description of transitivity is proposed and the algorithm of checking this property during forming the DM's preference model is developed. After the consistent system of the DM's estimates is obtained the classification algorithm is used. As a result of this algorithm running the DM's preferences on a set of criteria become known.

The methods of the following information processing that integrate the DM's quantitative and qualitative estimates in a system of his preferences are also discussed in the paper. The adaptive interactive procedure developed provides a suitability of the DM's preference system model proposed to various applications. The model is suitable for solving the multicriteria problems both with quantitative and qualitative criteria. This procedure is implemented as a software and it can be used as an adaptive component of decision support systems.

Karl Halvor Teigen
Department of General Psychology
University of Bergen
Norway

When two misfortunes are better than one

Studies of "lucky" events in daily life show that what did not happen (the counterfactual outcome) is at least equally important as what actually took place. Thus people feel lucky when something terrible "could" have happened, and more lucky the worse and the closer it is. For instance, people exposed to dangerous situations are typically characterized as more lucky than unlucky (despite the general aversiveness of the situation). The idea of counterfactual comparison as a determinant of good and bad luck is tested in the present study by using two strategies:

- (1) By having subjects explain what is implied in statements saying that somebody is "lucky" (fortunate) or "unlucky" (unfortunate).
- (2) By varying the psychological distance between what happened and what could have happened. For instance, experiences of bad luck are alleviated when the positive outcome is seen as two steps rather than one step away. Also the person who have had a choice is seen as closer (and hence more lucky or unlucky) to achieving an alternative outcome than if his actions would have been inconsequential. This makes it possible to predict whether for instance a situation involving both a winner and a loser will be seen as an example of a "good luck" or a "bad luck" situation.

JUDGING PROBABILITIES USING CAUSAL MODELS: EFFECTS OF BASERATE INFORMATION AND AMBIGUITY

Manfred Thüring

Gesellschaft für Mathematik und Datenverarbeitung (GMD), Institut für Integrierte Informations- und Publikationssysteme (IPSI), Dolivostr. 15,
D-6100 Darmstadt, F.R.G.

Judging the probability of an event people often rely on their causal knowledge. Under these circumstances, at least two factors might influence their judgments:

(1) Information about baserates: While in some situations people neglect or even ignore a given baserate (Tversky & Kahneman, 1982) they seem to use it as soon as they recognize its causal relevance (Bar-Hillel, 1980).

(2) Information about potential causes or effects: Data about causes or effects can contribute to the perceived ambiguity of a situation. Ambiguity in turn can influence subjective probabilities (Einhorn & Hogarth, 1985). For example, imagine a person who knows that *both* A and B are necessary to cause C. If she is asked to judge the probability of C given A without knowing anything about B the situation appears highly ambiguous.

The ambiguity theory proposed by Einhorn and Hogarth (1985) can principally account for the effects of both, baserate information and ambiguity. In this paper, it is adapted to situations in which people judge probabilities on the grounds of their causal knowledge. The resulting hypotheses are tested in an experiment: Subjects learn causal models of different structures (Mackie, 1965) which describe artificial diseases. In a subsequent test, they are asked to estimate the probability of each disease for a variety of patients. To do so, they are given baserate information together with causally relevant data. Depending on the subjects' models, the data should lead to the perception of diverse degrees of ambiguity. These degrees and the given baserates are used to predict differences between mean probability judgments.

The results of the experiment confirm most of the predictions. They support the assumption that baserate information and ambiguity are two important factors determining subjective probabilities which are generated on the basis of causal knowledge.

EFFECTIVENESS OF DECISION AIDS AS A FUNCTION OF PROBLEM COMPLEXITY

Daniëlle Timmermans
Medical Decision Analysis Unit, University of Leiden

The effectiveness of decision aids depends on the ability to counterbalance the weaknesses of human reasoning. When solving complex problems, people simplify the problem and use shortcuts in order to reduce the cognitive effort needed to solve the problem. One of the aims of decision support is to compensate for these limited information processing capacities and to reduce cognitive effort. A decision aid based on the MAU-model might be helpful in this. Whether this is the case is an empirical issue.

In this paper I will present the results of an experiment in which the effectiveness of a MAU decision aid (DECAID, Pitz, 1987) is evaluated for problems differing in complexity. Seventy-two subjects were asked to solve two decision problems dealing with the selection of candidates for a job opening. The complexity of the problem was operationalized as number of alternatives (3, 6 or 9 candidates) and number of attributes (5 or 12). Subjects were asked to decide on the most suitable candidate for the job, while using DECAID or having no support. Results showed that decision support was most successful for problems of medium complexity, not really necessary for simple decision problems, and rather inadequate for more complex problems. There seems to be a reversed U-shaped relationship between the effectiveness of a MAU decision aid and problem complexity. This is probably due to the overload of information resulting from using the decision program.

Although the principle of decomposition, which is central to MAU decision aids, is generally effective, people might lose insight into the problem when it becomes too complex. To test this we also conducted a process analysis of the unaided decision process. Moreover, a process analysis may provide ideas to improve decision aids which are better adapted to the human decision maker. The paper concludes with some requirements for the design of user-friendly decision aids.

AN INTEGRATIVE APPROACH TO COOPERATIVE DECISION MAKING

Paul A. M. Van Lange

Free University, Amsterdam

In a diversity of interpersonal situations we see people sharing, aiding, sacrificing themselves as well as exploiting or taking advantage of others. How can this variety of social behaviors, or social decisions be understood? We assume that large differences in social decisions can be understood by taking an integrative approach which considers not only situational and intrapersonal influences, but also interactionistic influences.

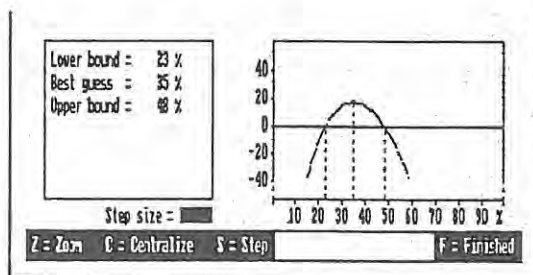
The present paper focuses on cooperative decision making, defined as the pursuit of collective welfare at the expense of one's own immediate welfare. In addition to situational factors, we consider the role of individual differences in social value orientations, defined as preferences for particular distributions of outcomes to oneself and others. The consistent finding here is that people concerned with both own and other's welfare, pro-socials, are more likely to make cooperative decisions than people primarily concerned with own welfare, individualists, or relative differences between own and other's welfare in favour of self, competitors.

More recently, researchers have become increasingly interested in examining interactionistic influences on both cooperative decisions, and the cognitive processes that may mediate such decisions. As regards cooperative decisions, there is evidence that the effects of other's decisions are moderated by individual differences in social value orientations. For example, pro-socials' decisions appear to be much more dependent upon variations in other's decisions than those of individualists and competitors. Research focusing on cognitive processes has revealed that personality impressions of others are formed not only on the basis of other's behavior but also in combination with the perceiver's social value orientation. Relevant empirical research to substantiate the theoretical claims underlying the integrative approach will be discussed.

Development and Evaluation of ELI, an Interactive Elicitation Technique for Subjective Probability Distributions

Jelle van Lenthe
University of Groningen, The Netherlands

Uncertain knowledge about continuous quantities is usually formalized through Subjective Probability Distributions (SPD's). Results of past research often point to the rather poor quality of SPD's; especially *overconfidence* appears to be an exceptionally persistent phenomenon. Usually cognitive- or item-induced biases are considered responsible for poor SPD quality. In this paper it will be argued that the particular technique used to elicit the SPD's affects their quality and that the poor quality of SPD's is --at least in part-- an artifact of the technique used (*method induced bias*). In our view, human beings are not only 'information processing systems with limited capacities', they also are outstanding users of all kinds of tools; equipped with a more appropriate elicitation tool they might prove to be more competent probability assessors than research thus far suggested. This notion motivated the development of the new ELI elicitation technique ELI. One of the central features of ELI is the *direct realization of a proper scoring rule in a graphically oriented interactive computer program*.



Lower part of an ELI screen

In this paper considerations which guided the development of ELI and a more detailed operational description of the technique will be presented. Furthermore, the paper will discuss the outcomes of an evaluation study. An experiment was carried out, to evaluate ELI performance and to compare it with the

performance of other elicitation techniques. The techniques were evaluated and compared with respect to their contribution to reliable and valid SPD's and with respect to their practical usefulness. ELI performance appeared to be superior. In particular, only with ELI support overconfidence appeared to be almost absent.

The results were promising and together with the *Interuniversity expertise center ProGAMMA*, we will continue developing ELI. In view of the potential advantages of using proper scoring rules and in view of the positive experimental results thus far, it is expected that ELI will prove to be an efficient elicitation technique that (a) yields reliable and valid SPD's, (b) is acceptable for different groups of assessors, and (c) is suitable for different applied and experimental settings.

INDIVIDUAL DIFFERENCES IN DECISION MAKING STRATEGIES:
EFFECTS OF NEED FOR COGNITION AND TIME PRESSURE ON
EXTERNAL INFORMATION SEARCH

Bas Verplanken

University of Nijmegen,
Department of Social Psychology,
PO Box 9104, 6500 HE Nijmegen, The Netherlands

In the process tracing tradition of research on decision making relatively little attention has been given to individual differences in external information search strategies and decision making (Ford et al., 1989). The present study focused on effects of 'need for cognition' on external search. Need for cognition refers to relatively stable individual differences in people's motivation to engage in cognitive endeavors in general.

An experiment was performed using the information display board paradigm with subjects who varied widely in age and educational background. Subjects performed an information acquisition task either under conditions of time pressure or without time constraints. As expected, differences in search behavior were found as function of need for cognition and time pressure. Measures of cognitive responses and self-reports concerning the search task confirm an explanation of these results in terms of cognitive effort subjects were motivated to expend to the task.

Ford, J.K., Schmitt, N., Schechtman, S.L., Hults, B.M. & Doherty, M.L. (1989). Process tracing methods: contributions, problems, and neglected research questions. Organizational Behavior and Human Decision Processes, 43, 75-117.

RESPONSE MODES AND DECISION PROCESSES

Mirjam R. M. Westenberg
Department of Clinical Psychology
University of Amsterdam

Until recently research in the field of riskless multiattribute decision making has generally ignored a distinction between response modes, i.e. the type of response required from the decision maker.

This neglect persisted in spite of the results of early studies on preference reversals in gambles, which showed a distinct difference between the decision outcomes of choice compared with those of judgment. Moreover, in riskless decision research on the decision process, the choice-judgment distinction can be used to explain the apparently contradictory findings of the two major research paradigms: structural modeling, which typically requires a numerical judgment, and process tracing, which usually asks for a choice. In general, choice seems to induce noncompensatory information processing, and judgment compensatory information processing.

We will present a classification of response modes according to their place on a choice-judgment continuum. This classification is based on the number of scale points involved in the response mode, and will be discussed in relation to the so-called prominence hypothesis (Tversky et al., 1988). According to this hypothesis, attributes are weighted more equally in judgment than in choice; in other words, judgment induces more compensatory processing than choice does. In addition, empirical support for the proposed classification of response modes will be presented.

Tversky, A., Sattath, S. and Slovic, P. (1988). Contingent weighting in judgment and choice. *Psychological Review*, 95, 371-384.

Why is "prospect theory" still fascinating?

Andreas Wilm

Institut für Psychologie, Christian-Albrechts-Universität zu Kiel

In 1979, Kahneman and Tversky presented their prospect theory (PT) as a descriptive model to account for preferences in decision making under risk. PT can describe some effects in experiments on choice behavior violating the axioms of expected utility theory, especially "framing effects" (violating the invariance principle) and the "certainty effect" (violating the independence axiom). Until today, PT has been cited in numberless papers, including some studies by other authors in which PT has been critically examined (e.g. Hershey & Schoemaker 1980; Lehner 1980; Cohen, Jaffray & Said 1987).

In my paper I present some of the empirical results (including my own experimental work) violating PT. Meanwhile the contradictions refer to all components of PT, i.e. not only to the "reflection effect". Hence PT cannot be correct. An analysis of the contradictions shows that especially the assumed form of the "value function" is problematic. This may be a hint that the more general forms of the "value function" and the "weighting function" in the sources of PT (Markowitz 1952 and Edwards 1962) could be a better basis for the formal model of PT (see Yilmaz 1983 for a generalization of a "value function" with three inflection points and the notion of a "subjective probability function book" in Edwards 1962).

In contrast to the empirical results against PT there are papers referring to PT positively in the field of applied decision making (e.g. Puto 1987; Rowe & Puto 1987; Klein & Oglethorpe 1987). Reasons for this psychological view are presumable the point of reference and his variability which are apparently the most fascinating aspects of PT (see e.g. Van Schlie & Van der Pligt 1990). But these and related studies show that even a modified PT cannot really account for preferences unless there is a theory making assertions regarding the variability of the point of reference. For example, such a theory might reveal the "certainty effect" as a "framing effect" (Grossberg & Gutowski 1987)—contradicting the assumptions of PT.

COHERENCE, CALIBRATION AND EXPERTISE IN
JUDGEMENTAL PROBABILITY FORECASTING

George Wright
Strathclyde Graduate Business School

Fergus Bolger
University College London

Gene Rowe
Bristol Business School

John Gormack
Bristol Business School

This study investigates the relationship between judgemental probability forecasting performance, self-rated expertise and degree of coherence with the probability laws. Self-rated expertise was found to be a good predictor of subsequent forecasting performance whereas measures of individual coherence were less predictive. However, recompositions of individual subjects' marginal and conditional assessments into intersections, disjunctions and unions proved a useful mechanical procedure since those recomposed assessments showed improved performance relative to their holistically-assessed counterparts. The act of recombination produced the most marked improvement in forecasting performance for novices. These results provide evidence that decomposition of compound assessments into "simpler" assessments which are then subsequently recomposed is a useful mechanical debiasing procedure.

Evidence of consistent individual differences in forecasting ability was found across item sets, suggesting that it may be possible to select those individuals who possess differential abilities in judgmental probability forecasting.

AUTHORS LIST

- Atman, Cynthia (Pittsburgh) 159
 Ayton, Peter (London) 150
 Bar-Hillel, Maja (Jerusalem) 149, 151
 Barlas, Sema (Chicago) 155
 Barron, Hutton F. (Alabama) 149, 152
 Bartczak, Gizela (Warsaw) 153
 Beach Lee Roy (Tucson) 154
 Beattie, Jane (Chicago) 155
 Benartzi, Shlomo (Ithaca) 156
 Benson III, Lehman (Sweden) 145
 Bettman, James R. (Durham) 80
 Bohanec, Marko (Ljubljana) 157
 Böhm, Gisela (Berlin) 195
 Bolger, Fergus (London) 149, 173, 199, 214
 Bonini, Nicolao (Trieste) 158
 Borcharding, Katrin (Darmstadt)
 Bostrom, Ann (Pittsburgh) 146, 159
 Braspenning, Jozé (Amsterdam) 160
 Brehmer, Berndt (Uppsala) 3
 Brown, Rex V. (Reston) 144
 Brun, Wibecke (Norway) 146
 Budescu, David V. (Haifa) 161
 Burgemeister, K. (Germany) 146
 Camerer, Colin (Philadelphia) 162
 Cavemi, Jean-Paul (Aix-en-Provence) 163
 Chapman, P.R. (Cambridge) 164
 Coupey, Eloise (Durham) 80
 Crozier, W.R. (Cardiff) 165
 D'Ydewalle, G. (Leuven) 178
 De Bruyn, Eric (Nijmegen) 149, 184
 Edland, Ann (Sweden) 145
 Englander, Tibor (Hungary) 149
 Eppel, Thomas (West Lafayette) 166
 Erb, Hans-Peter (Mannheim) 175
 Emst, Andreas (Freiburg i.Br.) 167
 Evers-Kiebooms, G. (Leuven) 178
 Farago, Klara (Budapest)
 Femers, Susanne (Germany) 146
 Fischhoff, Baruch (Pittsburgh) 159
 Frumkina, R.M. (Moscow) 168
 Gallhofer, I.N. (Amsterdam) 169
 Gambara, Hilda (Madrid) 187
 Garber, Ilya (Saratov) 170
 Gärling, Tommy (Umea) 171, 190
 Garvill, Jörgen (Umea) 171
 Gertzen, Heiner (Heidelberg) 172
 Gigerenzer, Gerd (Salzburg) 176
 Gormack, John (Bristol) 214
 Groeger, J.A. (Cambridge) 164
 Harvey, Nigel (London) 150, 173
 Hendrickx, Laurie (Groningen) 174
 Henneken, Oscar (Amsterdam) 182
 Hilton, Denis (Toulouse) 175
 Hoffrage, Ulrich (Salzburg) 176
 Hogarth, Robin (Pennsylvania) 186
 Holtgrave, David R. (Oklahoma) 177
 Huber, Oswald (Fribourg) 147
 Huys, J. (Leuven) 178
 Jakubowski, Andrzej (Warsaw) 179
 Janetzko, D. (Germany) 148
 Johnson, Eric J. (Durham) 80, 147, 162
 Jungermann, H. (Germany) 144
 Juslin, Peter (Uppsala) 180
 Karbach, W. (Germany) 148
 Keren, Gideon (Utrecht) 124
 Kerstholt, José H. (Soesterberg) 181
 Koele, Pieter (Amsterdam) 182
 Krasnenker, Alexander (Moscow) 183
 Krol, Nicole (Nijmegen) 184
 Kühberger, Anton (Salzburg) 185
 Kunreuther, Howard (Pennsylvania) 186
 Larichev, Oleg (Moscow) 144, 148
 Leclerc, F. (USA) 147
 Léon, Orfelio (Madrid) 187
 Levchenkov, S. (USSR) 146
 Lewicka, Maria (Poland) 147
 Liebrand, Wim B.G. (Amsterdam) 28
 Lindberg, Erik (Umea) 171, 190
 Louviere, Jordan L. (Edmonton) 193
 Lundberg, U. (Sweden) 145
 Macchi, Laura (Milano) 188
 Manago, M. (France) 148
 Mano, H. (USA) 145
 Maule, John A. (Leeds) 145
 McClelland, Alastair (London) 173
 Messick, David M. (Santa Barbara) 189
 Meszaros, Jacqueline (Pennsylvania) 186
 Mitchell, Deborah (Pennsylvania) 147, 191
 Montgomery, Henry (Göteborg) 147, 171, 190
 Morgan, Granger (Pittsburgh) 159
 Moshkovich, Helen (Moscow) 44
 Mumpower, Jeryl L. (Albany) 149
 Neter, Efrat (Jerusalem) 151
 Omodei, Mary (Melbourne) 192
 Oppewal, Harmen (Eindhoven) 193
 Otten, Wilma (Amsterdam) 194
 Payne, John W. (Durham) 80
 Peris, Jean-Luc (Aix-en-Provence) 163
 Pfister, H.-Rüdiger (Berlin) 195
 Phillips, Lawrence (London) 144, 196
 Phillips, Maryann (London) 196
 Pidgeon, N. (UK) 146
 Polister, P. (USA) 144

- Potter, Richard (Tucson) 154
 Purkitt, H. (USA) 149
 Rajkovic, Vladislav (Maribor) 157
 Ranyard, R.H. (Bolton) 147, 165
 Rapoport, Amnon (Haifa) 161
 Rebric, S. (USSR) 148
 Renn, Ortwin (USA) 146
 Retschitzki, Jean (Fribourg) 197
 Richard, René (Amsterdam) 198
 Rohrbaugh, J.W. (USA) 144
 Rohrmann, Bernd (Mannheim) 146
 Rowe, Gene (Bristol) 199, 214
 Ruhrmann, Georg (Germany) 146
 Russo, Edward (Ithaca) 147
 Sarin, R. (USA) 144
 Saris, W.E. (Amsterdam) 169
 Schell, Terry (Santa Barbara) 189
 Schmalhofer, Franz (Kaiserslautern) 148
 Schütz, Holger (Germany) 146
 Selart, Marcus (Göteborg) 190
 Shanteau, Jim (Kansas) 148, 149
 Shapira, Zur (New York) 144, 200
 Sokolowska, Joanna (Warsaw) 201
 Spranca, Mark (Pennsylvania) 186
 Stokmans, Mia (Tilburg) 202
 Stouppe, Melinda S. (Cambridge) 203
 Strube, Gerhard (Germany) 148
 Suleiman, Ramzi (Tucson) 161
 Svenson, Ola (Lund) 145, 146, 204
 Tanaevsky, V.G. (Minsk) 205
 Teigen, Karl Halvor (Bergen) 206
 Terekhina, A. Y. (USSR) 149
 Thaler, Richard (Ithaca) 156
 Thüring, Manfred (Darmstadt) 207
 Timmermans, Daniëlle (Leiden) 208
 Timmermans, Harry (Eindhoven) 193
 Tinsley, Barbara (California) 177
 Tyszk, Tadeusz (Warsaw) 201
 Urh, Bozo (Ljubljana) 157
 Van den Bercken, John (Nijmegen) 149, 184
 Van der Pligt, Joop (Amsterdam) 194
 Van Lange, Paul (Amsterdam) 209
 Van Lenthe, Jelle (Groningen) 210
 Van Schie, Els (Amsterdam) 158
 Vari, Anna (Albany) 144
 Verplanken, Bas (Nijmegen) 145, 211
 Vlek, Charles (Groningen) 124, 146, 174
 Voogt, Robert (Amsterdam) 169
 Wagenaar, W. (Netherlands) 149
 Waller, W. (USA) 149
 Watson, Stephen R. (Cambridge) 125, 144
 Wearing, Alexander J. (Melbourne) 145, 192
 Westenberg, Mirjam (Amsterdam) 212
 Wiedemann, P. (Germany) 146
 Wilm, Andreas (Kiel) 213
 Wright, George (London) 149, 199, 214
 Wright, A. (USA) 149
 Yaniv, I. (USA) 147